The Exact Marked Bridge Between the Erdős–Straus Divisor Selector and the S^6 Terminal Presentation with the Modulo-840 Lattice, Triality, and Exact Shell Maps

Independent reconstruction note

6 September 2026

1 The exact marked Erdős–Straus–S^6 bridge

Current lattice and modular continuation

The shared continuation now proves the full cubic image $2\sqrt{2}(18\mathbb{Z} \cup (6\mathbb{Z}+2) \cup (6\mathbb{Z}-2))$ of the literal common Niemeier–Leech lattice, with explicit integral and local sections. The marked $A_5^4 D_4$ root system maps to the Cheng–Duncan–Harvey shadow $(5I_{12}+J)S_6$, where $J_{rs} = \mathbf{1}_{s=5r \ (12)}$. The proof calculates the original glue-group action on twisted shadows, the complete theta determinant character, and the coefficient map to $T_6 = \eta(\tau)^5\eta(3\tau)/(\eta(2\tau)\eta(6\tau)^5)$, including $T_6(-1/(6\tau)) = 72/T_6(\tau)$. See theorem 8.1 and section 9.

Result and scope

For each prime $p \equiv 1 \pmod{12}$, the complete first-half Erdős–Straus divisor selector is in explicit bijection with a normalized, positively oriented, divisor-supported, and shell-marked slice of the terminal parameter lattice occurring in the candidate complex structure on S^6. The coordinate change is integral in both directions. More strongly, the selector relation matrix and the S6 terminal relation matrix have exactly the same row lattice: they differ by a displayed element of $\mathrm{SL}_2(\mathbb{Z})$. Composing the bijection with the shell reconstruction gives positive denominators solving $4/p = 1/x + 1/y + 1/z$, and every positive solution is detected by at least one of the two tagged slices.

The nonlinear condition $u \mid a_j^2$, together with the external labels (p, j, ε), is indispensable. If it is removed, each normalized S6 fibre is an infinite affine line and carries no Erdős–Straus existence information. A particularly natural unmarked S6 point produces precisely the already-solved congruence class $p \equiv 13 \pmod{24}$ and produces no selector state when $p \equiv 1 \pmod{24}$. Thus the bridge is an exact arithmetic equivalence of marked presentations, not a proof of universal selector nonemptiness and not a geometric morphism from the reciprocal cubic to the complex threefold.

The separate continuation in Section 2 constructs the index-72 modulo-840 lattice $A_5^4 \perp D_4 \subset N$ and its exact marked D_4 triality maps. It realizes the known Weyl quotient by the coordinate splitting $O(N) = W(A_5^4 D_4) \rtimes L(G)$, $G \cong GL_2(\mathbb{F}_3)$, and places the residue six-cycle inside the Weyl group, centralizing this chosen complement (Theorems 2.15 and 2.16; [1, Table 3]). Its divisor and Fourier maps retain the prime, shell, and finite sample; in particular the shell count does not descend to the prime residue alone.

1.1 All objects and conventions

Fix throughout a prime

$$p \equiv 1 \pmod{12}, \qquad h = \frac{p-1}{12} \in \mathbb{Z}_{>0},$$

and write

$$L = \frac{p-1}{4} = 3h, \qquad J_p = \{j \in \mathbb{Z} : 0 \le j < L\}.$$

For $j \in J_p$, retain the four integers

$$R_j = 4j + 3, \qquad a_j = 3h + j + 1 = \frac{p + R_j}{4}, \qquad N_j = pa_j, \qquad 4a_j = p + R_j. \tag{1.1}$$

No label in (p, j) will be suppressed. The inequalities

$$3 \le R_j \le p - 2, \qquad \frac{p+3}{4} \le a_j \le \frac{p-1}{2} < p$$

give

$$\gcd(p, a_j) = \gcd(p, R_j) = \gcd(a_j, R_j) = 1. \tag{1.2}$$

Indeed p is prime and $0 < a_j < p$; then $R_j = 4a_j - p$ gives the other two equalities.

Let

$$\mathscr{S}(p) = \left\{(x, y, z) \in \mathbb{Z}^3_{>0} : 4xyz = p(xy + xz + yz)\right\}.$$

The equation is symmetric, but whenever completeness is invoked we choose the ordered representative $x \le y \le z$. Write $\mathrm{ES}(p)$ for the assertion $\mathscr{S}(p) \neq \varnothing$.

For the S6 terminal parameters put

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2 \tag{1.3}$$

and define the abstract terminal group

$$G(\ell_0, \ell_1, \ell_2) = \left\langle c, x, y \;\middle|\; c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \right\rangle. \tag{1.4}$$

After eliminating y, and using the ordered generator basis (x, c), its ordered row-relation matrix is

$$\mathsf{R}(\ell_0, \ell_1, \ell_2) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R} = -\lambda(\ell_0, \ell_1, \ell_2). \tag{1.5}$$

The letter p in this note always denotes the Erdős–Straus prime. The signed terminal value is denoted

$$q_\varepsilon = p^\varepsilon \in \{1, p\}, \qquad \varepsilon \in \{0, 1\}.$$

Thus no identification of the external prime p with the terminal value is made in the $\varepsilon = 0$ channel.

For any integral 2×2 matrix A, define its row-relation lattice and row-presented group by

$$\mathrm{Rel}(A) = \{v^{\mathsf{T}} A : v \in \mathbb{Z}^2\} \subseteq \mathbb{Z}^2, \qquad \mathcal{G}(A) = \mathbb{Z}^2 / \mathrm{Rel}(A) = \mathrm{coker}(A^{\mathsf{T}}). \tag{1.6}$$

This convention makes every matrix direction in the bridge explicit.

1.2 The complete first-half shell selector

Lemma 1.1 (first-half denominator)**.** *Let $p \equiv 1 \pmod 4$ be an odd prime. If $(x, y, z) \in \mathscr{S}(p)$ and $x \le y \le z$, then*

$$\frac{p}{4} < x \le \frac{p}{2}.$$

Proof. Positivity of the two reciprocals other than $1/x$ gives $1/x < 4/p$, hence $x > p/4$. Suppose $x > p/2$. Clearing denominators gives

$$4xyz = p(xy + xz + yz),$$

so $p \mid xyz$. If $p \mid x$, then $x, y, z \ge p$ and the reciprocal sum is at most $3/p$. If $p \mid y$, then $y, z \ge p$, while $1/x < 2/p$, so the sum is strictly less than $4/p$. Both cases are impossible. Therefore $z = kp$ for some $k \in \mathbb{Z}_{>0}$, while neither x nor y is divisible by p. Substitution yields

$$(4k - 1)xy = kp(x + y).$$

Reduction modulo p gives $p \mid 4k - 1$. Write $4k - 1 = rp$. Since $p \equiv 1 \pmod 4$, one has $r \equiv 3 \pmod 4$, hence $r \ge 3$. After cancelling p,

$$rxy = k(x + y).$$

As $x \le y$, division by xy gives

$$x \le \frac{2k}{r} = \frac{rp + 1}{2r} = \frac{p}{2} + \frac{1}{2r} < \frac{p + 1}{2}.$$

The integer $x > p/2$ must be at least $(p + 1)/2$, a contradiction. □

For fixed $j \in J_p$, define the ordered residual shell

$$\mathscr{P}_{p,j} = \{(y, z) \in \mathbb{Z}^2_{>0} : R_j yz = N_j(y + z)\} \tag{2.1}$$

and the oriented divisor shell

$$\mathscr{D}_{p,j} = \{d \in \mathbb{Z}_{>0} : d \mid N_j^2,\ d \equiv -N_j \pmod{R_j}\}. \tag{2.2}$$

Proposition 1.2 (oriented shell bijection)**.** *The maps*

$$\begin{aligned} \delta_{p,j} &: \mathscr{P}_{p,j} \longrightarrow \mathscr{D}_{p,j}, & (y, z) &\longmapsto R_j y - N_j, \\ \rho_{p,j} &: \mathscr{D}_{p,j} \longrightarrow \mathscr{P}_{p,j}, & d &\longmapsto \left(\frac{N_j + d}{R_j}, \frac{N_j + N_j^2/d}{R_j}\right) \end{aligned}$$

are mutually inverse. The involution

$$\iota_{N_j}(d) = \frac{N_j^2}{d}$$

corresponds exactly to $(y, z) \mapsto (z, y)$.

Proof. Abbreviate $R = R_j$ and $N = N_j$. Equation (2.1) is equivalent to

$$(Ry - N)(Rz - N) = N^2. \tag{2.3}$$

Both factors are positive: for example, division of $Ryz = N(y+z)$ by z gives $Ry = N(1+y/z) > N$, and similarly for $Rz - N$. Therefore $d = Ry - N$ is a positive divisor of N^2 and $d \equiv -N \pmod{R}$.

Conversely, let $d \in \mathscr{D}_{p,j}$ and put $e = N^2/d$. By (1.2), d and N are units modulo R; hence

$$e \equiv N^2(-N)^{-1} \equiv -N \pmod{R}.$$

Thus both components of $\rho_{p,j}(d)$ are positive integers. Their two shifted factors are d and e, so (2.3), and therefore (2.1), holds. The formulas visibly compose to the identity. Replacing d by $e = N^2/d$ exchanges the two reconstructed components. □

For $\varepsilon \in \{0,1\}$, define the fixed-shell selector and the fully marked selector by

$$\mathcal{W}_\varepsilon(p,j) = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p^\varepsilon\}, \tag{2.4}$$

$$\mathcal{W}_\varepsilon(p) = \{(j,u) : j \in J_p,\ u \in \mathcal{W}_\varepsilon(p,j)\}. \tag{2.5}$$

The tag $\varepsilon = 0$ is the exterior, or E, channel; $\varepsilon = 1$ is the middle, or M, channel. Whenever a disjoint union of the two channels is written below, it means the tagged coproduct

$$\{0\} \times \mathcal{W}_0(p) \ \sqcup\ \{1\} \times \mathcal{W}_1(p).$$

This convention is necessary because the two raw subsets can intersect.

Proposition 1.3 (exact two-channel decomposition)**.** *The maps*

$$\iota_E : \mathcal{W}_0(p,j) \longrightarrow \mathscr{D}_{p,j}, \qquad u \longmapsto p^2 u,$$
$$\iota_M : \mathcal{W}_1(p,j) \longrightarrow \mathscr{D}_{p,j}, \qquad u \longmapsto pu$$

are bijections onto the subsets on which the p-adic valuations are 2 and 1, respectively. The valuation-zero subset is $\iota_{N_j}(\iota_E(\mathcal{W}_0(p,j)))$. Consequently

$$\mathscr{D}_{p,j} = \iota_E(\mathcal{W}_0(p,j)) \ \sqcup\ \iota_M(\mathcal{W}_1(p,j)) \ \sqcup\ \iota_{N_j}(\iota_E(\mathcal{W}_0(p,j))), \tag{2.6}$$

and

$$|\mathscr{D}_{p,j}| = 2|\mathcal{W}_0(p,j)| + |\mathcal{W}_1(p,j)|. \tag{2.7}$$

Proof. Because $p \nmid a_j$, every divisor $d \mid p^2 a_j^2$ has $v_p(d) \in \{0,1,2\}$.

If $d = p^2 u$, then $u \mid a_j^2$. The shell congruence is

$$R_j \mid d + N_j = p(pu + a_j).$$

After cancelling the unit p modulo R_j and multiplying by 4, the identity $4a_j = p + R_j$ makes this equivalent to $R_j \mid 4u + 1$. This proves the E-channel assertion in both directions. If $d = pu$, the same calculation gives

$$R_j \mid d + N_j \iff R_j \mid u + a_j \iff R_j \mid 4u + p,$$

which proves the M-channel assertion. Complementation preserves the shell congruence by Proposition 1.2, exchanges valuations 0 and 2, and preserves valuation 1. This proves the disjoint decomposition and its cardinality formula. □

For $u \in \mathcal{W}_\varepsilon(p,j)$, define the two exact quotients

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4}. \tag{2.8}$$

Since $R_j \equiv 3 \pmod 4$ and $p^\varepsilon \equiv 1 \pmod 4$, the first equation gives $T \equiv 3 \pmod 4$. Hence $K \in \mathbb{Z}_{>0}$. Direct calculation, without dropping the tag, gives

$$R_j K = a_j + p^{1-\varepsilon}u, \qquad T a_j = u + p^\varepsilon K. \tag{2.9}$$

Indeed, multiplication by 4 changes these identities into

$$\begin{aligned} 4R_jK &= p^{1-\varepsilon}(4u + p^\varepsilon) + R_j = 4p^{1-\varepsilon}u + p + R_j, \\ 4Ta_j &= T(p + R_j) = pT + 4u + p^\varepsilon = 4u + 4p^\varepsilon K. \end{aligned}$$

Proposition 1.4 (denominator reconstruction)**.** *The map*

$$\Psi_{p,j,\varepsilon} : \mathcal{W}_\varepsilon(p,j) \longrightarrow \mathscr{S}(p), \qquad u \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u}\right) \tag{2.10}$$

is well-defined. Its two oriented shell factors are

$$R_j(pK) - N_j = p^{2-\varepsilon}u, \qquad R_j\left(\frac{p^\varepsilon a_j K}{u}\right) - N_j = p^\varepsilon \frac{a_j^2}{u}. \tag{2.11}$$

Moreover,

$$\mathrm{ES}(p) \quad \Longleftrightarrow \quad (\{0\} \times \mathcal{W}_0(p)) \sqcup (\{1\} \times \mathcal{W}_1(p)) \neq \varnothing. \tag{2.12}$$

Proof. The last coordinate in (2.10) is integral. To see this without assuming it, reduce the second identity in (2.9) modulo u, multiply by a_j, and use $u \mid a_j^2$: this gives $u \mid p^\varepsilon a_j K$. The two identities (2.11) now follow from the first identity in (2.9). Their product is N_j^2, so Proposition 1.2 shows that (2.10) lies in $\mathscr{S}(p)$.

Conversely, suppose $\mathscr{S}(p) \neq \varnothing$, and sort a solution. Lemma 1.1 shows that its smallest denominator is uniquely of the form $x = a_j$ with $j \in J_p$: explicitly

$$R_j = 4x - p \equiv 3 \pmod 4, \qquad j = \frac{4x - p - 3}{4}.$$

Proposition 1.2 supplies a divisor $d \in \mathscr{D}_{p,j}$, and Proposition 1.3 places either d or its complement in the image of ι_E, or places d in the image of ι_M. Hence one of the two marked selector sets is nonempty. The forward implication follows from (2.10). □

1.3 The exact selector–terminal-lattice equivalence

Define the normalized, positively oriented, divisor-supported terminal slice

$$\mathcal{H}_\varepsilon(p) = \left\{ (j, \ell_0, \ell_1) \in \mathbb{Z}^3 : \begin{array}{l} j \in J_p, \quad \lambda(\ell_0, \ell_1, 1) = p^\varepsilon, \\ T(\ell_0) := 4\ell_0 - 1 > 0, \\ u(j,\ell_0,\ell_1) := \ell_1 + jT(\ell_0) > 0, \\ u(j,\ell_0,\ell_1) \mid a_j^2 \end{array} \right\}. \tag{3.1}$$

Let $\mathcal{H}_\varepsilon(p,j)$ be the fibre with the displayed marker j.

For a selector state define its relation matrix

$$\mathsf{A}_\varepsilon(p;j,u)=\begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix}, \qquad R_jT=4u+p^\varepsilon. \tag{3.2}$$

For the marker j, retain both inverse unimodular shears

$$\mathsf{U}_j=\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{L}_j=\mathsf{U}_j^{-1}=\begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}. \tag{3.3}$$

Theorem 1.5 (strong marked bridge theorem)**.** *For every $\varepsilon\in\{0,1\}$, the map*

$$\Theta_{\varepsilon,p}:\mathcal{W}_\varepsilon(p)\longrightarrow\mathcal{H}_\varepsilon(p), \qquad (j,u)\longmapsto\left(j,\frac{T+1}{4},u-jT\right) \tag{3.4}$$

is a bijection. Its inverse is

$$\Theta_{\varepsilon,p}^{-1}(j,\ell_0,\ell_1)=(j,\ell_1+j(4\ell_0-1)). \tag{3.5}$$

For corresponding points, with $\ell=(\ell_0,\ell_1,1)$, one has the exact matrix identities

$$\boxed{\mathsf{U}_j\mathsf{A}_\varepsilon(p;j,u)=\mathsf{R}(\ell_0,\ell_1,1), \qquad \mathsf{A}_\varepsilon(p;j,u)=\mathsf{L}_j\mathsf{R}(\ell_0,\ell_1,1).} \tag{3.6}$$

In particular,

$$\mathrm{Rel}(\mathsf{A}_\varepsilon(p;j,u))=\mathrm{Rel}(\mathsf{R}(\ell_0,\ell_1,1)), \qquad \det\mathsf{A}_\varepsilon=\det\mathsf{R}=-p^\varepsilon. \tag{3.7}$$

Thus the bridge intertwines the entire integral row presentation, not only its determinant.

Proof. Take $(j,u)\in\mathcal{W}_\varepsilon(p)$. Section 2 proves that T is a positive integer satisfying $T\equiv 3 \pmod 4$, so $\ell_0=(T+1)/4\in\mathbb{Z}$. Put $\ell_1=u-jT$. Since $R_j=4j+3$,

$$\begin{aligned}\lambda(\ell_0,\ell_1,1)&=3(T+1)-4(u-jT)-3\\&=(4j+3)T-4u=p^\varepsilon.\end{aligned}$$

The positivity and divisor-support conditions in (3.1) hold because $u=\ell_1+jT$. Thus (3.4) is well-defined.

Conversely, take $(j,\ell_0,\ell_1)\in\mathcal{H}_\varepsilon(p)$, and set

$$T=4\ell_0-1, \qquad u=\ell_1+jT.$$

Expanding $\lambda(\ell_0,\ell_1,1)=p^\varepsilon$ gives

$$(4j+3)T-4u=p^\varepsilon,$$

so $R_j\mid 4u+p^\varepsilon$. All other selector conditions occur explicitly in (3.1). The two constructions recover T,u,ℓ_0,ℓ_1 literally, which proves bijectivity and (3.5).

Finally, direct multiplication gives

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 4j+3 & -u \\ 4 & -T \end{pmatrix}=\begin{pmatrix} 3 & -u+jT \\ 4 & -T \end{pmatrix}=\begin{pmatrix} 3 & -\ell_1 \\ 4 & 1-4\ell_0 \end{pmatrix}.$$

Both shears have determinant 1, so their integral row operations are invertible and the row lattices are equal. The signed determinant is $-R_jT+4u=-p^\varepsilon$, proving (3.7). □

Corollary 1.6 (presented groups and the Erdős–Straus equation)**.** *For every corresponding selector/S6 point,*

$$\mathcal{G}(\mathsf{A}_\varepsilon(p;j,u)) = \mathcal{G}(\mathsf{R}(\ell_0,\ell_1,1)) \cong G(\ell_0,\ell_1,1) \cong \mathbb{Z}/p^\varepsilon\mathbb{Z}. \tag{3.8}$$

Here $\mathbb{Z}/1\mathbb{Z}$ is the trivial group. Moreover the direct map

$$\widehat{\Psi}_{\varepsilon,p} : \mathcal{H}_\varepsilon(p) \longrightarrow \mathscr{S}(p) \tag{3.9}$$

is obtained by (3.5), (2.8), and

$$(j,\ell_0,\ell_1) \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u}\right), \quad \begin{cases} T = 4\ell_0 - 1, \\ u = \ell_1 + jT, \\ K = (p^{1-\varepsilon}T+1)/4. \end{cases} \tag{3.10}$$

It satisfies

$$\boxed{\mathrm{ES}(p) \Longleftrightarrow (\{0\}\times\mathcal{H}_0(p)) \sqcup (\{1\}\times\mathcal{H}_1(p)) \neq \varnothing.} \tag{3.11}$$

Proof. The relation $xy = c^{\ell_0}$ in (1.4) gives $y = x^{-1}c^{\ell_0}$; since c is central, x and y commute. The remaining two relations are the rows of (1.5). The greatest common divisor of the four entries of R is 1, because it divides both 3 and 4. Its Smith form is therefore $\mathrm{diag}(1, p^\varepsilon)$, proving (3.8). Formula (3.10) is the composition of Theorem 1.5 with Proposition 1.4, and (3.11) follows from (2.12). □

Remark 1.7 (the selector tag and the solution map). The channel tag is not redundant. For $p = 13$, $j = 0$, and $u = 2$, one has $u \in \mathcal{W}_0(p,j) \cap \mathcal{W}_1(p,j)$, but the two tags give different terminal values and different marked terminal points. Nor is the map to unordered denominator triples injective: in the M-channel at $p = 13, j = 0$, the complementary divisors $u = 2$ and $u = 8 = a_0^2/2$ both give the unordered solution $\{4, 26, 52\}$. The bijection in Theorem 1.5 is a bijection of marked selector states with marked terminal states, not a bijection with unordered solutions.

Corollary 1.8 (fixed-fibre affine coordinates and the intrinsic gate)**.** *Put*

$$C_\varepsilon = \frac{p^\varepsilon + 3}{4}, \qquad D_{\varepsilon,j} = j + C_\varepsilon = \frac{R_j + p^\varepsilon}{4}.$$

Every integral point with $\ell_2 = 1$ and $\lambda(\ell_0,\ell_1,1) = p^\varepsilon$ is uniquely

$$(\ell_0,\ell_1,\ell_2) = (n, 3n - C_\varepsilon, 1), \qquad n \in \mathbb{Z}, \tag{3.12}$$

and, for each fixed marker $j \in J_p$, define

$$T = 4n - 1, \qquad u = R_j n - D_{\varepsilon,j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4}. \tag{3.13}$$

Then $T \equiv 3 \pmod 4$, so $K \in \mathbb{Z}$, and the congruence equation $R_j T = 4u + p^\varepsilon$ holds identically. On the locus $T > 0$ and $u > 0$, one has

$$\gcd(R_j, u) = 1, \qquad u \mid a_j^2 \Longleftrightarrow u \mid K^2. \tag{3.14}$$

Thus the nonlinear divisor gate can be written entirely in the S6 coordinates, but it does not disappear.

Proof. Equation (3.12) is the direct solution of $12\ell_0 - 4\ell_1 - 3 = p^\varepsilon$. Substitution gives the first two formulas in (3.13), and $T \equiv 3 \pmod 4$ proves that the displayed integer K is well defined. The definitions give

$$4u = 4R_j n - (R_j + p^\varepsilon) = R_j T - p^\varepsilon,$$

so the congruence equation holds for every integral point in (3.12). If a positive integer divides both R_j and u, then the congruence equation makes it divide p^ε; (1.2) therefore gives $\gcd(R_j, u) = 1$. Without invoking the selector hypothesis, direct substitution gives

$$4\big(R_j K - p^{1-\varepsilon} u - a_j\big) = p^{1-\varepsilon}(R_j T - 4u) - p = 0.$$

Thus $a_j \equiv R_j K \pmod u$. Since R_j is a unit modulo u, $a_j^2 \equiv R_j^2 K^2 \pmod u$, which proves (3.14) in both directions. □

The normalization automatically lies in the admissible residue class used by the S6 terminal presentation:

$$\ell_2 = 1 \text{ is odd}, \qquad \ell_1 \equiv 2 \pmod 3. \tag{3.15}$$

Indeed reduction of $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon \equiv 1 \pmod 3$ gives $-\ell_1 \equiv 1 \pmod 3$.

Remark 1.9 (orientation and the constructed S6 candidate). The geometrically reconstructed candidate elsewhere in the audited corpus uses the oriented terminal parameter $\lambda(0, 1, -1) = -1$. The two arithmetic channels in this note use the positive terminal values $+1$ and $+p$. No audited map identifies the -1 terminal datum with either positive datum. Consequently the row shear proved here does not, by itself, place any selector state on the particular complex threefold constructed in the -1 branch.

1.4 The exact remaining arithmetic incidence

The bridge can be expressed without an untyped phrase such as "some divisor of a square." The next factor coordinates are canonical and retain their orientation.

Lemma 1.10 (canonical ordered factor coordinates)**.** *For $a, u \in \mathbb{Z}_{>0}$, the condition $u \mid a^2$ is equivalent to the existence of a unique ordered triple $(r, s, t) \in \mathbb{Z}_{>0}^3$ such that*

$$u = r^2 s, \qquad a = rst, \qquad \gcd(r, t) = 1. \tag{4.1}$$

Explicitly, with $g = \gcd(a, u)$,

$$r = \frac{u}{g}, \qquad s = \frac{g^2}{u}, \qquad t = \frac{a}{g}. \tag{4.2}$$

Proof. Write $v_q(a) = \alpha$ and $v_q(u) = \beta \le 2\alpha$ prime by prime. Then

$$v_q(g) = \min(\alpha, \beta), \qquad v_q(g^2/u) = 2\min(\alpha, \beta) - \beta \ge 0.$$

Thus (4.2) consists of positive integers. If $\beta \le \alpha$, then $q \nmid r$; if $\beta \ge \alpha$, then $q \nmid t$. Hence $\gcd(r, t) = 1$, and direct multiplication gives (4.1).

Conversely, (4.1) gives $a^2/u = st^2 \in \mathbb{Z}_{>0}$. It also gives $\gcd(a, u) = rs$, because $\gcd(r, t) = 1$. Formula (4.2) then recovers (r, s, t), proving uniqueness. □

Proposition 1.11 (factor form of both selector channels)**.** *For the unique factorization $u = r^2 s$, $a_j = rst$, $\gcd(r,t) = 1$, one has*

$$u \in \mathcal{W}_0(p,j) \iff R_j \mid pr + t, \tag{4.3}$$

$$u \in \mathcal{W}_1(p,j) \iff R_j \mid r + t. \tag{4.4}$$

Proof. All of r, s, t are units modulo R_j, because their product is a_j and $\gcd(a_j, R_j) = 1$. For the E-channel,

$$t(4u+1) = 4r^2 st + t = (p + R_j)r + t = (pr + t) + R_j r.$$

Cancelling t modulo R_j proves (4.3). For the M-channel,

$$4u + p \equiv 4(u + a_j) = 4rs(r + t) \pmod{R_j},$$

and cancellation of the unit $4rs$ proves (4.4). □

Theorem 1.12 (necessary-and-sufficient counterexample criterion)**.** *A prime $p \equiv 1 \pmod{12}$ is a counterexample to the Erdős–Straus equation if and only if, for every $j \in J_p$ and every ordered $(r,s,t) \in \mathbb{Z}^3_{>0}$ satisfying*

$$a_j = rst, \qquad \gcd(r,t) = 1,$$

one has simultaneously

$$R_j \nmid pr + t, \qquad R_j \nmid r + t. \tag{4.5}$$

Equivalently, with the tag retained, it is necessary and sufficient that

$$R_j \nmid p^{1-\varepsilon} r + t \tag{4.6}$$

for every $j \in J_p$, every $\varepsilon \in \{0,1\}$, and every such factor triple.

Proof. Lemma 1.10 is a bijection between all allowed divisors in the selector (2.4) and all displayed ordered factor triples. Proposition 1.11 translates the two selector congruences without loss. The assertion now follows directly from (2.12), or equivalently (3.11), by negating the finite existential statement. □

This theorem isolates the unresolved problem exactly: the S6 row shear has not removed the bounded factor-incidence condition (4.5).

1.5 Why the labels and support cannot be forgotten

Lemma 1.13 (full terminal fibres are affine lattices)**.** *The kernel of $\lambda : \mathbb{Z}^3 \to \mathbb{Z}$ is*

$$\ker \lambda = \mathbb{Z}(1,3,0) \oplus \mathbb{Z}(1,0,4). \tag{5.1}$$

If $q \equiv 1 \pmod{12}$ and $m_q = (q+11)/12$, then

$$\lambda^{-1}(q) = \{(m_q + r + s,\ 2 + 3r,\ 1 + 4s) : r, s \in \mathbb{Z}\}. \tag{5.2}$$

In the normalized slice $\ell_2 = 1$, this is an infinite affine line.

Proof. If $12a - 4b - 3c = 0$, reduction modulo 4 gives $c = 4s$. The remaining equation gives $b = 3(a - s)$; putting $r = a - s$ yields $(a,b,c) = r(1,3,0) + s(1,0,4)$, uniquely. Since $\lambda(m_q, 2, 1) = 12m_q - 11 = q$, translation by (5.1) proves (5.2). □

The lemma proves that the mere existence of a terminal parameter with signed value 1 or p is automatic. It cannot imply selector nonemptiness. The following examples show separately why the marker and the divisor-support condition are both part of the theorem's domain.

Example 1.14 (forgetting j and p is noninjective)**.** For each eligible prime q, define the fixed-prime forgetful map

$$\mathfrak{f}_q : \mathcal{H}_0(q) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{f}_q(j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

At $p = 13$, the two E-states

$$(j, u, T) = (0, 2, 3), \qquad (1, 5, 3)$$

satisfy

$$\Theta_{0,13}(0, 2) = (0, 1, 2), \qquad \Theta_{0,13}(1, 5) = (1, 1, 2),$$

and both images are sent by $\mathfrak{f}_{13}$ to $(1, 2, 1)$. To compare different prime fibres, form the tagged coproduct and the global forgetful map

$$\mathfrak{F} : \bigsqcup_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{12}}} (\{q\} \times \mathcal{H}_0(q)) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{F}(q; j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

The state $(p, j, u) = (73, 1, 5)$ has $\Theta_{0,73}(1, 5) = (1, 1, 2)$, so its tagged image also maps under $\mathfrak{F}$ to $(1, 2, 1)$. Therefore neither j nor the external prime p is recoverable after applying the explicitly displayed forgetful maps.

Example 1.15 (ambient positivity does not imply divisor support)**.** For $p = 13$, $j = 2$, one has $R_j = 11$ and $a_j = 6$. The positive normalized E-line has

$$(\ell_0, \ell_1) = (n, 3n - 1), \qquad u = 11n - 3,$$

and the positive normalized M-line has

$$(\ell_0, \ell_1) = (n, 3n - 4), \qquad u = 11n - 6.$$

Both lines are infinite. The only positive $u \le a_j^2 = 36$ are

$$E : \ 8, 19, 30, \qquad M : \ 5, 16, 27,$$

and none divides 36. Hence $\mathcal{H}_0(13, 2) = \mathcal{H}_1(13, 2) = \varnothing$, even though both ambient positive S6 lines are nonempty. The prime 13 is solved at other markers, so this is a fixed-shell counterexample to forgetting support, not an Erdős–Straus counterexample.

1.6 The natural unmarked S^6 point and its complete divisor gate

The most direct attempted use of the S6 presentation starts from one point in the signed p-fibre and shears it through every shell. Its exact content can be classified completely.

Write

$$p = 12m - 11, \qquad m = h + 1 \ge 2, \qquad J_p = \{0, \dots, 3m - 4\}. \tag{6.1}$$

Retain the unmarked S6 point

$$\ell^{\mathrm{aff}} = (m, 2, 1), \qquad \lambda(\ell^{\mathrm{aff}}) = p, \tag{6.2}$$

and for $j \in J_p$ put

$$T_0 = 4m - 1, \quad u_j = 2 + jT_0, \quad a_j = 3m + j - 2, \quad R_j = 4j + 3. \tag{6.3}$$

Then

$$R_j T_0 = 4u_j + p, \qquad R_j m = a_j + u_j, \qquad T_0 a_j - u_j = pm, \tag{6.4}$$

and

$$\mathsf{U}_j \begin{pmatrix} R_j & -u_j \\ 4 & -T_0 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 4 & 1 - 4m \end{pmatrix} = \mathsf{R}(m, 2, 1). \tag{6.5}$$

Theorem 1.16 (complete affine-line gate)**.** *For prime $p = 12m - 11$, $m \geq 2$, and $0 \leq j < 3m - 3$,*

$$u_j \mid a_j^2 \iff (j = 0 \text{ and } 2 \mid m) \text{ or } (j = 2 \text{ and } 8 \mid m). \tag{6.6}$$

Consequently the single unmarked point (6.2), with all markers restored, produces an Erdős–Straus witness if and only if $p \equiv 13 \pmod{24}$. In that case $j = 0, u = 2, K = m$ gives

$$\frac{4}{p} = \frac{1}{3m - 2} + \frac{1}{pm} + \frac{1}{pm(3m - 2)/2}. \tag{6.7}$$

For $p \equiv 1 \pmod{24}$, this affine S6 point supplies no supported selector state at any $j \in J_p$.

Proof. The identities (6.4) and (6.5) follow by direct expansion. Assume first that $u_j \mid a_j^2$, and put $g = \gcd(u_j, a_j)$. Write $u_j = gu'$ and $a_j = ga'$ with $\gcd(u', a') = 1$. The assumed divisibility gives $u' \mid g(a')^2$, hence $u' \mid g$, and therefore

$$u_j \mid g^2. \tag{6.8}$$

The last identity in (6.4) gives $g \mid pm$. Since $0 < a_j < p$, the primality of p and $g \mid a_j$ imply $p \nmid g$; therefore $g \mid m$. Reducing $u_j = 2 + j(4m - 1)$ modulo m now gives

$$g \mid j - 2. \tag{6.9}$$

Suppose $j \neq 0, 2$. Then $j \geq 1$, and (6.8)–(6.9) imply $u_j \leq g^2 \leq (j - 2)^2$. But

$$u_j - (j - 2)^2 = j(4m + 3 - j) - 2 > 0 \tag{6.10}$$

For $m \geq 2$ and $1 \leq j < 3m - 3$, indeed $4m + 3 - j \geq m + 7$, so the right-hand side of (6.10) is at least $m + 5 > 0$. This is a contradiction. Thus only $j = 0$ or $j = 2$ can occur.

At $j = 0$, $u_0 = 2$ and $a_0 = 3m - 2$, so $u_0 \mid a_0^2$ exactly when m is even. At $j = 2$, $u_2 = 8m$ and $a_2 = 3m$, so

$$u_2 \mid a_2^2 \iff 8m \mid 9m^2 \iff 8 \mid m.$$

These calculations also prove both converse implications in (6.6).

Now $p = 12m - 11$ is congruent to 13 (mod 24) exactly when m is even. The $j = 0$ state lies in the M-channel, and (2.8) gives $T = T_0$ and $K = (T_0 + 1)/4 = m$. Formula (2.10) is precisely (6.7). If m is odd, (6.6) excludes every marker on this particular S6 point. □

The last sentence is deliberately local to (6.2). For example, $p = 73$ has $m = 7$ odd, so the affine point contributes nothing; nevertheless $(j, u, T; \ell_0, \ell_1) = (1, 1, 11; 3, -10)$ is an M-state. Thus failure of the affine point is not failure of the full selector.

1.7 Exact descendant maps transported through the bridge

The bridge also transports a verified selector witness along multiplicative descendant rays. This is a map from an already nonempty domain, not a nonemptiness theorem.

Fix a supported state at (p, j), retain $R = R_j$, and let $b \in \mathbb{Z}_{>0}$. Define

$$a_b = ba_j, \qquad p_b = 4a_b - R = bp + (b-1)R. \tag{7.1}$$

Whenever p_b is again a prime congruent to 1 (mod 12), the same marker j belongs to J_{p_b}, since $R < p \le p_b$, and the following maps have the displayed domains and codomains:

$$\Phi^E_{b;p} : \mathcal{H}_0(p, j) \longrightarrow \mathcal{H}_0(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, \ell_0, \ell_1), \tag{7.2}$$

$$\Phi^M_{b;p} : \mathcal{H}_1(p, j) \longrightarrow \mathcal{H}_1(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, b\ell_0, b\ell_1 - j(b-1)). \tag{7.3}$$

Write

$$\theta_{\varepsilon,p,j} : \mathcal{W}_\varepsilon(p, j) \longrightarrow \mathcal{H}_\varepsilon(p, j), \qquad \theta_{\varepsilon,p,j}(u) = \Theta_{\varepsilon,p}(j, u)$$

for the fixed-marker restriction of the marked bridge.

Proposition 1.17 (intertwining of descendant actions)**.** *The selector maps corresponding to (7.2) and (7.3) are the typed maps*

$$\psi^E_{b;p} : \mathcal{W}_0(p, j) \longrightarrow \mathcal{W}_0(p_b, j), \qquad u \longmapsto u,$$

$$\psi^M_{b;p} : \mathcal{W}_1(p, j) \longrightarrow \mathcal{W}_1(p_b, j), \qquad u \longmapsto bu.$$

They intertwine the fixed-marker bridges exactly:

$$\Phi^\bullet_{b;p} \circ \theta_{\varepsilon,p,j} = \theta_{\varepsilon,p_b,j} \circ \psi^\bullet_{b;p}, \qquad (\bullet, \varepsilon) \in \{(E, 0), (M, 1)\}. \tag{7.4}$$

Equivalently,

$$\theta^{-1}_{0,p_b,j} \Phi^E_{b;p} \theta_{0,p,j}(u) = u, \qquad \theta^{-1}_{1,p_b,j} \Phi^M_{b;p} \theta_{1,p,j}(u) = bu.$$

On the accompanying arithmetic metadata these maps replace (a_j, p, u) by (a_b, p_b, u) and (a_b, p_b, bu), respectively. For the latter,

$$T_b = bT + b - 1, \qquad u_b = bu, \qquad K_b = bK, \tag{7.5}$$

and

$$\lambda(b\ell_0, b\ell_1 - j(b-1), 1) = p_b. \tag{7.6}$$

For either tag the maps are injective. For $c \in \mathbb{Z}_{>0}$, put

$$a_{bc} = bc\, a_j, \qquad p_{bc} = 4a_{bc} - R = bc\, p + (bc-1)R.$$

Whenever p_b and p_{bc} are primes congruent to 1 (mod 12)*, the displayed maps are composable and satisfy*

$$\Phi^\bullet_{c;p_b} \circ \Phi^\bullet_{b;p} = \Phi^\bullet_{bc;p} \tag{7.7}$$

Proof. In the E-channel, $u \mid a_j^2$ implies $u \mid a_b^2$, and the congruence $R \mid 4u + 1$ and quotient T are unchanged. Hence ℓ_0, ℓ_1 are unchanged by (3.4).

In the M-channel, $bu \mid b^2 a_j^2 = a_b^2$, and

$$4bu + p_b = b(4u + p) + (b-1)R.$$

Thus the descendant is a selector state and its quotient is $T_b = bT + b - 1$. Since $K = (T+1)/4$ in this channel, $K_b = (T_b + 1)/4 = bK$. Applying (3.4) gives

$$\ell_{0,b} = \frac{T_b + 1}{4} = b\ell_0, \qquad \ell_{1,b} = bu - jT_b = b\ell_1 - j(b-1),$$

which proves (7.3)–(7.6), including the intertwining identity (7.4). Each map is visibly injective for fixed b. Substitution into (7.1)–(7.3), with the source prime retained, proves the typed composition law (7.7). □

Reverse descent need not preserve divisor support: the factor b can introduce divisors absent from a proposed parent. Therefore (7.2)–(7.7) propagate witnesses but cannot create the first witness on an empty ray.

1.8 Sharp verdict

The strongest theorem in this selector/terminal section is Theorem 1.5 together with the exhaustive equivalence (3.11): the complete tagged Erdős–Straus selector for a fixed prime $p \equiv 1 \pmod{12}$ is exactly the marked divisor-supported S6 terminal slice, and the two integral relation matrices have the same row lattice. The exact remaining obstruction is (4.5).

Three stronger readings are false or unsupported.

1. The abstract group $G(\ell)$, its determinant, or an unmarked terminal triple does not determine selector support. Every E-state has the same trivial presented group, every M-state has group $\mathbb{Z}/p\mathbb{Z}$, and Examples 1.14 and 1.15 exhibit the information lost by forgetting the marker and divisor gate.

2. The row shear is not a geometric map from the reciprocal cubic $4xyz = p(xy + xz + yz)$ to the candidate complex threefold X. No map of varieties or analytic spaces, no cycle map, no attaching map, and no cohomological correspondence is defined by left multiplication of a relation matrix.

3. The existence of the full S6 affine fibre cannot prove universal Erdős–Straus positivity. Those fibres are always nonempty by (5.2), whereas selector nonemptiness is exactly the additional finite nonlinear incidence in (4.5). The natural affine point is completely exhausted by Theorem 1.16 and contributes nothing to the unresolved $p \equiv 1 \pmod{24}$ half.

Accordingly, the bridge is a genuine theorem and is stronger than a shared determinant or a heuristic analogy. It is also exact enough to prove its own boundary: without a new argument forcing one of the bounded factor congruences in (4.5), it neither proves nor disproves the Erdős–Straus conjecture.

The additional lattice and shell theorems in Section 2 supply further exact morphisms without changing this incidence question. In particular, Theorem 2.10 gives $A_3(2521) = 0$ and $A_3(3361) = 6$ although both primes are 1 modulo 840; Theorem 2.12 retains the actual counts in the integral augmentation vector $Q_{R,P} = 6W_{R,P} - M_{R,P}u \in N$.

2 The modulo-840 lattice, triality, and exact shell maps

This section constructs the original modulo-840 lattice N with root system $A_5^4 D_4$, its full integral isometry group, and the exact maps from denominator shells and finite prime samples to the retained

residue coordinates. The original data and source labels are recorded in [5]. The shared continuation proves the index-six Leech neighbour Λ and the marked isometry $\mathcal{L}$ into the three octonion sectors. In the specified Cayley frame, put $\tau(q) = \mathrm{Re}((q_1q_2)q_3)$ and $F = \tau \circ \mathcal{L}$; the Albert determinant is $\lambda^3 - \lambda(y,y) + 2F(y)$. The value-generated groups have quotient $\mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/12$ (Theorem 4.3). The three residue-cycled neighbours meet in J, with determinant 81 and index 9 in each; the fixed fourth extension has exactly 72 roots of type A_2^{12} (Theorems 5.3 and 5.5). The common lattice and this fourth extension have the exact same value-generated group $\frac{1}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{216}\mathbb{Z} \oplus \frac{\sqrt{3}}{18}\mathbb{Z}$. The inclusions from $\mathcal{V}_F(N)$ and into $\mathcal{V}_F(\Lambda)$ have matrices $\mathrm{diag}(1,4,1)$ and $\mathrm{diag}(1,1,3)$ and quotients $\mathbb{Z}/4$, $\mathbb{Z}/3$ (Theorems 6.2 and 6.3). Wilson's construction [8] is realized with both metrics and the direction $c_{\mathrm{W}} = c^{-1}$ retained. Its fixed Gram matrix is $3G_{E_8}$, its complement is explicitly isometric to $A_2 \otimes E_8$, and the index is 3^8, with quotient $\mathbb{F}_3^8$. Its literal intersections with $\mathcal{L}(N)$ and $\mathcal{L}(\Lambda)$ are the same rank-four lattice of Gram matrix $6G_{D_4}$, minimum squared norm 12 and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for τ ($8\sqrt{2}\mathbb{Z}$ for 2τ). This is not the actual image: $12\sqrt{2}$ is not attained by τ on that intersection. All coordinates, multiplication conventions, domain lattices and maps are specified in the complete proofs below.

2.1 The finite multiplicative carrier and its four fibres

For an odd prime q and a unit r modulo q, write $\left(\frac{r}{q}\right) = 1$ when r is a square modulo q, and -1 otherwise. Let

$$\begin{aligned} V_{840} &= \{r \in (\mathbb{Z}/840\mathbb{Z})^\times : r \equiv 1 \pmod{24}\}, \\ \beta(r) &= \left(\frac{1-\left(\frac{r}{5}\right)}{2}, \frac{1-\left(\frac{r}{7}\right)}{2}\right) \in \mathbb{F}_2^2, \\ S_{840} &= \ker\beta, \qquad F_\eta = \beta^{-1}(\eta) \quad (\eta \in \mathbb{F}_2^2). \end{aligned} \tag{1}$$

Multiplication is used in V_{840} and addition in $\mathbb{F}_2^2$.

Theorem 2.1 (The marked cyclic fibre)**.** *The map β is an onto group homomorphism, each F_η has six elements, and*

$$\begin{aligned} S_{840} &= \{1, 121, 169, 289, 361, 529\} = \langle 289 \rangle, \\ (289^j)_{j=0}^5 &= (1, 289, 361, 169, 121, 529) \pmod{840}. \end{aligned} \tag{2}$$

Each fibre is a torsor under multiplication by S_{840}. In particular the marking

$$\kappa : S_{840} \xrightarrow{\cong} \mathbb{Z}/6\mathbb{Z}, \qquad r = 289^{\kappa(r)} \pmod{840}, \tag{3}$$

is an isomorphism of groups.

Proof. The Chinese remainder map modulo $24, 5, 7$, whose moduli are pairwise coprime, identifies V_{840} with $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/7)^\times$. The powers of 2 modulo 5 are $1, 2, 4, 3$, and the powers of 3 modulo 7 are $1, 3, 2, 6, 4, 5$; these prove the respective cyclic orders four and six. In a cyclic group of either even order, the squares are exactly the even powers. Therefore each Legendre character is the homomorphism recording exponent parity and is onto. Their product is onto $\mathbb{F}_2^2$. Its kernel has $(4/2)(6/2) = 6$ elements and every fibre has that same cardinality.

Successive multiplication of the six displayed residues by 289 gives the next entry and then returns from 529 to 1. They are distinct. Moreover $289 \equiv 1 \pmod{24}$, $289 \equiv 4 \pmod{5}$, and $289 \equiv 2 \pmod{7}$, so it belongs to the simultaneous square kernel. Its six powers therefore exhaust that kernel. If $r, s \in F_\eta$, then $sr^{-1} \in S_{840}$ and $s = (sr^{-1})r$, uniquely. This proves the torsor assertion and all claims about κ. □

Fix the coordinate order

$$(\eta_1, \eta_2, \eta_3, \eta_4) = (00, 10, 01, 11)$$

and choose one origin $f_i \in F_{\eta_i}$ in each fibre, with $f_1 = 1$. The six ordered elements of its fibre are $f_i 289^j$, $0 \le j \le 5$. These origins are part of the coordinate identifications used below; no selection of them is inferred from an individual denominator certificate.

2.2 The root lattice and its exact discriminant form

For a finite set F, give the free abelian group $\mathbb{Z}[F] = \bigoplus_{r\in F} \mathbb{Z}e_r$ the metric $(e_r, e_s) = 1$ for $r = s$ and zero otherwise. Put

$$A(F) = \left\{ \sum_{r\in F} n_r e_r : \sum_{r\in F} n_r = 0 \right\}.$$

The ordered fibre coordinates identify each $A(F_{\eta_i})$ isometrically with

$$A = \{(n_0, \dots, n_5) \in \mathbb{Z}^6 : \textstyle\sum_j n_j = 0\}.$$

In a separate Euclidean space with orthonormal basis e_1, e_2, e_3, e_4, define

$$D = \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2\}, \qquad R_{\mathrm{ES}} = \left(\bigoplus_{i=1}^{4} A(F_{\eta_i}) \right) \perp D. \tag{4}$$

All five summands are orthogonal with precisely their displayed metrics. The symbol R_{ES} denotes this lattice; the unadorned R later used for an integer shell modulus has a different declared domain.

For a full integral lattice L in its real span, let

$$L^\vee = \{x : (x, L) \subset \mathbb{Z}\}, \qquad D_L = L^\vee / L.$$

If L is even integral, its discriminant quadratic form is $q_L(x + L) = (x, x) \pmod{2\mathbb{Z}}$. This is well-defined because replacing x by $x + \ell$ changes its squared norm by $2(x, \ell) + (\ell, \ell) \in 2\mathbb{Z}$.

Theorem 2.2 (Root metric and marked discriminant)**.** *The lattice R_{ES} is even integral of rank 24 and has Gram matrix*

$$G_{R_{\mathrm{ES}}} = G_A^{\oplus 4} \oplus G_D,$$

where

$$G_A = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \qquad G_D = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}.$$

Their determinants are 6 and 4, respectively, and $\det R_{\mathrm{ES}} = 6^4 \cdot 4 = 5184$.

In each identified A summand use $\lambda = (5, -1, -1, -1, -1, -1)/6$ as the discriminant generator. For D, use the representatives

$$v = (1, 0, 0, 0), \qquad s = \tfrac{1}{2}(1, 1, 1, 1), \qquad c = \tfrac{1}{2}(-1, 1, 1, 1) \tag{5}$$

and the exact marking $10 \mapsto v + D$, $01 \mapsto s + D$, $11 \mapsto c + D$. Then

$$\begin{aligned} D_{R_{\mathrm{ES}}} &= (\mathbb{Z}/6\mathbb{Z})^4 \oplus \mathbb{F}_2^2, \\ q_{R_{\mathrm{ES}}}(x_1, x_2, x_3, x_4; y) &= \frac{5}{6} \sum_{i=1}^{4} x_i^2 + \mathbf{1}_{y\neq 0} \pmod{2\mathbb{Z}}. \end{aligned} \tag{6}$$

Proof. For a six-element ordered fibre the five differences $e_j - e_{j+1}$, $0 \le j \le 4$, form a basis of its augmentation lattice: the coefficients of a zero-sum vector in this basis are its first five successive partial sums. Dot products give G_A. The determinant d_n of the analogous tridiagonal n-by-n matrix satisfies $d_n = 2d_{n-1} - d_{n-2}$, $d_0 = 1, d_1 = 2$, by first-row expansion; induction gives $d_n = n + 1$. The four vectors $e_1 - e_2, e_2 - e_3, e_3 - e_4, e_3 + e_4$ lie in D and their coordinate determinant has absolute value 2. Their integer span therefore has index 2 in $\mathbb{Z}^4$. The parity homomorphism $\mathbb{Z}^4 \to \mathbb{F}_2$, $x \mapsto \sum x_i \pmod 2$, is onto and has kernel D, so this span equals D. Their dot products give G_D, whose determinant is the square of the coordinate determinant, namely 4. Integer coordinates make all pairings integral. In either lattice the squared norm is congruent modulo two to the coordinate sum, hence is even. Rank and determinant now follow from the orthogonal direct sum.

The vector λ has sum zero and integral pairing with each augmentation root. Its norm is $5/6$, and $k\lambda \in A$ holds exactly when 6 divides k. Thus it has exact order six in D_A. In an integral basis of an integral lattice with Gram matrix G, the dual coordinates are $G^{-1}\mathbb{Z}^n$; therefore $|L^\vee/L| = |\det G|$. Since $|D_A| = 6$, this element generates the full discriminant group and gives its stated form.

A vector pairing integrally with all $e_i - e_j$ has common fractional parts in its four coordinates. Pairing also with $e_i + e_j$ forces this common fractional part to be either zero or one half. Conversely every entirely integral or entirely half-integral vector pairs integrally with every root of D. There are two integer classes according to sum parity, and two half-integer classes according to the parity of their difference from s. The four representatives $0, v, s, c$ are therefore all the classes, and their squared norms are $0, 1, 1, 1$. Their addition is as marked because $v + s - c = (2, 0, 0, 0) \in D$. The dual and quadratic form of an orthogonal sum are the direct sums of those of its summands, which proves (6). □

2.3 The index-72 glue and all glue-class minima

Define the binary and ternary codes

$$\begin{aligned} C_2 &= \operatorname{span}_{\mathbb{F}_2}\{(1111;00), (1100;10), (1010;01)\}, \\ C_3 &= \operatorname{span}_{\mathbb{F}_3}\{z_1 = (1,1,1,0),\ z_2 = (1,2,0,1)\}. \end{aligned} \tag{7}$$

The semicolon distinguishes the four binary A5 coordinates from the two D4 discriminant coordinates. Let

$$\begin{aligned} \mathcal{H} &= \{(3u + 2z; y) : (u; y) \in C_2, z \in C_3\} \subset D_{R_{\mathrm{ES}}}, \\ N &= \{x \in R_{\mathrm{ES}}^\vee : x + R_{\mathrm{ES}} \in \mathcal{H}\}. \end{aligned} \tag{8}$$

Here $3u + 2z$ is computed coordinatewise modulo six using the integer representatives of u and z.

Theorem 2.3 (Even unimodular completion with exactly 144 roots)**.** *The subgroup $\mathcal{H}$ is isotropic of order 72. Its inverse image N is a positive-definite even unimodular lattice of rank 24, with $[N : R_{\mathrm{ES}}] = 72$. Among the 71 nonzero glue classes, 46 have minimum squared norm 4 and 25 have minimum squared norm 6. The roots of N are exactly those of R_{ES}, with root system $A_5^4D_4$: 30 roots in each A5 summand and 24 in D4.*

Proof. The binary generators are independent, hence give eight words. Listing their sums shows that the seven nonzero words comprise one word $(1111; 00)$ and six words whose first part has weight two and whose second part is nonzero. A ternary word is

$$az_1 + bz_2 = (a + b, a + 2b, a, b).$$

Its last two coordinates prove injectivity of the parameter pair (a, b), so there are nine words. If $a = 0 \neq b$ the third coordinate alone vanishes; if $b = 0 \neq a$ the fourth alone vanishes; and if both are nonzero exactly one of $a + b, a + 2b$ vanishes. Thus every nonzero word has weight three, and each of the four coordinates is the unique vanishing coordinate for two words.

The map $(u_i, z_i) \mapsto 3u_i + 2z_i \pmod 6$ is bijective: reduction modulo two recovers u_i, and reduction modulo three recovers $2z_i$, from which multiplication by 2 recovers z_i. It is a homomorphism of additive groups, since changing binary or ternary representatives by their moduli changes $3u_i + 2z_i$ by a multiple of six. Its product with the retained binary y-coordinate therefore maps $C_2 \times C_3$ injectively onto the subgroup $\mathcal{H}$. It follows that $|\mathcal{H}| = 8 \cdot 9 = 72$. For $u_i \in \{0, 1\}$ and $z_i \in \{0, 1, 2\}$,

$$\frac{5}{6}(3u_i + 2z_i)^2 = \frac{15}{2}u_i + 10u_i z_i + \frac{10}{3}z_i^2 \equiv \frac{3}{2}u_i + \frac{4}{3}z_i^2 \pmod{2\mathbb{Z}}.$$

The cross-term is even; a nonzero z_i has square 1 or 4, whose contributions in the last expression differ by 4. The binary contribution including D4 is therefore zero, 6, or $3 + 1 = 4$, and the ternary contribution is zero or $(4/3)3 = 4$ modulo $2\mathbb{Z}$. This proves isotropy.

The inverse image is a lattice because it is a subgroup between the full lattices R_{ES} and $R_{\mathrm{ES}}^\vee$. Isotropy gives $(x, x) \in 2\mathbb{Z}$ for every $x \in N$, and

$$(x, y) = \frac{(x + y, x + y) - (x, x) - (y, y)}{2} \in \mathbb{Z} \qquad (x, y \in N).$$

Thus it is even integral. Its index is $|\mathcal{H}| = 72$. For a finite-index inclusion of equal-rank lattices, the covolume changes by that index, and the Gram determinant is the square of the covolume in orthonormal coordinates. Hence

$$\det N = \frac{6^4 \cdot 4}{72^2} = 1.$$

The dual-index formula proved above implies $N = N^\vee$. Its rank and positive definiteness are inherited from the unchanged real span and metric.

For $0 \leq k \leq 5$, every vector in the A5 class $k\lambda + A$ has the form

$$m - \frac{k}{6}(1, 1, 1, 1, 1, 1), \qquad m \in \mathbb{Z}^6, \quad \sum m_i = k.$$

Its squared norm is $\sum m_i^2 - k^2/6$. Since $m_i^2 \geq m_i$ for every integer m_i, the minimum is at least $k - k^2/6$. It is attained by choosing exactly k of the six entries equal to one and the others zero. Thus

$$\mu_A(k) = \frac{k(6 - k)}{6}. \tag{9}$$

Every nonzero integral D4 class has minimum one, attained by v; every half-integral vector has norm at least $4(1/2)^2 = 1$, attained by s, c in their respective classes. The four A5 minima and the D4 minimum add and are simultaneously attained because the metric is an orthogonal sum.

For a coordinate with $k_i = 3u_i + 2z_i \pmod 6$, its minimum contribution is 0 for $(u_i, z_i) = (0, 0)$, 3/2 for $(1, 0)$, 4/3 for $(0, z_i \neq 0)$, and 5/6 for $(1, z_i \neq 0)$. The complete nonzero case table is

$\mathrm{wt}(u)$	$\mathrm{wt}(z)$	$\lvert\mathrm{supp}(u) \cap \mathrm{supp}(z)\rvert$	classes	minimum squared norm
2	0	0	6	$2(3/2) + 1 = 4$
4	0	0	1	$4(3/2) = 6$
0	3	0	8	$3(4/3) = 4$
2	3	1	24	$(3/2) + 2(4/3) + (5/6) + 1 = 6$
2	3	2	24	$(4/3) + 2(5/6) + 1 = 4$
4	3	3	8	$(3/2) + 3(5/6) = 4.$

(10)

For each of the six weight-two binary words, four nonzero ternary words omit a coordinate inside its support, and four omit a coordinate outside. This proves the two counts $6 \cdot 4 = 24$; all other counts follow from the enumerated binary and ternary words. Thus the total number at minimum 4 is $6 + 8 + 24 + 8 = 46$, and the total at minimum 6 is $1 + 24 = 25$. Their sum is 71, leaving exactly the zero glue class.

No nonzero glue class can contain a norm-two vector. In the zero class, an A5 norm-two vector has exactly one coordinate 1 and another -1, so is $e_i - e_j$, with $6 \cdot 5 = 30$ possibilities. A D4 norm-two vector has exactly two coordinates of absolute value one, so is $\pm e_i \pm e_j$, with $4\binom{4}{2} = 24$ possibilities. An orthogonal sum of two nonzero root-lattice vectors has norm at least 4 because each summand is positive even. These are therefore all $4 \cdot 30 + 24 = 144$ roots of N. □

A Niemeier lattice means a positive-definite even unimodular lattice of rank 24. The theorem constructs one with root system $A_5^4 D_4$ directly. Its root type appears with lambency 6 in the primary umbral table [1, Table 3, p. 21]. This table reference is not needed to prove the inverse-image construction or any minimum above.

For comparison with the earlier labeled glue presentation in the source [4, 5], put

$$h_1 = (1, 3, 1, 1; 00), \qquad h_2 = (1, 4, 2, 3; 11), \qquad h_3 = (3, 0, 3, 0; 01),$$

and let $\mathcal{H}_{\mathrm{old}}$ be their subgroup. The exact transport is

$$P(x_1, x_2, x_3, x_4; y) = (x_3, x_4, x_1, x_2; y), \qquad P(\mathcal{H}_{\mathrm{old}}) = \mathcal{H}. \tag{11}$$

Indeed, if $rh_1 + th_2 + wh_3 = 0$ with $r, t \in \mathbb{Z}/6$ and $w \in \mathbb{Z}/2$, the final two coordinates imply that t is even and $w = 0$. The first and third coordinates then give $r + t = 0$ and $r + 2t = 0$ modulo six, whence $t = r = 0$. Thus $|\mathcal{H}_{\mathrm{old}}| = 6 \cdot 6 \cdot 2 = 72$. Directly,

$$\begin{aligned} P(h_1) &= 3(1111; 00) + 2(2, 2, 2, 0; 00), \\ P(h_2) &= 3(0110; 11) + 2(1, 0, 2, 2; 00), \\ P(h_3) &= 3(1010; 01). \end{aligned}$$

The displayed ternary parts are $2z_1$ and $2z_1 + 2z_2$. Consequently $P(\mathcal{H}_{\mathrm{old}}) \subset \mathcal{H}$, and the equal orders prove equality. This transporter is realized by the corresponding isometry permuting the full identified A5 summands.

2.4 The common D4 root lattice, shortest weights, and triality

Use orthonormal coordinates $\varepsilon_1, \ldots, \varepsilon_4$ for the ordinary F4 root realization. Its long roots and the three parts of its short roots are

$$\begin{aligned} \Phi_D &= \{\pm\varepsilon_i \pm \varepsilon_j : 1 \le i < j \le 4\}, \\ \mathcal{V} &= \{\pm\varepsilon_i : 1 \le i \le 4\}, \\ \mathcal{S}_\pm &= \left\{ \frac{1}{2} \sum_{i=1}^{4} \delta_i \varepsilon_i : \delta_i \in \{1, -1\},\ \prod_i \delta_i = \pm 1 \right\}, \\ \Phi_F &= \Phi_D \cup \mathcal{V} \cup \mathcal{S}_+ \cup \mathcal{S}_-. \end{aligned} \tag{12}$$

For a nonzero vector a, reflection means the exact map $s_a(x) = x - 2(x, a)a/(a, a)$. Write W_D and W_F for the groups generated by the reflections in Φ_D and Φ_F, respectively.

Theorem 2.4 (Integral D4 and the marked triality quotient)**.** *The map*

$$\iota_D : D \longrightarrow \mathbb{Z}\langle \Phi_D \rangle, \qquad (x_1, x_2, x_3, x_4) \longmapsto \sum_{i=1}^{4} x_i \varepsilon_i \tag{13}$$

is an integral isometry onto the long-root lattice. It extends to duals and discriminant forms, sending the shortest-vector sets of the marked classes v, s, c respectively to $\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-$.

The two matrices

$$R_D = \operatorname{diag}(-1, 1, 1, 1), \qquad H_D = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \tag{14}$$

are the short-root reflections s_{ε_1} and s_c, where $c = (-1, 1, 1, 1)/2$ is read in the ε coordinates. They induce $(s\ c)$ and $(v\ s)$ on the three nonzero discriminant classes. There is a split exact sequence

$$1 \longrightarrow W_D \longrightarrow W_F \longrightarrow S_3 \longrightarrow 1, \qquad W_D \cong \{\textit{even signed permutations of four coordinates}\}, \tag{15}$$

whose displayed section is $\langle R_D, H_D \rangle \cong S_3$. In particular $|W_D| = 192$ and $|W_F| = 1152$.

Proof. The four D4 basis vectors in Theorem 2.2 are among Φ_D, and every vector of Φ_D has even coordinate sum. Thus the displayed identity of coordinates has exactly the stated image, preserves the metric, and extends to duals by the integer-pairing definition. In a nonzero integral dual class, norm one forces one coordinate ± 1 and all others zero. In a half-integral class, norm one forces all four coordinates to be $\pm 1/2$. The difference from s has even coordinate sum exactly when the number of negative signs is even. This proves the three shortest-vector identifications with the stated marking.

The reflection identity for R_D is immediate, and $H_D = I - 2cc^T$ follows by multiplying the four coordinates of c. Both are orthogonal involutions. The first preserves the parity of an integer coordinate sum. For $x \in D$, the four coordinates of $H_D x$ are integers because each numerator is congruent to $\sum x_i$ modulo two, and their sum is $2x_1$. Thus both preserve D. Their effects on the representatives show that R_D fixes the class v and exchanges s, c, whereas $H_D v = s$, $H_D s = v$, and $H_D c = -c \equiv c \pmod{D}$. They therefore preserve all of Φ_F by the shortest-vector description. On the plane with ordered basis ε_1, c, their product has matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

whose cube is the identity, while both reflections fix the orthogonal complement. Consequently $R_D^2 = H_D^2 = (R_D H_D)^3 = 1$. Every word in two such involutions reduces to $(R_D H_D)^k$ or $(R_D H_D)^k R_D$, $0 \le k \le 2$. Their discriminant permutations are the six elements of S_3, so these six words are distinct and give the stated section.

Reflection in $\varepsilon_i - \varepsilon_j$ swaps coordinates i, j. Reflection in $\varepsilon_i + \varepsilon_j$ swaps them and changes both signs. Hence the long-root reflections generate every permutation and every even sign change, and each long-root reflection is of this form. This proves the assertion about W_D and its order $2^3 \cdot 4! = 192$. Such a reflection acts trivially on $D^\vee/D$: for a root a of squared norm two, $s_a(x) - x = -(x, a)a \in D$ for all $x \in D^\vee$.

Let $K = \langle W_D, R_D, H_D \rangle$. Each of its generators preserves the long roots, so W_D is normal in K by $g s_a g^{-1} = s_{ga}$. The even signed permutations act transitively on each of $\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-$, and the

displayed section permutes these three sets transitively. Thus every short root is $g\varepsilon_1$ for some $g \in K$, and its reflection gR_Dg^{-1} belongs to K. All generators of W_F therefore belong to K; the reverse inclusion follows from their definitions, proving $W_F = K$. An element in the kernel of its permutation action on the three short sets preserves $\mathcal{V}$, so it is a signed permutation matrix. Preservation of $\mathcal{S}_+$ forces an even number of sign changes, so the kernel is W_D. This proves exactness, splitting, and the order $6 \cdot 192 = 1152$. □

The exact geometric datum and weight marking. Let $\mathbb{O}$ denote the real normed octonion algebra and let $J = \mathfrak{h}_3(\mathbb{O})$ be the real vector space of Hermitian three-by-three matrices, with Jordan product $a \circ b = (ab + ba)/2$. Define

$$\mathbb{OP}^2 = \{a \in J : a^2 = a,\ \operatorname{tr} a = 1\}, \qquad \mathrm{Fl}(\mathbb{O}) = \{(a, b) \in (\mathbb{OP}^2)^2 : \operatorname{Re}\operatorname{tr}(ab) = 0\}.$$

For the compact group $F_4 = \mathrm{Aut}_{\mathbb{R}}(J, \circ)$, Mare and Willems prove the homogeneous identification $\mathrm{Fl}(\mathbb{O}) = F_4/\operatorname{Spin}(8)$ and identify its tangent representation as $V_8 \oplus S_8^+ \oplus S_8^-$ [2, Definition 2.2, Proposition 2.4, Proposition 2.7, and pp. 491–492]. These are the prior geometric results used here. With their torus coordinate forms L_i, their equations (5.4)–(5.6) assign the vector, positive half-spin, and negative half-spin weights respectively the patterns $\pm L_i$, even half-sign sums, and odd half-sign sums [2, pp. 510–511]. The coordinate map $L_i \mapsto \varepsilon_i$ retains that sign marking and identifies these three weight sets with those of (12).

For completeness the half-spin weight patterns admit the following direct calculation. In the complex exterior algebra $\Lambda^*\mathbb{C}^4$, let a_i be exterior multiplication by the i-th basis vector and b_i contraction by its dual. On a basis wedge, insertion and deletion give

$$a_ia_j + a_ja_i = b_ib_j + b_jb_i = 0, \qquad a_ib_j + b_ja_i = \delta_{ij}I.$$

Thus $C_i = a_i + b_i$ and $D_i = \mathrm{i}(b_i - a_i)$ square to one and mutually anticommute for distinct Clifford generators. Writing $N_i = a_ib_i$, direct multiplication gives

$$\tfrac{1}{2}C_iD_i = \mathrm{i}(N_i - \tfrac{1}{2}I).$$

The four commuting infinitesimal torus operators therefore have weights $\sum_i(\mathbf{1}_{i\in I} - 1/2)\varepsilon_i$ on the basis wedge indexed by $I \subset \{1, 2, 3, 4\}$. The even exterior part has an even number of negative signs, and the odd part has an odd number. There are eight one-dimensional weight spaces in each part. For the vector representation the four real coordinate planes complexify into their two rotation eigenspaces, with weights $\varepsilon_i, -\varepsilon_i$. This proves the displayed weight formulas directly in the stated marking.

The map proved here is consequently the actual chain

$$D \xrightarrow{\iota_D} \mathbb{Z}\langle\Phi_D\rangle, \qquad \{v, s, c\} \longleftrightarrow \{\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-\} = \{\mathrm{wt}(V_8), \mathrm{wt}(S_8^+), \mathrm{wt}(S_8^-)\}.$$

The same S_3 acts through the explicitly proved permutations of these three sets. The companion higher-rung treatment records the geometric datum and root-coordinate comparison at the respective labels

`h24:thm:octonionic-triality`
`h24:prop:two-Coxeter-systems`.

The exact source pointers and passages are included in [5]. The integral metric map in this section is the map on D4 and its dual. The definitions make no map from the whole rank-24 lattice N into the flag tangent space.

2.5 The full indicated glue stabilizer and its triality quotient

All vector spaces over finite fields in this subsection retain their displayed bases. Write

$$j : \mathbb{F}_3^2 \longrightarrow \mathbb{F}_3^4, \qquad j(a,b) = (a+b, a+2b, a, b), \qquad C_3 = j(\mathbb{F}_3^2).$$

The four row forms defining this injection are

$$\ell_1 = (1,1), \quad \ell_2 = (1,2), \quad \ell_3 = (1,0), \quad \ell_4 = (0,1).$$

They represent all four one-dimensional subspaces of the dual of $\mathbb{F}_3^2$. Indeed each nonzero row has either first coordinate zero, in which case its line is represented by ℓ_4, or a unique scalar multiple with first coordinate one, namely one of $(1,0), (1,1), (1,2)$.

Let $E \subset \mathbb{F}_2^4$ be the even-weight subspace and define

$$\tau : E \longrightarrow \mathbb{F}_2^2, \qquad \tau(u_1,u_2,u_3,u_4) = (u_2+u_4, u_3+u_4).$$

The binary code in the index-72 glue is exactly

$$C_2 = \{(u, \tau(u)) : u \in E\}.$$

To verify this, the independent vectors $1111, 1100, 1010$ form a basis of the three-dimensional space E, and their images under τ are respectively $00, 10, 01$. These are precisely the three stated binary code generators. In particular τ is onto and

$$\ker \tau = \langle 1111 \rangle.$$

Explicitly, $\tau(u) = 0$ gives $u_2 = u_3 = u_4$, and the even-weight equation then gives $u_1 = u_4$.

For a permutation p of $\{1,2,3,4\}$, use the coordinate convention

$$(P_p u)_i = u_{p(i)}.$$

The permutation preserves E and $\langle 1111 \rangle$, so it induces a unique invertible map B_p satisfying

$$B_p \tau(u) = \tau(P_p u) \qquad (u \in E). \tag{16}$$

For an entirely explicit formula, the two columns of B_p are $\tau(P_p(1100))$ and $\tau(P_p(1010))$. The convention implies $P_p P_q = P_{q \circ p}$. Accordingly $B_p B_q = B_{q \circ p}$; no reversal is suppressed in what follows.

The subgroup of (8) can therefore be written

$$\mathcal{H} = \{(3u + 2z; \tau(u)) : u \in E,\ z \in C_3\} \subset (\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2.$$

Let G be its stabilizer among the transformations

$$g = (\varepsilon, p, B), \qquad (x;y) \longmapsto \big((\varepsilon_i x_{p(i)})_{i=1}^4; By\big),$$

where $\varepsilon_i \in \{1,-1\}$, $p \in S_4$, and $B \in GL_2(\mathbb{F}_2)$. Denote the signed permutation matrix on the first four coordinates by M_g.

Theorem 2.5 (An explicit isomorphism to $GL_2(\mathbb{F}_3)$)**.** *Restriction to the ternary code gives an isomorphism*

$$\Theta : G \xrightarrow{\cong} GL_2(\mathbb{F}_3), \qquad M_g j = j\Theta(g).$$

For every $A \in GL_2(\mathbb{F}_3)$, its inverse image is specified uniquely by

$$\ell_i A = \varepsilon_i \ell_{p(i)} \quad (1 \le i \le 4), \qquad B = B_p. \tag{17}$$

In particular $|G| = 48$.

Proof. The elements of $\mathcal{H}$ of order dividing two are precisely $(3u;\tau(u))$, and those of order dividing three are precisely $(2z;0)$. Every indicated ambient transformation preserves these primary parts, so an element of G satisfies $M_gC_3 = C_3$. The injection j therefore yields one and only one $\Theta(g) \in GL_2(\mathbb{F}_3)$. The binary part is preserved exactly when $B\tau(u) = \tau(P_pu)$, because all signs become one modulo two. Surjectivity of τ forces $B = B_p$.

If $\Theta(g) = I$, the equality $M_gj = j$ implies $\varepsilon_i\ell_{p(i)} = \ell_i$ for every i. The four row lines are distinct, so $p(i) = i$. Since each ℓ_i is nonzero and $1 \neq -1$ in $\mathbb{F}_3$, also $\varepsilon_i = 1$ for every i. It follows that $B_p = I$ and $g = 1$. Thus Θ is injective.

Conversely fix $A \in GL_2(\mathbb{F}_3)$. The rows ℓ_iA are nonzero and belong to four distinct row lines. Since the ℓ_i represent every row line, each ℓ_iA has a unique expression $\varepsilon_i\ell_{p(i)}$, where the nonzero scalar in $\mathbb{F}_3$ is represented by the exact integer sign $\varepsilon_i = 1$ or -1. Distinctness of the row lines makes p a permutation. The resulting signed permutation satisfies $M_gj = jA$, hence preserves C_3. Choosing $B = B_p$ preserves C_2 as proved above, hence preserves $\mathcal{H}$. This constructs an element of G mapping to A, proving surjectivity and the stated inverse formula.

Finally

$$M_{gh}j = M_gM_hj = M_gj\Theta(h) = j\Theta(g)\Theta(h),$$

so $\Theta(gh) = \Theta(g)\Theta(h)$. An invertible 2-by-2 matrix over $\mathbb{F}_3$ has eight choices for its first nonzero column and six choices for its second column outside the span of the first. Thus its group order is $8 \cdot 6 = 48$. □

The map to the permutation of the four component labels is more precisely $g \mapsto p^{-1}$, because a vector supported on component k is sent to component $p^{-1}(k)$. Its kernel under Θ is $\{I, -I\}$. In fact a matrix preserving all four row lines is diagonal by preservation of ℓ_3 and ℓ_4, and its diagonal entries are equal by preservation of ℓ_1. Conversely scalar matrices preserve all four lines. Therefore the component permutation image has order $48/2 = 24$, so is all S_4, with

$$G/\Theta^{-1}(\{\pm I\}) \xrightarrow{\overline{\Theta}} PGL_2(\mathbb{F}_3) \cong S_4.$$

The center of $GL_2(\mathbb{F}_3)$ is exactly $\{\pm I\}$: a central matrix commutes with $\mathrm{diag}(1,-1)$, forcing its off-diagonal entries to vanish, and commuting with $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ forces the diagonal entries to agree.

Theorem 2.6 (The exact triality quotient and its quaternion kernel)**.** *The projection $\pi : G \to GL_2(\mathbb{F}_2)$, $(\varepsilon, p, B) \mapsto B$, is onto, and under Θ its kernel is the quaternion subgroup generated by*

$$I_q = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad J_q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

There is a split exact sequence

$$1 \longrightarrow Q_8 \longrightarrow GL_2(\mathbb{F}_3) \xrightarrow{\pi\Theta^{-1}} GL_2(\mathbb{F}_2) \longrightarrow 1.$$

The splitting is the subgroup generated by

$$A_R = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A_H = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}.$$

It identifies $GL_2(\mathbb{F}_3)$ with $Q_8 \rtimes S_3$ for the conjugation action of this displayed subgroup on this displayed Q_8.

Proof. The nonzero elements of $E/\langle 1111\rangle$ are represented by the complementary pairs of two-element subsets

$$\{12 \mid 34\}, \qquad \{13 \mid 24\}, \qquad \{14 \mid 23\},$$

marked respectively by $10, 01, 11$ via τ. Thus B_p is exactly the permutation action on these three partitions, expressed in the displayed binary marking.

Using (17) gives

$$M(A_R)x = (x_2, x_1, x_3, -x_4), \quad B_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

$$M(A_H)x = (x_1, x_3, x_2, -x_4), \quad B_H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

These binary matrices act respectively as the transpositions (01 11) and (10 01), so they generate all permutations of the three nonzero binary vectors. There are $(4-1)(4-2) = 6$ invertible binary matrices; a matrix fixing all nonzero vectors fixes the basis. Hence this permutation action identifies $GL_2(\mathbb{F}_2)$ with S_3 and proves surjectivity of π. Its kernel therefore has order eight.

For I_q the row rule gives

$$p = (2, 1, 4, 3), \qquad \varepsilon = (-1, 1, 1, -1),$$

and for J_q it gives

$$p = (3, 4, 1, 2), \qquad \varepsilon = (-1, -1, 1, 1).$$

Both permutations fix each of the three displayed complementary partitions, so both matrices lie in the kernel. Direct matrix multiplication over $\mathbb{F}_3$ gives

$$I_q^2 = J_q^2 = -I, \qquad I_q J_q = -J_q I_q, \qquad I_q J_q = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}.$$

The eight matrices $I, -I, \pm I_q, \pm J_q, \pm I_q J_q$ are pairwise distinct by their displayed entries. The relations show this set is a group: powers reduce using $I_q^2 = J_q^2 = -I$, and factors can be reordered using $J_q I_q = -I_q J_q$. These relations identify it with the quaternion group Q_8. Since the kernel has order eight, this is the whole kernel.

The matrices A_R, A_H satisfy

$$A_R^2 = A_H^2 = I, \qquad A_R A_H = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad (A_R A_H)^3 = I.$$

Every word in the two involutions reduces to one of $(A_R A_H)^k$ or $(A_R A_H)^k A_R$, $k = 0, 1, 2$, by alternation and the cubic relation. Their images under π are the six distinct permutations generated by B_R, B_H. Hence the subgroup has exactly six elements and maps isomorphically onto $GL_2(\mathbb{F}_2)$. This gives the splitting. For any g, the unique splitting element with image $\pi(g)$ leaves g times its inverse in the kernel; the intersection of kernel and splitting is trivial. Multiplication therefore realizes the asserted semidirect product with the indicated conjugation action. □

The matrices in the kernel have an additional marked interpretation: their projective component permutations are the four permutations $1, (12)(34), (13)(24), (14)(23)$. Each fixes all three partitions, and these are exactly the kernel of the onto action $S_4 \to S_3$, whose kernel has order $24/6 = 4$. Thus the preceding exact sequence is the precise map from the order-48 ternary-code symmetry to the three D_4 discriminant classes.

Theorem 2.7 (A faithful action on the actual index-72 lattice)**.** *Fix the four identifications of the A_5 coordinate lattices used in the glue presentation, and the discriminant marking $10 \mapsto v, 01 \mapsto s, 11 \mapsto c$ of D_4. The group $G \cong GL_2(\mathbb{F}_3)$ has a faithful action by integral isometries of the index-72 overlattice. Write $\varphi_i : A \longrightarrow A(F_{\eta_i})$ for the restriction of the fixed free-coordinate isometry $e_j \mapsto e_{f_i 289^j}$. For $X_i \in A(F_{\eta_i})$, the i-th output component is exactly $\varepsilon_i \varphi_i \varphi_{p(i)}^{-1} X_{p(i)}$. In the identified A coordinates this is the action*

$$L_g(X_1, X_2, X_3, X_4; d) = \big(\varepsilon_1 X_{p(1)}, \ldots, \varepsilon_4 X_{p(4)}; \delta(B)d\big),$$

where $\delta : GL_2(\mathbb{F}_2) \to O(D_4)$ is the section given by

$$\delta(B_R) = R_D = \mathrm{diag}(-1,1,1,1), \qquad \delta(B_H) = H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Under Θ, the two displayed full triality lifts are exactly A_R and A_H.

Proof. The four identified A_5 summands carry the same Euclidean metric, so the displayed signed permutation is a root-lattice isometry. The matrices R_D, H_D preserve D_4: R_D negates one integral coordinate and preserves parity of the sum; for $x \in D_4$, each coordinate of $H_D x$ is integral because its numerator has the same parity as $\sum x_i$, and the coordinate sum of $H_D x$ is $2x_1$. Both matrices are orthogonal involutions by multiplication. Furthermore $R_D = s_{e_1}$, $H_D = s_c$, with $c = (-1,1,1,1)/2$, and $(e_1, c) = -1/2$. On the plane with basis e_1, c, their product has matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ and has cube the identity; both reflections fix the orthogonal complement pointwise. The discriminant actions are the two distinct transpositions B_R, B_H. The same word reduction as in the preceding proof therefore shows that $\langle R_D, H_D \rangle$ has exactly six elements and that its discriminant action is an isomorphism onto $GL_2(\mathbb{F}_2)$. Its inverse is the homomorphism δ.

Thus $g \mapsto L_g$ is a homomorphism of root-lattice isometries. It extends to the dual lattice because an isometry preserves the integer-pairing condition defining the dual. Its discriminant action is the specified (ε, p, B), which preserves $\mathcal{H}$. Consequently it preserves the inverse image

$$N = \{x \in R_{\mathrm{ES}}^{\vee} : x + R_{\mathrm{ES}} \in \mathcal{H}\}.$$

It is therefore an integral isometry of N. If L_g is the identity on N, it is the identity on the full real span, because N contains the full-rank lattice R_{ES}. It is therefore also the identity on $R_{\mathrm{ES}}^{\vee}/R_{\mathrm{ES}}$, in particular on the ternary code. Hence $\Theta(g) = I$, and injectivity of Θ gives $g = 1$. This proves faithfulness. The exact formulas for A_R, A_H were established above, so their lifts are the stated full triality transformations. □

This constructs an explicit finite subgroup of $O(N)$, including its action on all five root summands and on their glue, and its exact D_4-triality quotient. The group and its projective quotient agree with the $A_5^4 D_4$ row of [1, Table 3, p. 21]; the isomorphisms here have been proved by their actual code actions.

For clarity, the two full triality elements, denoted $\widehat{R} = L_{\Theta^{-1}(A_R)}$ and $\widehat{H} = L_{\Theta^{-1}(A_H)}$, act in the fixed coordinates by

$$\widehat{R}(X_1, X_2, X_3, X_4; d) = (X_2, X_1, X_3, -X_4; R_D d),$$
$$\widehat{H}(X_1, X_2, X_3, X_4; d) = (X_1, X_3, X_2, -X_4; H_D d).$$

Their squares and the cube of their product are the identity on the entire lattice, by the proved faithful group homomorphism. The product on the first three A5 summands is the indicated three-cycle; its two negations on the fourth summand cancel.

2.6 The exact reduced-shell divisor bijection

Let p be a positive prime with $p \equiv 1 \pmod{24}$, and let the positive integer shell R satisfy $R \equiv 3 \pmod{4}$. Retain

$$a = a_R(p) = \frac{p+R}{4}, \qquad N_{\mathrm{sh}} = pa, \qquad g = \gcd(R, N_{\mathrm{sh}}), \qquad R_0 = \frac{R}{g}, \quad N_0 = \frac{N_{\mathrm{sh}}}{g}. \tag{18}$$

The integer N_{sh} is the source's scalar $N = pa$; the subscript distinguishes it from the lattice N without changing either object. In particular a, R_0, N_0 are positive integers, $\gcd(R_0, N_0) = 1$, and the prescribed first denominator equation is

$$\frac{4}{p} - \frac{1}{a} = \frac{4a-p}{pa} = \frac{R}{N_{\mathrm{sh}}} = \frac{R_0}{N_0} = \frac{1}{b} + \frac{1}{c}. \tag{19}$$

Here b, c are positive integer denominators, and their order is initially retained. Define the finite set

$$\mathcal{D}_{R,p} = \{d \in \mathbb{Z}_{>0} : d \mid N_0^2,\ d \equiv -N_0 \pmod{R_0}\}, \qquad A_R(p) = |\mathcal{D}_{R,p}|.$$

Theorem 2.8 (Certificates, ordered pairs, and complementary pairs)**.** *There is an exact bijection*

$$\begin{aligned} \mathcal{D}_{R,p} &\longrightarrow \{(b,c) \in \mathbb{Z}_{>0}^2 : 4/p = 1/a + 1/b + 1/c\}, \\ d &\longmapsto \left(\frac{N_0 + d}{R_0}, \frac{N_0 + N_0^2/d}{R_0} \right), \end{aligned} \tag{20}$$

with inverse $d = R_0 b - N_0$. The involution $d \mapsto N_0^2/d$ corresponds exactly to $(b,c) \mapsto (c,b)$. Consequently the number of unordered pairs of remaining denominators is

$$U_R(p) = \frac{A_R(p) + \delta_{R,p}}{2}, \qquad \delta_{R,p} = \begin{cases} 1, & R_0 \mid 2, \\ 0, & R_0 \nmid 2. \end{cases} \tag{21}$$

The unique possible fixed certificate is $d = N_0$, giving $b = c = 2N_0/R_0$.

Proof. For a positive solution of (19), $R_0 b/N_0 = 1 + b/c > 1$, so $d = R_0 b - N_0 > 0$; likewise $e = R_0 c - N_0 > 0$. Multiplying the fraction equation by $N_0 bc$ gives $R_0 bc = N_0(b+c)$, and therefore

$$(R_0 b - N_0)(R_0 c - N_0) = N_0^2.$$

Thus $d \mid N_0^2$, $d \equiv -N_0 \pmod{R_0}$, and $e = N_0^2/d$.

Conversely, let $d \in \mathcal{D}_{R,p}$ and $e = N_0^2/d$. For $R_0 > 1$, the coprimality $\gcd(R_0, N_0) = 1$ makes $d \equiv -N_0$ a unit, and hence

$$e \equiv N_0^2(-N_0)^{-1} \equiv -N_0 \pmod{R_0}.$$

For $R_0 = 1$, this congruence is vacuous and holds automatically. Thus both expressions in (20) are positive integers. Substitution into $(R_0 b - N_0)(R_0 c - N_0) = N_0^2$ and expansion give the original fraction equation. Each construction recovers exactly $d = R_0 b - N_0$ and $e = R_0 c - N_0$, so they are inverse, proving the bijection and the involution statement.

For a positive divisor, $d = N_0^2/d$ holds exactly when $d = N_0$. This divisor is a certificate exactly when $N_0 \equiv -N_0 \pmod{R_0}$, that is, $R_0 \mid 2N_0$. Coprimality makes this equivalent to $R_0 \mid 2$. Every other orbit contains two certificates, so counting the fixed orbits and the two-element orbits gives $U_R(p) = \delta_{R,p} + (A_R(p) - \delta_{R,p})/2$, exactly (21). □

Factor the exact integer N_0 as $N_0 = \prod_{j=1}^{m} q_j^{e_j}$, with distinct positive primes q_j and positive exponents e_j, and define

$$\begin{aligned} B(N_0) &= \prod_{j=1}^{m} \{-e_j, -e_j+1, \ldots, e_j\}, \\ \lambda_{R_0,N_0} : B(N_0) &\longrightarrow (\mathbb{Z}/R_0\mathbb{Z})^{\times}, \qquad \lambda_{R_0,N_0}(\beta) = \prod_{j=1}^{m} q_j^{\beta_j} \pmod{R_0}. \end{aligned} \tag{22}$$

The group for modulus one is taken to be the one-element group, so that the same notation also expresses the vacuous congruence in that case. For $R_0 > 1$, every q_j is a unit by coprimality, so all negative powers in this formula are defined.

Theorem 2.9 (The exact exponent-box map and the two raw channels)**.** *The map*

$$\beta \longmapsto d = \prod_{j=1}^{m} q_j^{e_j+\beta_j} \tag{23}$$

bijects $\lambda_{R_0,N_0}^{-1}(-1)$ with $\mathcal{D}_{R,p}$. It retains each valuation and both original integers R_0, N_0.

In addition, suppose $\gcd(R, pa) = 1$ *and* $p \nmid a$*. Write*

$$a = \prod_{j=1}^{s} q_j^{e_j}, \qquad B(a) = \prod_{j=1}^{s} \{-e_j, \ldots, e_j\},$$

and define, for each unit t modulo R,

$$c_R(a;t) = \#\left\{\beta \in B(a) : \prod_j q_j^{\beta_j} \equiv t \pmod{R}\right\}.$$

Then the number of raw divisor certificates, equivalently ordered denominator pairs, is

$$A_R(p) = c_R(a;-1) + 2c_R(a;-p^{-1}). \tag{24}$$

Its complementary-divisor quotient is the count in (21)*, with $R_0 = R$ under these additional coprimality hypotheses.*

Proof. A divisor of N_0^2 has one and only one exponent vector $\alpha_j \in \{0, \ldots, 2e_j\}$. The exact change $\beta_j = \alpha_j - e_j$ gives the entire displayed box and $dN_0^{-1} \equiv \prod_j q_j^{\beta_j} \pmod{R_0}$. Multiplication of $d \equiv -N_0$ by the unit N_0^{-1} gives precisely the target -1; modulo one both statements are vacuous. This proves the first bijection and its inverse.

Under the additional hypotheses, $R_0 = R$, $N_0 = pa$, and the exponent of p in pa is exactly one. Every positive integer divisor d of p^2a^2 therefore has exactly one valuation-index representation

$$d = p^{t+1} \prod_j q_j^{e_j+\beta_j}, \qquad (t, \beta) \in \{-1, 0, 1\} \times B(a).$$

After division by pa, its unit-group residue is

$$d(pa)^{-1} \equiv p^t \prod_j q_j^{\beta_j} \pmod{R}.$$

Distinct valuation-index pairs may have the same unit-group residue; they remain distinct integer divisors in the count. The three conditions for this to equal -1 are

t	$\prod_j q_j^{\beta_j} \pmod R$
-1	$-p$
0	-1
1	$-p^{-1}$.

The negation $\beta \mapsto -\beta$ is a bijection of $B(a)$ and inverts the unit-group value, so it bijects the fibres at $-p$ and $-p^{-1}$. Summing the sizes of these three disjoint raw exponent classes proves (24). The complementary divisor sends (t, β) to $(-t, -\beta)$, so identifying it changes the raw count as proved in Theorem 2.8; it does not leave the factor two in (24) unchanged. □

This counting convention corrects the wording attached to the source's two-channel formula: the displayed sum is a raw count, whereas a count after identifying complementary divisors is (21). The exact source locator and formula are retained in [3, 5].

2.7 A fixed-shell count does not descend to the residue fibre

Theorem 2.10 (Two primes in the same residue with different shell counts)**.** *The two primes* 2521 *and* 3361 *both reduce to* $1 \in S_{840}$*, but for the same shell* $R = 3$ *their raw and unordered counts are*

$$A_3(2521) = 0, \qquad A_3(3361) = 6, \qquad U_3(2521) = 0, \quad U_3(3361) = 3. \tag{25}$$

One actual denominator certificate in the second case is

$$\frac{4}{3361} = \frac{1}{841} + \frac{1}{974690} + \frac{1}{28266010}. \tag{26}$$

Thus the raw count, its unordered quotient, and the indicator of completion of this fixed shell each fail to factor through prime reduction modulo 840.

Proof. The residue equations are the literal equalities $2521 = 3 \cdot 840 + 1$ and $3361 = 4 \cdot 840 + 1$. For an explicit finite primality verification, the primes at most 57, in increasing order, are

$$\mathcal{P} = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53).$$

The remainders against its first 15 entries for 2521, all 16 entries for 3361, and its first nine entries for 631, are respectively

$$(1, 1, 1, 1, 2, 12, 5, 13, 14, 27, 10, 5, 20, 27, 30),$$
$$(1, 1, 1, 1, 6, 7, 12, 17, 3, 26, 13, 31, 40, 7, 24, 22),$$
$$(1, 1, 1, 1, 4, 7, 2, 4, 10).$$

All are nonzero, and the respective integer square roots are 50, 57, and 25. The prime 29 has remainders $1, 2, 4$ against $2, 3, 5$, the primes at most its integer square root 5. Every composite positive integer greater than one has a prime divisor at most its square root: in a factorization into two factors greater than one, the smaller factor is at most the square root and has a prime divisor no larger than itself. These nonzero remainder lists therefore prove each stated primality.

For $R = 3$,

$$a_3(2521) = 631, \qquad a_3(3361) = 841 = 29^2.$$

Both primes are 1 modulo three; also $631 \equiv 1 \pmod 3$ and $841 \equiv 1 \pmod 3$. Thus $\gcd(3, pa) = 1$ in both cases. Both a values are smaller than their respective primes, so $p \nmid a$. Every additional hypothesis of Theorem 2.9 is now verified. The two targets -1 and $-p^{-1}$ coincide modulo three.

For $a = 631$, all exponents $-1, 0, 1$ give residue one, so the target -1 has count zero. For $a = 29^2$, among the five exponents $-2, -1, 0, 1, 2$, exactly $-1, 1$ give -1 modulo three because $29 \equiv -1 \pmod 3$. Both target counts are therefore 2 and the raw count is $2 + 2 \cdot 2 = 6$. Here $R_0 = 3 \nmid 2$, so both unordered counts are half the corresponding raw counts. This proves (25).

For the second prime, retain

$$N_{\mathrm{sh}} = 3361 \cdot 841 = 2826601, \qquad d = 3361 \cdot 29 = 97469, \qquad \frac{N_{\mathrm{sh}}^2}{d} = 3361 \cdot 29^3 = 81971429.$$

Then

$$\frac{N_{\mathrm{sh}} + d}{3} = 974690, \qquad \frac{N_{\mathrm{sh}} + N_{\mathrm{sh}}^2/d}{3} = 28266010.$$

The bijection of Theorem 2.8 now proves (26). The first prime has no completion with this prescribed first denominator by its zero certificate count. Two arguments with the same residue and unequal function values cannot be the pullback of a single residue function: if $f(p) = h(p \bmod 840)$, equality of the residues forces equality of the values. Applying this observation to the three displayed functions proves the claimed failures of factorization. □

2.8 An actual residue cocycle and a retained-sample Fourier map

For $r \in S_{840}$, multiplication by r permutes each fibre. Let $\rho(r)$ act by

$$\rho(r)e_s = e_{rs} \quad (s \in V_{840}), \qquad \rho(r)|_D = I, \qquad b(r) = e_r - e_1 \in A(S_{840}) \subset N. \tag{27}$$

The two free coordinates in $b(r)$ belong to the principal fibre.

Theorem 2.11 (The full-lattice residue action and its exact cocycle)**.** *The permutations $\rho(r)$ extend to a faithful homomorphism $\rho : S_{840} \to O(N)$. The map $b : S_{840} \to N$ satisfies*

$$b(rs) = b(r) + \rho(r)b(s) \qquad (r, s \in S_{840}), \qquad b(1) = 0, \qquad (b(r), b(r)) = 2 \quad (r \neq 1). \tag{28}$$

Thus reduction of any prime into S_{840}, followed by b, is an explicit lattice-valued map on that prime domain.

Proof. Multiplication permutes the orthonormal free coordinates, preserves their augmentation kernels, and obeys $\rho(r)\rho(s) = \rho(rs)$. On the i-th fibre its dual discriminant generator is

$$\lambda_i = e_{f_i} - \frac{1}{6} \sum_{t \in F_{\eta_i}} e_t.$$

Its image differs from it by $e_{rf_i} - e_{f_i} \in A(F_{\eta_i})$. Hence $\rho(r)$ acts trivially on the entire A5 discriminant group, since λ_i generates it. It fixes D and therefore the whole discriminant group, in particular $\mathcal{H}$. It follows from the inverse-image definition that it preserves N. If $\rho(r)$ were the identity, applying it to $e_1 - e_{289}$ would give $e_r - e_{289r} = e_1 - e_{289}$. Since $289 \neq 1$, the positive coordinate on the left is exactly r, so comparison of free-coordinate coefficients forces $r = 1$. This proves faithfulness.

Finally

$$b(r) + \rho(r)b(s) = (e_r - e_1) + (e_{rs} - e_r) = e_{rs} - e_1 = b(rs).$$

For $r \neq 1$, the two basis vectors in the difference are orthogonal unit vectors, giving squared norm two. The stated prime map is the composition $p \mapsto p \bmod 840 \mapsto b(p \bmod 840)$, whose domains have all been specified. □

Fix a positive shell $R \equiv 3 \pmod 4$, and let P be any finite set of positive primes whose reduction lies in S_{840}. For every $p \in P$, use its actual count $A_R(p)$ defined by the reduced divisor set, including the full data of (18). Define

$$\begin{aligned} w_{R,P}(r) &= \sum_{\substack{p \in P \\ p \equiv r \pmod{840}}} A_R(p), \\ \widehat{w}_{R,P}(k) &= \sum_{r \in S_{840}} w_{R,P}(r) \exp\left(-\frac{2\pi \mathrm{i} k \kappa(r)}{6}\right), \qquad k \in \mathbb{Z}/6\mathbb{Z}. \end{aligned} \tag{29}$$

This is the pushforward along prime reduction, followed by the finite Fourier transform with the already fixed generator 289.

Theorem 2.12 (Inversion and an integral augmentation-valued count map)**.** *The function $w_{R,P}$ is recovered by*

$$w_{R,P}(r) = \frac{1}{6} \sum_{k=0}^{5} \widehat{w}_{R,P}(k) \exp\left(\frac{2\pi \mathrm{i} k \kappa(r)}{6}\right). \tag{30}$$

Let $u = \sum_{r \in S_{840}} e_r$, $W_{R,P} = \sum_r w_{R,P}(r) e_r$, and $M_{R,P} = \sum_r w_{R,P}(r) = \widehat{w}_{R,P}(0)$. Then

$$Q_{R,P} = 6W_{R,P} - M_{R,P} u \in A(S_{840}) \subset N \tag{31}$$

is a well-defined integral lattice-valued count map. Put $\zeta_6 = e^{2\pi \mathrm{i}/6}$ and

$$f_k = \sum_{j=0}^{5} \zeta_6^{kj} e_{289^j} \qquad (0 \le k \le 5).$$

The five vectors $f_1, \dots, f_5$ form a basis of $A(S_{840}) \otimes \mathbb{C}$, with

$$\rho(289) f_k = \zeta_6^{-k} f_k, \qquad Q_{R,P} = \sum_{k=1}^{5} \widehat{w}_{R,P}(k) f_k. \tag{32}$$

Both the prime sample P and the shell R are retained in the domain of these maps.

Proof. For an integer m, the sum $\sum_{k=0}^{5} \zeta_6^{km}$ equals 6 if $6 \mid m$. Otherwise it is the geometric sum $(1 - \zeta_6^{6m})/(1 - \zeta_6^m) = 0$. Substituting (29) into the right side of (30) therefore gives

$$\sum_{s \in S_{840}} w_{R,P}(s) \frac{1}{6} \sum_{k=0}^{5} \zeta_6^{k(\kappa(r) - \kappa(s))} = w_{R,P}(r),$$

because κ is bijective. This proves inversion.

The coefficient sum of $6W_{R,P} - M_{R,P}u$ is $6M_{R,P} - 6M_{R,P} = 0$, and its coefficients are integers, so it belongs to the exact augmentation lattice. The same geometric sum shows that f_k has coefficient sum zero for $1 \le k \le 5$ and $f_0 = u$. Moreover the six f_k are independent: the coefficient Fourier transform of f_k is six in coordinate k and zero in all other coordinates, by that same identity. Since the augmentation space has dimension five, its five nonzero-frequency vectors form a basis. Replacing j by $j + 1$ in $\rho(289) f_k$ gives the multiplier ζ_6^{-k}, including its displayed sign.

Inversion in each free coordinate gives $W_{R,P} = (1/6) \sum_{k=0}^{5} \widehat{w}_{R,P}(k) f_k$. Multiplication by 6 and subtraction of $M_{R,P}u = \widehat{w}_{R,P}(0) f_0$ prove the second identity in (32). All sums were taken over the specified finite set P and its residue image. The transform therefore recovers these aggregate counts; it makes no identification between the separate counts of distinct primes with the same residue. Their possible inequality is proved in Theorem 2.10. □

2.9 The entire lattice automorphism group and its Weyl quotient

Cheng, Duncan and Harvey define the umbral group as the quotient of the Niemeier lattice automorphism group by its root-reflection subgroup; their Table 3 identifies this quotient as $GL_2(3)$ for $A_5^4D_4$ [1, Section 2.4 and Table 3]. The argument here proves that identification in the present residue, metric, discriminant and glue coordinates, and locates the arithmetic six-cycle in the full group. It does not claim the abstract group as new. The displayed complement is the particular lift already constructed here; no equality of its representation with a simple-root-preserving section is assumed.

Retain the exact lattice N, the root lattice R_{ES}, the four fibre identifications φ_i, the discriminant marking, the glue subgroup $\mathcal{H}$, and the lift $g \mapsto L_g$ constructed above. For a lattice K in a Euclidean space, $O(K)$ means the group of real linear isometries of its real span that map K onto itself. For the present root lattice define

$$W(R_{\mathrm{ES}}) = \langle s_\alpha : \alpha \in R_{\mathrm{ES}},\ (\alpha,\alpha) = 2\rangle, \qquad s_\alpha(x) = x - (x,\alpha)\alpha.$$

All actions below retain the previously fixed six-coordinate order in each A_5 summand and the marking $10 \mapsto v$, $01 \mapsto s$, $11 \mapsto c$ in $D^\vee/D$.

Theorem 2.13 (All isometries of the displayed A_5 lattice)**.** *Let $A = \{x \in \mathbb{Z}^6 : \sum_{j=0}^{5} x_j = 0\}$ with its displayed Euclidean metric, and for a permutation σ of $\{0,1,2,3,4,5\}$ let $Q_\sigma e_j = e_{\sigma(j)}$. Every element of $O(A)$ has exactly one expression $\epsilon Q_\sigma|_{\mathbb{R}A}$, where $\epsilon \in \{1,-1\}$. Thus*

$$O(A) = \langle Q_\sigma|_{\mathbb{R}A} : \sigma \in S_6\rangle \times \{I, -I\}, \qquad W(A) = \langle Q_\sigma|_{\mathbb{R}A} : \sigma \in S_6\rangle \cong S_6.$$

On the marked discriminant group $D_A = (\mathbb{Z}/6)\lambda$, $\lambda = e_0 - \frac{1}{6}\sum_{j=0}^{5} e_j$, this isometry acts by $k\lambda + A \mapsto \epsilon k\lambda + A$. The kernel of this discriminant action is exactly $W(A)$.

Proof. The norm-two vectors are precisely $\alpha_{ij} = e_i - e_j$ for $i \neq j$. For two distinct such roots,

$$(\alpha_{ij}, \alpha_{kl}) = \delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl}.$$

Their inner product is one exactly when they have the same first index or the same second index. Indeed equality of first indices, with distinct second indices, makes the displayed expression one; equality of second indices, with distinct first indices, does the same. If both positive Kronecker terms vanish the expression is nonpositive, and if both are one then the roots are equal, the excluded case.

Put

$$C_i = \{\alpha_{ij} : j \neq i\}, \qquad -C_i = \{\alpha_{ji} : j \neq i\} \qquad (0 \le i \le 5).$$

These twelve sets are exactly the maximal sets of roots in which every two distinct members have inner product one. To prove this, a singleton can be enlarged by keeping its first index and choosing a third index, so a maximal such set contains a pair. If the pair is α_{ij}, α_{ik}, with $j \neq k$, a further root α_{lm} has inner product one with both only if

$$(l = i \text{ or } m = j) \quad \text{and} \quad (l = i \text{ or } m = k).$$

Since $j \neq k$, these conditions force $l = i$. Thus the whole set is contained in C_i, and maximality makes it equal to C_i. For a pair with common second index the same argument, after negating both roots and every possible further member, gives $-C_i$. Each C_i and $-C_i$ has five roots and is maximal by this argument.

Form a graph whose vertices are these twelve maximal sets, with an edge when the two sets have nonempty intersection. Its intersections are exactly

$$C_i \cap C_j = (-C_i) \cap (-C_j) = \varnothing \quad (i \neq j), \qquad C_i \cap (-C_j) = \begin{cases} \{\alpha_{ij}\}, & i \neq j, \\ \varnothing, & i = j. \end{cases}$$

Consequently it is the bipartite graph on the two displayed six-element families with every cross-edge except the six pairs $(C_i, -C_i)$. It is connected: two distinct vertices on the same side have a common neighbor indexed by any index different from both; vertices on opposite sides with different indices are adjacent; and C_i and $-C_i$ are joined by the path $C_i, -C_j, C_k, -C_i$ for distinct i, j, k. A connected bipartite graph has only two choices for its ordered bipartition, because the side of a fixed vertex determines the side of each other vertex along any connecting path. Hence an isometry of A, which permutes roots and preserves their inner products and set intersections, either preserves both displayed families or exchanges them.

In the first case write $C_i \mapsto C_{\sigma(i)}$ and $-C_i \mapsto -C_{\tau(i)}$. The unique opposite-family vertex not adjacent to C_i is $-C_i$. Preservation of that missing edge forces $\tau(i) = \sigma(i)$ for each i. The singleton intersection formula then gives $\alpha_{ij} \mapsto \alpha_{\sigma(i)\sigma(j)}$. In the case exchanging the families, composition with $-I$ reduces to the first case, so the root action is $\alpha_{ij} \mapsto -\alpha_{\sigma(i)\sigma(j)}$. The roots $e_j - e_5$, $0 \leq j \leq 4$, span $\mathbb{R}A$, so the root action determines the whole linear isometry. These cases give precisely the asserted expressions.

The permutation σ is determined by the permutation of the six vertices C_i, after the sign is determined by whether the two families are preserved or exchanged. Thus the expression is unique. Every Q_σ and $-I$ preserves A, and $-I$ commutes with every linear map; uniqueness proves the direct product. Reflection in $e_i - e_j$ interchanges free coordinates i, j and fixes the others, as follows on substituting this root in $x \mapsto x - (x, e_i - e_j)(e_i - e_j)$. Every permutation is a product of transpositions: each cycle $(i_1\ i_2\ \cdots\ i_m)$ is $(i_1\ i_m)(i_1\ i_{m-1}) \cdots (i_1\ i_2)$. These reflections therefore generate precisely the displayed S_6. Finally

$$Q_\sigma \lambda - \lambda = e_{\sigma(0)} - e_0 \in A.$$

Thus the discriminant action is the stated sign. Since $\lambda + A$ has exact order six, its negative is distinct from itself; the action is trivial exactly when $\epsilon = 1$. □

Theorem 2.14 (All isometries of the displayed D_4 lattice)**.** *For the exact lattice $D = \{x \in \mathbb{Z}^4 : \sum_i x_i \equiv 0 \pmod 2\}$, the action on its marked discriminant group gives the split exact sequence*

$$1 \longrightarrow W_D \longrightarrow O(D) \xrightarrow{d_D} GL_2(\mathbb{F}_2) \longrightarrow 1, \qquad O(D) = W_D \rtimes \delta(GL_2(\mathbb{F}_2)).$$

Here W_D consists of all signed coordinate permutations with an even number of minus signs, and δ is exactly the section generated by the earlier matrices R_D, H_D. In particular $O(D) = W_F$ in the root and weight coordinates of Theorem 2.4.

Proof. Every isometry of D preserves its dual: if $T \in O(D)$, $x \in D^\vee$, and $a \in D$, then $(Tx, a) = (x, T^{-1}a) \in \mathbb{Z}$. It therefore induces an additive automorphism of $D^\vee/D \cong \mathbb{F}_2^2$, giving the homomorphism d_D. The three nonzero classes have the exact norm-one representative sets

$$\begin{aligned} \mathcal{V} &= \{\pm e_i : 1 \leq i \leq 4\}, \\ \mathcal{S}_+ &= \{\tfrac{1}{2}\textstyle\sum_i \delta_i e_i : \prod_i \delta_i = 1\}, \\ \mathcal{S}_- &= \{\tfrac{1}{2}\textstyle\sum_i \delta_i e_i : \prod_i \delta_i = -1\}, \qquad \delta_i \in \{1, -1\}. \end{aligned}$$

in the respective marked classes v, s, c. For the integral nonzero class, norm one forces exactly one coordinate ± 1. A half-integral vector has norm at least $4(1/2)^2 = 1$, with equality exactly when every coordinate is $\pm 1/2$. The difference of such a vector from $s = (1,1,1,1)/2$ has one coordinate -1 per minus sign, so it belongs to D exactly when their number is even. This proves the three sets and their marking directly.

If $d_D(T) = I$, then T preserves $\mathcal{V}$. The vectors Te_i are orthonormal members of $\mathcal{V}$, so $Te_i = \delta_i e_{\sigma(i)}$ for a permutation σ of four indices and signs δ_i. Preservation of $\mathcal{S}_+$ forces $Ts \in \mathcal{S}_+$, hence $\prod_i \delta_i = 1$. Thus the kernel consists only of even signed permutations.

Conversely, reflection in $e_i - e_j$ interchanges coordinates i, j, and reflection in $e_i + e_j$ interchanges them and negates both. Their composition produces a sign change in exactly those two coordinates. Any even set of sign changes is a product of such disjoint pairs, and transpositions generate all permutations. The root reflections therefore generate every even signed permutation, and each root reflection is itself such a matrix. If $\alpha \in D$ has norm two and $x \in D^\vee$, then $s_\alpha x - x = -(x,\alpha)\alpha \in D$. Every root reflection acts trivially on the discriminant group. This proves $\ker d_D = W_D$ in both directions.

For completeness the section has the original exact matrices

$$R_D = \operatorname{diag}(-1,1,1,1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

They preserve D, are orthogonal involutions, and act on (v, s, c) by $(s\ c)$ and $(v\ s)$, respectively, as calculated in Theorem 2.4. Their binary matrices in the retained marking are

$$B_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B_H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The same earlier coordinate calculation gives $R_D^2 = H_D^2 = (R_D H_D)^3 = I$; thus their subgroup has at most six elements, by reducing words to $(R_D H_D)^k$ or $(R_D H_D)^k R_D$, $k = 0, 1, 2$. The induced two transpositions generate the six permutations of the three nonzero binary vectors. There are exactly six invertible binary matrices, by the $3 \cdot 2$ choices of their independent columns, and their action on the three nonzero vectors is faithful. Hence the six words have distinct discriminant actions, and the restriction of d_D to $\langle R_D, H_D \rangle$ is an isomorphism onto $GL_2(\mathbb{F}_2)$. Its inverse is the specified δ. For every $T \in O(D)$, $T\delta(d_D(T))^{-1} \in \ker d_D = W_D$. The two factors intersect trivially because $d_D \delta = I$. This proves the asserted splitting and exhausts $O(D)$. The earlier theorem proved $W_F = \langle W_D, R_D, H_D \rangle$, so it also proves the last equality. □

Theorem 2.15 (The exact full automorphism quotient of N)**.** *Every isometry of N has a unique factorization*

$$T = wL_g, \qquad w \in W(R_{\mathrm{ES}}), \quad g \in G.$$

The Weyl group is normal, and the induced discriminant action gives the split exact sequence and the indicated isomorphisms

$$\begin{aligned} 1 \longrightarrow W(R_{\mathrm{ES}}) &\longrightarrow O(N) \xrightarrow{d_{\mathrm{ES}}} G \longrightarrow 1, \\ O(N) &= W(R_{\mathrm{ES}}) \rtimes L(G), \\ L(G) &\xrightarrow{\cong} O(N)/W(R_{\mathrm{ES}}) \cong G \xrightarrow{\Theta} GL_2(\mathbb{F}_3), \\ W(R_{\mathrm{ES}}) &\cong S_6^4 \times W_D, \qquad |O(N)| = 48 \cdot (6!)^4 \cdot (2^3 \cdot 4!). \end{aligned} \tag{33}$$

In identified coordinates, if $w = (Q_{\sigma_1}, Q_{\sigma_2}, Q_{\sigma_3}, Q_{\sigma_4}; w_D)$ *and* $g = (\varepsilon, p, B)$*, the exact semidirect-product action is*

$$L_g w L_g^{-1} = (Q_{\sigma_{p(1)}}, Q_{\sigma_{p(2)}}, Q_{\sigma_{p(3)}}, Q_{\sigma_{p(4)}}; \delta(B) w_D \delta(B)^{-1}). \tag{34}$$

In the original fibre coordinates the i-th permutation factor on the right means $\varphi_i Q_{\sigma_{p(i)}} \varphi_i^{-1}$*.*

Proof. Theorem 2.3 proves that the norm-two vectors of N are exactly the 144 roots of its five original orthogonal summands. Join two distinct roots by an edge when their inner product is nonzero. The five summands are exactly the connected components of this graph. There are no edges between different summands by orthogonality. Within an A_5 summand, the simple roots $e_0 - e_1, e_1 - e_2, e_2 - e_3, e_3 - e_4, e_4 - e_5$ have consecutive inner product -1, forming a connected graph, and span its real space. Every root not already among them has nonzero inner product with at least one of them; otherwise it is orthogonal to its entire real space and has norm zero, contrary to norm two. Thus its whole root graph is connected. In D, the four roots

$$e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_3 + e_4$$

span $\mathbb{R}D$, and the second has inner product -1 with each of the other three. The same spanning argument proves connectedness of the D_4 root graph.

An element $T \in O(N)$ permutes the norm-two vectors and preserves these edges, hence permutes their connected components. Their cardinalities are $30, 30, 30, 30, 24$, so T preserves the D_4 component and permutes the four A_5 components. The roots generate each original lattice summand: the five and four displayed simple roots are their integral bases by Theorem 2.2. Consequently T preserves their integer span R_{ES}. The same component argument applies to every element of $O(R_{\mathrm{ES}})$, since its norm-two vectors are the same 144 roots. It follows that every such element has, in the fixed identified coordinates, the form

$$T(X_1, X_2, X_3, X_4; d) = \left(\varepsilon_i Q_{\sigma_i} X_{p(i)}\right)_{i=1}^{4}; T_D d, \tag{35}$$

where $p \in S_4$, $\varepsilon_i \in \{1, -1\}$, $\sigma_i \in S_6$, and $T_D \in O(D)$. This follows by transporting each isometry from its input A_5 summand to its output summand using the fixed φ_i, then applying Theorem 2.13. All choices in (35) are unique. The convention is precisely the earlier convention: a vector supported on component k is sent to component $p^{-1}(k)$.

Let $\mathcal{A}_{\mathrm{ES}}$ be the group of transformations of the exact discriminant group $(\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2$ given by

$$(x; y) \longmapsto ((\varepsilon_i x_{p(i)})_{i=1}^{4}; By), \qquad \varepsilon_i \in \{1, -1\}, \quad p \in S_4, \quad B \in GL_2(\mathbb{F}_2).$$

Each tuple gives a distinct transformation: the image of the order-six generator in each individual summand determines its output summand and its sign, and the images of the two binary basis vectors determine B. The discriminant action of (35), calculated by the two preceding theorems, is exactly this tuple with $B = d_D(T_D)$. This defines a homomorphism

$$d_{\mathrm{ES}} : O(R_{\mathrm{ES}}) \longrightarrow \mathcal{A}_{\mathrm{ES}}.$$

It is a homomorphism because extending an isometry to the dual and then passing to the quotient commutes with composition. In particular the component permutation rule remains $P_p P_q = P_{q \circ p}$; there is no change of the earlier convention.

This homomorphism is onto. Given any tuple $a = (\varepsilon, p, B) \in \mathcal{A}_{\mathrm{ES}}$, define

$$\widetilde{L}_a(X_1, X_2, X_3, X_4; d) = \left(\varepsilon_i X_{p(i)}\right)_{i=1}^{4}; \delta(B) d.$$

It is a root-lattice isometry and has discriminant action a. The map $a \mapsto \widetilde{L}_a$ is a homomorphism: if $a = (\varepsilon, p, B)$ and $a' = (\varepsilon', q, B')$, its composite has i-th output $\varepsilon_i \varepsilon'_{p(i)} X_{q(p(i))}$ and D-component $\delta(B)\delta(B')d = \delta(BB')d$, which are exactly the tuple product in the stated convention.

The kernel of d_{ES} consists exactly of the tuples in (35) with $p = 1$, every $\varepsilon_i = 1$, and $T_D \in W_D$. It is therefore the direct product $S_6^4 \times W_D$. Every root reflection acts on its own orthogonal summand and fixes the other four pointwise, and the preceding two theorems identify the groups generated inside those summands. Hence this direct product is exactly $W(R_{\mathrm{ES}})$.

Every root reflection also preserves N. Indeed $\alpha \in N$ and integrality of N give $s_\alpha x = x - (x, \alpha)\alpha \in N$ for every $x \in N$. The reflection is an involution, so inclusion is equality. Thus $W(R_{\mathrm{ES}}) \subset O(N)$. It is normal there, since for $T \in O(N)$

$$T s_\alpha T^{-1}(x) = x - (x, T\alpha)T\alpha = s_{T\alpha}(x),$$

and $T\alpha$ is again a norm-two root.

It remains to impose the actual glue, which completely determines the remaining isometries. For $T \in O(R_{\mathrm{ES}})$ and $x \in R_{\mathrm{ES}}^\vee$, the inverse-image definition gives

$$Tx \in N \quad \Longleftrightarrow \quad d_{\mathrm{ES}}(T)(x + R_{\mathrm{ES}}) \in \mathcal{H}.$$

Consequently $T(N) = N$ if and only if $d_{\mathrm{ES}}(T)(\mathcal{H}) = \mathcal{H}$. For the forward implication, represent each class of $\mathcal{H}$ by a vector of N and apply both T and T^{-1}. For the reverse implication, the displayed equivalence applies to T and T^{-1} because the action on $\mathcal{H}$ is bijective. The group preserving $\mathcal{H}$ inside precisely $\mathcal{A}_{\mathrm{ES}}$ is the group G of Theorem 2.5, proved there to be exactly $GL_2(\mathbb{F}_3)$ through the code map Θ. We have therefore proved the equality of subgroups

$$O(N) = d_{\mathrm{ES}}^{-1}(G) \quad \text{inside } O(R_{\mathrm{ES}}).$$

The section $\widetilde{L}$ restricts on G to the already specified lift L. This proves surjectivity of the restricted map $d_{\mathrm{ES}} : O(N) \to G$, with exactly the asserted kernel. For $T \in O(N)$, set $g = d_{\mathrm{ES}}(T)$. Then $w = TL_g^{-1}$ lies in the kernel and $T = wL_g$. If $wL_g = w'L_{g'}$, applying d_{ES} gives $g = g'$, then cancellation gives $w = w'$. In particular $L(G) \cap W(R_{\mathrm{ES}}) = \{I\}$. Normality and this unique factorization prove the semidirect product and the quotient isomorphism, including its surjectivity. The four S_6 factors have order $(6!)^4$; W_D has 2^3 sign choices and $4!$ permutation choices, and $|G| = 48$. Unique factorization proves the displayed order.

Finally put $Z = L_g^{-1}X$. From the i-th output equation $X_i = \varepsilon_i Z_{p(i)}$ follows $Z_{p(i)} = \varepsilon_i X_i$, with each vector read in the fixed identified A coordinates. Therefore the i-th output of $L_g w L_g^{-1} X$ is

$$\varepsilon_i Q_{\sigma_{p(i)}} Z_{p(i)} = \varepsilon_i Q_{\sigma_{p(i)}} \varepsilon_i X_i = Q_{\sigma_{p(i)}} X_i.$$

The D-component is $\delta(B) w_D \delta(B)^{-1} d$. It belongs to W_D by normality of $\ker d_D$. These are exactly the components of (34), proving the asserted action with all signs and component indices retained. □

Theorem 2.16 (The residue six-cycle in the full automorphism group)**.** *For $r \in S_{840} = \langle 289 \rangle$, let $\rho(r)e_s = e_{rs}$ on each of the four fibres and $\rho(r)|_D = I$, with the exact fibre action used in* (27)*. Then*

$$\rho(S_{840}) \subset W(R_{\mathrm{ES}}), \qquad d_{\mathrm{ES}}(\rho(r)) = 1, \qquad L_g \rho(r) L_g^{-1} = \rho(r) \quad (g \in G,\ r \in S_{840}). \tag{36}$$

In particular the map

$$S_{840} \times G \longrightarrow O(N), \qquad (r, g) \longmapsto \rho(r)L_g$$

is an injective homomorphism with image of order $6 \cdot 48 = 288$, isomorphic to $C_6 \times GL_2(\mathbb{F}_3)$.

Proof. Put $Ce_j = e_{j+1 \bmod 6}$ on the free six-coordinate space and restrict it to $\mathbb{R}A$. If $r = 289^m$, with $m \in \{0,1,2,3,4,5\}$, the exact fibre identification gives

$$\rho(r)e_{f_i289^j} = e_{f_i289^{j+m}}, \qquad \rho(r)|_{A(F_{\eta_i})} = \varphi_i C^m \varphi_i^{-1}.$$

Thus in all four identified summands $\rho(r)$ is the same coordinate permutation C^m, and it is the identity on D. Theorem 2.13 puts it in the corresponding product of Weyl groups, giving the first two assertions. Each sign $\varepsilon_i I$ commutes with C^m, and the same operator C^m occurs in every component. Directly the i-th outputs of $L_g\rho(r)$ and $\rho(r)L_g$ are respectively $\varepsilon_i C^m X_{p(i)}$ and $C^m \varepsilon_i X_{p(i)}$, which agree; the D-outputs also agree since $\rho(r)|_D = I$. This proves the conjugation identity without changing the chosen origins.

The six restrictions $C^m|_{\mathbb{R}A}$ are distinct: if $C^m(e_0 - e_1) = e_0 - e_1$, comparison of the unique coordinate with coefficient $+1$ gives $m = 0 \pmod 6$. Thus ρ is faithful. Commutation makes the displayed product map a homomorphism. If $\rho(r)L_g = I$, then $L_g = \rho(r)^{-1}$ lies in $L(G) \cap W(R_{\mathrm{ES}}) = \{I\}$. Faithfulness of both factors gives $r = 1$ and $g = 1$. This proves injectivity and the exact image order. □

3 The exact Leech neighbor of the retained index-72 lattice

Retain the lattice, metric, discriminant marking, and glue literally:

$$\begin{aligned}
A &= \{(x_0,\dots,x_5) \in \mathbb{Z}^6 : \textstyle\sum x_j = 0\},\\
D &= \{(d_1,\dots,d_4) \in \mathbb{Z}^4 : \textstyle\sum d_i \equiv 0 \pmod 2\}, \qquad R = A^{\perp 4} \perp D,\\
C_2 &= \langle b_0 = (1111;00), b_1 = (1100;10), b_2 = (1010;01)\rangle_{\mathbb{F}_2},\\
C_3 &= \langle z_1 = (1,1,1,0), z_2 = (1,2,0,1)\rangle_{\mathbb{F}_3},\\
H &= \{(3u + 2z; y) : (u;y) \in C_2,\ z \in C_3\},\\
N &= \{x \in R^\vee : x + R \in H\}.
\end{aligned}$$

The coordinates in each A remain the six ordered elements f_i289^j of its original residue fibre. The A discriminant generator is $\lambda = (5,-1,-1,-1,-1,-1)/6$. The D markings $10, 01, 11$ remain represented by $v_D = (1,0,0,0)$, $s_D = (1,1,1,1)/2$, $c_D = (-1,1,1,1)/2$. The preceding proof establishes that N is even unimodular, $[N : R] = 72$, $\det R = 5184$, and its roots are exactly those of $A_5^4 D_4$. Every root has squared norm two. No metric rescaling occurs anywhere below.

3.1 The Weyl vector and the exact neighbor

Use the original simple roots $a_j = e_j - e_{j+1}$, $0 \le j \le 4$, in each A component and

$$d_1 = e_1 - e_2, \quad d_2 = e_2 - e_3, \quad d_3 = e_3 - e_4, \quad d_4 = e_3 + e_4$$

in D. Their highest roots are $\theta_A = e_0 - e_5 = \sum_j a_j$ and $\theta_D = e_1 + e_2 = d_1 + 2d_2 + d_3 + d_4$. Both highest-root heights are five, so the common Coxeter number is six. Define

$$\rho_A = (5,3,1,-1,-3,-5)/2, \qquad \rho_D = (3,2,1,0), \qquad \rho = \rho_A^{\oplus 4} \perp \rho_D.$$

Each simple root pairs to one with ρ. Moreover $\rho_A - 3\lambda = (0,2,1,0,-1,-2) \in A$ and $\rho_D \in D$. Thus the exact glue class of ρ is $(3,3,3,3;00)$, the image of b_0, and $\rho \in N$. Directly

$$\rho_A^2 = 35/2, \quad \rho_D^2 = 14, \quad \rho^2 = 4(35/2) + 14 = 84 = 2 \cdot 6 \cdot 7.$$

Consequently $(\rho, N) \subset \mathbb{Z}$. Fix $\alpha = a_0$ in the first A component and put

$$\ell_\rho : N \to \mathbb{Z}/6, \quad x \mapsto (\rho, x) \bmod 6, \qquad K = \ker \ell_\rho, \quad R_0 = R \cap K, \quad v = \rho/6 - \alpha, \quad \Lambda = K + \mathbb{Z}v. \tag{NL1}$$

The pairing $(\rho, \alpha) = 1$ proves $[N : K] = [R : R_0] = 6$. For $x \in K$, the number $(v, x) = (\rho, x)/6 - (\alpha, x)$ is integral, and

$$v^2 = 84/36 - 2/6 + 2 = 4, \qquad 6v = \rho - 6\alpha \in K, \qquad (\rho, 6v) = 78.$$

Hence Λ is positive definite even integral of rank 24. If $mv \in N$, its pairing after adding $m\alpha$ with α is $m/6 \in \mathbb{Z}$, so $6 \mid m$. It follows that

$$N \cap \Lambda = K, \qquad [\Lambda : K] = 6, \qquad \det K = 36, \qquad \det \Lambda = 36/6^2 = 1. \tag{NL2}$$

Changing α to another root of ρ-pairing one changes v by an element of K, and so does not change Λ. We shall use the equivalent exact congruence description

$$\Lambda = \{x + k\rho/6 : x \in N,\ k \in \mathbb{Z},\ (\rho, x) + k \equiv 0 \pmod 6\}. \tag{NL3}$$

Indeed write $x + k\rho/6 = (x + k\alpha) + kv$; the first summand is in K exactly under this congruence. Changing k by six can be absorbed in x using $\rho \in N$, and preserves the congruence since $\rho^2 = 84$.

3.2 A complete rootlessness proof

Theorem 3.1 (The Leech neighbor in the retained coordinates)**.** *The lattice $\Lambda = K + \mathbb{Z}(\rho/6 - \alpha)$ is positive definite even unimodular of rank 24 and has minimum squared norm exactly four. It is the holy-construction Leech lattice of the retained $A_5^4 D_4$ glue, and $N \cap \Lambda = K$ with $[N : K] = [\Lambda : K] = 6$.*

Proof. The rank, parity, unimodularity and indices were proved in (NL1)–(NL2). The holy-presentation identification is proved in (NL8)–(NL11) below. We now prove the required minimum directly. For $k = 0$ in (NL3), a possible root is an original root of N. An A root pairs with ρ to one of $\pm 1, \ldots, \pm 5$. For D, all roots $\pm e_i \pm e_j$ have nonzero ρ_D-pairing of absolute value at most five. Thus none belongs to K. Negation and the reduction of k modulo six leave $k = 1, 2, 3$.

Define the exact shifted class minima by

$$m_A(k, t) = \min_{x \in t\lambda + A} |x + k\rho_A/6|^2, \qquad m_D(k, y) = \min_{x + D = y} |x + k\rho_D/6|^2.$$

Their values are

$k\backslash t$	0	1	2	3	4	5
1	35/72	35/72	35/72	35/72	35/72	35/72
2	11/18	4/9	11/18	4/9	11/18	4/9
3	3/8	17/24	17/24	3/8	17/24	17/24

$k\backslash y$	00	10	01	11
1	7/18	7/18	7/18	7/18
2	2/9	5/9	5/9	5/9
3	1/2	1/2	1/2	1/2

. (NL4)

We prove the table and its needed equality information. The shifted A coordinates are precisely

$$\left(n_j - \frac{t}{6} + \frac{k(5 - 2j)}{12}\right)_{j=0}^{5}, \qquad n_j \in \mathbb{Z}, \quad \sum n_j = t. \tag{NL5}$$

At $k = 1$, coordinatewise nearest representatives are the six numbers $\pm 1/12, \pm 3/12, \pm 5/12$ in an order depending on t. They sum to zero and their squares sum to 35/72. At $k = 2$ and even t, the nearest coordinates consist of two copies of each sign of $1/6$ and one copy of each sign of $1/2$; the

half-integer ties admit exactly the needed zero-sum choice and the squared sum is $11/18$. At odd t, there are two each of $0, 1/3, -1/3$, giving $4/9$. For $t = 1, 3, 5$ their actual ordered shifted vectors are

$$(-1, 1, 0, -1, 1, 0)/3, \quad (1, 0, -1, 1, 0, -1)/3, \quad (0, -1, 1, 0, -1, 1)/3. \tag{NL6}$$

Subtracting $\rho_A/3$ gives original squared norms $17/6, 3/2, 17/6$, respectively. At $k = 3, t = 0, 3$, the nearest coordinates alternate $\pm 1/4$, giving $3/8$ with zero sum. At each other t, the coordinatewise nearest representatives have three absolute values $1/12$ and three $5/12$, square sum $13/24$, and coordinate sum 1 or -1. Meeting the zero-sum constraint requires an integer change. Every nonzero coordinate change increases the squared sum by at least $(7/12)^2 - (5/12)^2 = 1/6$. Changing one suitable $5/12$ coordinate attains this increase and restores the zero sum. Thus the constrained minimum is $17/24$. At $t = 0, 3$, the original minimizing vectors are

$$(-1, -1, 0, 0, 1, 1), \qquad (-3, -1, -1, 1, 1, 3)/2, \tag{NL7}$$

of squared norms $4, 11/2$.

For D and $k = 1$ the shift is $(1/2, 1/3, 1/6, 0)$. The nearest integer coordinate distances are $(1/2, 1/3, 1/6, 0)$; the first-coordinate tie meets either integer parity. The nearest half-integer distances are $(0, 1/6, 1/3, 1/2)$; the last-coordinate tie meets either half-integer class. Both squared sums are $7/18$. At $k = 2$ the shift is $(1, 2/3, 1/3, 0)$. The unique nearest integer vector before shifting is $(-1, -1, 0, 0) \in D$, giving $2/9$. Changing parity costs at least $1/3$, attained by changing the second or third coordinate, giving $5/9$. The half-integer coordinate distances are $(1/2, 1/6, 1/6, 1/2)$, giving $5/9$; ties meet either marking. At $k = 3$ the shift is $(3/2, 1, 1/2, 0)$. Integer-class minimum distances are $(1/2, 0, 1/2, 0)$, and half-integer-class distances $(0, 1/2, 0, 1/2)$. Ties meet every marking and all square sums are $1/2$. In class 00, the two original minimizing vectors are

$$(-1, -1, 0, 0), \qquad (-2, -1, -1, 0),$$

of squared norms $2, 6$. In class 10, they are

$$(-1, -1, -1, 0), \qquad (-2, -1, 0, 0),$$

of squared norms $3, 5$. In either half-integer class they have the shapes $(-3/2, -1/2, -1/2, \pm 1/2)$ and $(-3/2, -3/2, -1/2, \pm 1/2)$. Exactly one last-coordinate sign in each shape meets each original marking; their squared norms are $3, 5$. This proves every entry and the relevant equality cases in (NL4).

For $k = 1$, every full shifted vector has squared norm at least $4(35/72) + 7/18 = 7/3 > 2$. For $k = 2$, the lower bound from (NL4) is two. Equality forces all t_i odd and $y = 00$. In the actual glue this forces $u = 1111$. If $z = 0$, all $t_i = 3$ and (NL6) gives $x^2 = 4(3/2) + 2 = 8$. If $z \neq 0$, its weight is three, so exactly one $t_i = 3$ and three $t_i \in \{1, 5\}$; then $x^2 = 3/2 + 3(17/6) + 2 = 12$. The norm-two equation is

$$2 = x^2 + \tfrac{2}{3}(\rho, x) + \tfrac{28}{3}, \qquad (\rho, x) = -11 - \tfrac{3}{2}x^2.$$

Its pairing is -23 or -29, always 1 modulo six. Condition (NL3) requires residue 4, so all nine equality candidates, one per ternary word, are excluded.

For $k = 3$, the lower bound two requires $z = 0$: every nonzero ternary word raises the minimum by $3(17/24 - 3/8) = 1$. If $u = 0$, the possible original squared norms are $18, 22$. If $u = 1111$, they are $24, 28$. If u has weight two, its D class is nonzero and they are $2(4) + 2(11/2) + 3 = 22$ or $2(4) + 2(11/2) + 5 = 24$. These are all sixteen equality candidates. But

$$2 = x^2 + (\rho, x) + 21, \qquad (\rho, x) = -19 - x^2,$$

and (NL3) would require $x^2 \equiv 2 \pmod 6$. None of $18, 22, 24, 28$ has that residue. Thus Λ has no roots. Together with (NL2), this proves that it is positive definite even unimodular of rank 24, with minimum squared norm exactly four, attained by v. □

3.3 The exact holy-construction presentation

For $0 \le t \le 5$ set

$$\omega_t = (\underbrace{1,\dots,1}_{t},\underbrace{0,\dots,0}_{6-t}) - \tfrac{t}{6}(1,\dots,1).$$

Then $\omega_t + A = t\lambda + A$ and $(\rho_A, \omega_t) = t(6-t)/2 = 3\omega_t^2$. For the D markings set

$$\omega_{00} = 0, \quad \omega_{10} = e_1, \quad \omega_{01} = (1,1,1,1)/2, \quad \omega_{11} = (1,1,1,-1)/2.$$

The last differs from the original c_D by $(1,0,0,-1) \in D$, so retains its exact marking. For $a = (t_1,t_2,t_3,t_4;y) \in H$ define

$$u_a = (\omega_{t_1},\omega_{t_2},\omega_{t_3},\omega_{t_4};\omega_y), \qquad g_a = u_a - \rho/6. \tag{NL8}$$

These give all 72 representatives of N/R. Each nonzero D representative has norm one and pairing three; hence

$$(\rho,u_a) = 3u_a^2 \in 6\mathbb{Z}, \quad u_a \in K, \quad g_a^2 = u_a^2 - (\rho,u_a)/3 + 84/36 = 7/3. \tag{NL9}$$

Let $\mathcal{F}$ be the 29 vectors consisting, in each original component, of the negative original simple roots and the positive highest root. Their pairings with ρ are -1 or 5. There are positive integral zero relations of coefficient sum six:

$$\theta_A + \sum_{j=0}^{4}(-a_j) = 0, \qquad \theta_D + (-d_1) + 2(-d_2) + (-d_3) + (-d_4) = 0. \tag{NL10}$$

Every component of g_a pairs to $1/6$ with the corresponding vertices except one tip, where the pairing is $-5/6$. For A this follows because ω_t has simple-root pairing one only at a_{t-1} when $t \ne 0$, and has $(\omega_t,\theta_A) = 1$ for $t \ne 0$. For D the three nonzero representatives have simple-root pairing one only at d_1,d_4,d_3, respectively; all have highest-root pairing one. The zero representative points to the highest root. These computations fix all orientations in the holy generator diagram.

The following coefficient presentations are exact equalities:

$$\begin{aligned} N &= \left\{ \sum_f a_f f + \sum_a b_a g_a : a_f, b_a \in \mathbb{Z},\ \sum_a b_a = 0 \right\}, \\ \Lambda &= \left\{ \sum_f a_f f + \sum_a b_a g_a : a_f, b_a \in \mathbb{Z},\ \sum_f a_f + \sum_a b_a = 0 \right\}. \end{aligned} \tag{NL11}$$

For the first, the f span R and $g_a - g_0 = u_a$ generate its given extension N; the zero coefficient sum cancels $\rho/6$. For the second, write $A = \sum a_f$, $B = \sum b_a$, $r = \sum a_f f$, and $x = r + \sum b_a u_a$. The vector is $x - B\rho/6$, and by (NL9) $(\rho,x) \equiv -A \equiv B$, exactly (NL3) with $k = -B$. Conversely, for any $x + k\rho/6$ in (NL3), write $x = r + \sum b_a u_a$. The coefficient of $u_0 = 0$ can be set so $B = -k$. Write r with the negative simple-root vertices. The congruence gives $A + B \equiv 0 \pmod 6$. Adding a suitable multiple of a zero relation (NL10) makes $A + B = 0$ exactly. Thus both directions of (NL11) are proved.

These are exactly the holy coefficient rules of Conway–Sloane [6, pp. 277–280]. Borcherds gives the uniform proof of the holy-construction theorem [7, Section 7, especially the coefficient rules (i)–(ii) and Lemma 7.3]. Thus (NL11) is an explicit realization of that Leech construction from the actual 72 marked glue words; all its metric, determinant, integrality and rootlessness assertions have also been independently proved here.

3.4 The indices, discriminant morphism, and cyclic order twelve

Reduction modulo R gives an isomorphism $K/R_0 \cong H$: its kernel is R_0, and (NL9) supplies representatives in K for every H class. Consequently

$$[K : R_0] = 72, \quad [N : R_0] = [\Lambda : R_0] = 432, \quad \det R_0 = 6^2 \cdot 5184 = 186624. \tag{NL12}$$

Thus the original and new inclusions, with every index, are

$$R_0 \subset R \subset N, \quad [R : R_0] = 6,\ [N : R] = 72; \qquad R_0 \subset K \subset \Lambda, \quad [K : R_0] = 72,\ [\Lambda : K] = 6.$$

In particular $R \cap \Lambda = R_0$, an exact common sublattice.

Theorem 3.2 (The exact order-twelve quotient generator)**.** *The quotient of Λ by the original height-zero root sublattice R_0 is $\Lambda/R_0 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/6)^2$. The class of $v = \rho/6 - \alpha$ has exact order twelve.*

Proof. Let $\bar{b}_i$ denote the image in H of the binary generator b_i, namely its first four coordinates multiplied by three and its last two coordinates retained. Since $6v = \rho - 6\alpha$ and $6\alpha \in R_0$,

$$6[v] = \bar{b}_0 \quad \text{in } \Lambda/R_0. \tag{NL13}$$

This is a nonzero order-two element. The classes $\bar{b}_0, \bar{b}_1, \bar{b}_2, 2z_1, 2z_2$ form the displayed independent binary and ternary generators of H. Eliminate $\bar{b}_0$ using (NL13). The resulting surjective presentation is

$$\Lambda/R_0 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/3)^2 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/6)^2. \tag{NL14}$$

There are no further relations: this presentation has order $12 \cdot 4 \cdot 9 = 432$, equal to (NL12). Thus $[v]$ has exact order twelve. This particular occurrence of twelve is proved from the Coxeter denominator six and the actual order-two Weyl-vector glue class. □

Set $M = N + \Lambda = N + \mathbb{Z}(\rho/6)$. Then

$$M = K^\vee, \qquad K^\vee/K = \langle [\alpha], [v] \rangle \cong (\mathbb{Z}/6)^2. \tag{NL15}$$

Indeed $M \subset K^\vee$, its index over N is six by pairing with α, and $[K^\vee : N] = [N : K] = 6$. The cyclic images of N/K and Λ/K have order six and trivial intersection, proving the stated generators. Their exact discriminant data are

$$q_K([\alpha]) = q_K([v]) = 0, \quad b_K([\alpha], [v]) = 1/6 \pmod{\mathbb{Z}}, \quad q_K(a[\alpha] + b[v]) = ab/3 \pmod{2\mathbb{Z}}. \tag{NL16}$$

These follow from $\alpha^2 = 2$, $v^2 = 4$ and $(\alpha, v) = 1/6 - 2 = -11/6$. The two original lattices are therefore the inverse images of the two indicated order-six isotropic subgroups in this specified discriminant form.

3.5 An exact Lorentzian quotient and symmetry transport

Take $\mathcal{I} = N \perp U$ in coordinates (x, m, n) with

$$\langle (x, m, n), (x', m', n') \rangle = (x, x') - mn' - m'n.$$

Thus U has Gram matrix $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. The vectors $z = (0, 0, 1)$ and $w = (\rho, 6, 7)$ are primitive isotropic: $w^2 = 84 - 2 \cdot 6 \cdot 7 = 0$, and six and seven are coprime. There are integral quotient isometries

$$\begin{aligned} z^\perp/\mathbb{Z}z &\longrightarrow N, \quad [(x, 0, n)] \mapsto x, \\ w^\perp/\mathbb{Z}w &\longrightarrow \Lambda, \quad [(x, m, n)] \mapsto x - m\rho/6. \end{aligned} \tag{NL17}$$

For the second map, orthogonality means $(\rho, x) = 7m + 6n$, giving (NL3) with $k = -m$. Conversely take $m = -k$ and $n = ((\rho, x) + 7k)/6$ for a vector in (NL3). If the image vanishes, $x = m\rho/6$; pairing with α forces $m = 6j$, and then $n = 7j$, so the kernel is exactly $\mathbb{Z}w$. Finally

$$|x - m\rho/6|^2 = x^2 - \tfrac{m}{3}(\rho, x) + \tfrac{7}{3}m^2 = x^2 - 2mn.$$

Polarization gives the full pairing, proving the isometry.

For every $T \in O(N)$ the exact transport laws are

$$T(K_\rho) = K_{T\rho}, \qquad T(\Lambda_\rho) = \Lambda_{T\rho}, \qquad (x, m, n) \mapsto (Tx, m, n) : w_\rho \mapsto w_{T\rho}. \tag{NL18}$$

They follow by substitution in (NL3). In particular the stabilizer of ρ acts on the fixed Λ_ρ. General isometries act on this specified family of neighbors; no assertion that an arbitrary chosen automorphism complement fixes this Weyl vector is required.

3.6 The exact F_4 Coxeter action of order twelve

The order twelve in (NL14) is a lattice quotient invariant of this explicitly marked construction. There is also the ambient F_4 Coxeter action on the same original D_4 real root space. Use the precise F_4 simple roots

$$\beta_1 = e_2 - e_3, \quad \beta_2 = e_3 - e_4, \quad \beta_3 = e_4, \quad \beta_4 = (e_1 - e_2 - e_3 - e_4)/2, \qquad C = s_{\beta_1} s_{\beta_2} s_{\beta_3} s_{\beta_4},$$

where $s_\beta(x) = x - 2(x, \beta)\beta/\beta^2$ and the rightmost factor acts first. The first two roots have squared norm two, and the last two squared norm one. No rescaling of the D_4 roots has occurred.

Theorem 3.3 (Coxeter twelve and its marked triality quotient)**.** *In the retained four-coordinate metric,*

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad C^3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{NL19}$$

It satisfies $C^4 - C^2 + I = 0$, $C^6 = -I$ and has exact order twelve. Its marked discriminant action is $v_D + D \mapsto c_D + D \mapsto s_D + D \mapsto v_D + D$. The restriction of the original triality quotient is the exact sequence

$$1 \longrightarrow \langle C^3 \rangle \cong \mathbb{Z}/4 \longrightarrow \langle C \rangle \cong \mathbb{Z}/12 \longrightarrow \langle (v_D\, c_D\, s_D) \rangle \cong \mathbb{Z}/3 \longrightarrow 1, \tag{NL20}$$

with $\langle C^3 \rangle \subset W(D_4)$.

Proof. The first reflection exchanges coordinates two and three; the second exchanges coordinates three and four; the third changes the sign of coordinate four; the last is $I - 2\beta_4\beta_4^T$. Their product gives the first matrix in (NL19). Multiplying it three times gives the second, whose square is $-I$. Multiplying the first matrix also gives $C^4 - C^2 + I = 0$, and hence $C^{12} = I$. Its first column

gives $Cv_D = c_D + (1,-1,0,0)$, with the displayed correction in D. Directly $Cs_D = v_D$ and $Cc_D = (0,0,-1,-1) + s_D$. The displayed integer correction lies in D and proves the stated three-cycle with the original v_D, s_D, c_D markings. The first two reflections preserve D as root reflections. The last two preserve D because a sign change preserves integer sum parity, and reflection in β_4 sends an integer even-sum vector to an integer vector of even sum. The inverse reflections have the same property, so these maps are isometries of D. Consequently their induced action on $D^\vee/D$ is defined and has the exact order-three image calculated above. The matrix C^3 is an even signed permutation of exact order four, and thus lies in the already identified group $W(D_4)$. If $C^m = I$, its discriminant image forces $3 \mid m$, and the exact order of C^3 then forces $4 \mid m/3$. Thus C has exact order twelve and the discriminant kernel on its cyclic subgroup is exactly $\langle C^3 \rangle$. This proves every map and group order in (NL20). □

In particular C^2 still has nontrivial order-three discriminant action, while C^3 is in the long-root Weyl group. The Coxeter denominator six in (NL1), the cyclic order twelve in (NL14), and the F_4 Coxeter twelve in (NL20) therefore have their stated, distinct domains and explicit relationships through the original D_4 marking.

3.7 An exact cyclic-triality isometry of the entire Niemeier space

Retain the ordered fibres $f_i 289^j$, $1 \le i \le 4$, $0 \le j \le 5$, the Euclidean augmentation coordinates of $A(F_{\eta_i})$, the ordered orthonormal coordinates of $D \subset \mathbb{R}^4$, and the 72-element glue $\mathcal{H}$ of (8). Thus the actual real span is

$$V_N = N \otimes_{\mathbb{Z}} \mathbb{R} = \left\{ (x_{i,j}; d_k) \in \mathbb{R}^{24} \oplus \mathbb{R}^4 : \sum_{j=0}^{5} x_{i,j} = 0 \ (1 \le i \le 4) \right\},$$

with squared norm $\sum_{i,j} x_{i,j}^2 + \sum_k d_k^2$. The four displayed linear constraints, rather than an implicit change of metric, give its dimension 24. Choose an ordered orthonormal octonion basis $(u_0, \dots, u_7)$ with $u_0 = 1$. Every octonion product below is the product of the chosen real octonion algebra, without a change of multiplication convention.

For $p \in \{0,1\}$ and $j \in \{0,1,2\}$, set

$$m_{i,p} = \frac{1}{3} \sum_{j=0}^{2} x_{i,p+2j}, \qquad t_{i,p,j} = x_{i,p+2j} - m_{i,p}, \tag{37}$$

$$\alpha_i = \frac{\sum_{j=0}^{2} x_{i,2j} - \sum_{j=0}^{2} x_{i,1+2j}}{\sqrt{6}}, \qquad (b_0, \dots, b_7) = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, d_1, d_2, d_3, d_4). \tag{38}$$

In particular $m_{i,0} = \alpha_i/\sqrt{6}$, $m_{i,1} = -\alpha_i/\sqrt{6}$, and $\sum_j t_{i,p,j} = 0$. The ordered map is

$$\begin{aligned} &\mathcal{L} : V_N \longrightarrow \mathbb{O}^3, \\ &(x; d) \longmapsto (q_1, q_2, q_3), \\ &q_{j+1} = \sum_{i=1}^{4} \sum_{p=0}^{1} \left(t_{i,p,j} + \frac{b_{2(i-1)+p}}{\sqrt{3}} \right) u_{2(i-1)+p}. \end{aligned} \tag{39}$$

Theorem 3.4 (Actual rank-24 map, inverse, pairing, and residue action)**.** *The map $\mathcal{L}$ is a real linear isometry onto $\mathbb{O}^3$ with its product norm. Write $q_{j+1} = \sum_{a=0}^{7} q_{j,a} u_a$, and define*

$$b_a = \frac{1}{\sqrt{3}} \sum_{j=0}^{2} q_{j,a}, \qquad t_{i,p,j} = q_{j,2(i-1)+p} - \frac{1}{3} \sum_{h=0}^{2} q_{h,2(i-1)+p}. \tag{40}$$

Its inverse is exactly

$$x_{i,p+2j} = t_{i,p,j} + \frac{(-1)^p b_{i-1}}{\sqrt{6}}, \qquad d_k = b_{k+3}. \tag{41}$$

If $r = \rho(289^2) = \rho(361)$*, with the residue action of Theorem 2.16, and* $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$*, then*

$$\mathcal{L}r = c\mathcal{L}, \qquad r^3 = c^3 = I, \qquad rN = N. \tag{42}$$

Thus $\mathcal{L}|_N$ *embeds the original even unimodular lattice, with every pairing and glue class retained, into the actual three octonion sectors and intertwines an actual order-three lattice isometry with cyclic triality.*

Proof. The two means sum to zero because each six-coordinate fibre has total sum zero. The equations for $m_{i,p}$ follow from $\alpha_i = 6m_{i,0}/\sqrt{6}$. Orthogonality to constant triplets gives

$$\sum_{j=0}^{5} x_{i,j}^2 = \sum_{p=0}^{1}\sum_{j=0}^{2} t_{i,p,j}^2 + 3m_{i,0}^2 + 3m_{i,1}^2 = \sum_{p,j} t_{i,p,j}^2 + \alpha_i^2.$$

For each $a = 2(i-1) + p$, the same zero-sum identity gives

$$\sum_{j=0}^{2} \left(t_{i,p,j} + \frac{b_a}{\sqrt{3}} \right)^2 = \sum_{j=0}^{2} t_{i,p,j}^2 + b_a^2.$$

Summing over all eight values of a proves $\sum_{h=1}^{3} |q_h|^2 = \sum_{i,j} x_{i,j}^2 + \sum_k d_k^2$. Polarization proves equality of all pairings. Equations (40) recover the means and residuals of each output triplet. Equation (41) then recovers the original means, the original fibre coordinates, and d. Conversely those formulas applied to arbitrary q yield $\sum_j x_{i,j} = 0$ and return the prescribed q. This proves bijectivity without a dimension-only argument.

The residue action sends the coordinate vector $e_{f_i 289^h}$ to $e_{f_i 289^{h+2}}$, so the coefficient in output position $p + 2j$ is the old coefficient in position $p + 2(j-1)$, with j read modulo three. Both means, all α_i, and all d_k are fixed. Hence the three output octonions become (q_3, q_1, q_2), proving the intertwining formula. Each coordinate permutation acts trivially on $A^\vee/A$: for $\lambda = e_0 - \frac{1}{6}\sum_h e_h$, its difference from λ is an augmentation root. The D-coordinates are fixed. Thus r acts trivially on the full discriminant group, preserves $\mathcal{H}$, and preserves its exact inverse image N. Its order is three, as seen on $e_0 - e_1$ in any fibre. □

The exact octonion-coordinate glue test. This embedding has the following fully explicit image, denoted $\Lambda_{\mathbb{O}} = \mathcal{L}(N)$. Given q, first recover $(x; d)$ by (40)–(41). For each fibre require $x_{i,h} - x_{i,k} \in \mathbb{Z}$ for every h, k, and put

$$k_i = -6x_{i,0} \pmod 6 \in \mathbb{Z}/6\mathbb{Z}.$$

This is defined because $6x_{i,0} = \sum_h (x_{i,0} - x_{i,h}) \in \mathbb{Z}$. Require also that the four d_k are all integral or all half-integral. Let $y(d) \in \mathbb{F}_2^2$ be the unique class with

$$d + D = 0 + D,\ v + D,\ s + D,\ c + D \quad \text{marked respectively by } 00, 10, 01, 11.$$

Then the final and exact test is

$$(k_1, k_2, k_3, k_4; y(d)) = \big(3u + 2(a+b, a+2b, a, b); \tau(u)\big) \tag{43}$$

for $u \in \mathbb{F}_2^4$ of even weight, $(a, b) \in \mathbb{F}_3^2$, with the first four coordinates read modulo six, and $\tau(u) = (u_2 + u_4, u_3 + u_4)$. Indeed the integral-difference conditions are precisely membership in $A^\vee$, and the common fractional part equals $-k_i/6 \pmod{\mathbb{Z}}$, which is the fractional part of $k_i\lambda$. The half-integral test is precisely membership in $D^\vee$, and (43) is the original $\mathcal{H}$. This proves both directions of the image description. No new lattice or rescaled glue is substituted.

The determinant and the factor two in the Albert metric. The matrix convention, also specifying this construction independently of its host document, is

$$Q_{\mathbb{O}}(\lambda; q) = \begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix} \in \mathfrak{h}_3(\mathbb{O}), \qquad X \circ Y = (XY + YX)/2.$$

Put

$$E(q) = Q_{\mathbb{O}}(0; q_1, q_2, q_3), \qquad \mathfrak{t}(q_1, q_2, q_3) = \operatorname{Re}((q_1 q_2) q_3), \qquad \mathfrak{t}_N(X) = \mathfrak{t}(\mathcal{L}X).$$

The original ordered Albert determinant becomes the exact polynomial

$$\det_{\mathbb{O}}\big(\lambda I + E(\mathcal{L}X)\big) = \lambda^3 - \lambda(X, X) + 2\mathfrak{t}_N(X). \tag{44}$$

Moreover

$$\operatorname{tr}\big(E(\mathcal{L}X) \circ E(\mathcal{L}Y)\big) = 2(X, Y). \tag{45}$$

For completeness, direct binary multiplication gives $\operatorname{tr} E = 0$, $\operatorname{tr}(E \circ E) = 2\sum_i |q_i|^2$, and $\operatorname{tr}(E \circ (E \circ E)) = 6\mathfrak{t}(q)$. For the last equality, the three cyclic off-diagonal coordinates of $E \circ E$ are $\overline{q_2 q_3}, \overline{q_3 q_1}, \overline{q_1 q_2}$; their trace pairings with the corresponding q_i each give $2\mathfrak{t}(q)$. The Jordan trace formula $\det X = \operatorname{tr}(X \circ (X \circ X))/3 - \operatorname{tr} X \operatorname{tr}(X \circ X)/2 + (\operatorname{tr} X)^3/6$ therefore gives the first identity after expanding $X = \lambda I + E$ and substituting the proved norm equality. For the second, each off-diagonal octonion occurs in both transposed matrix entries, giving exactly $2\sum_i \operatorname{Re}(q_i \overline{p_i})$. Real associativity and cyclicity of octonion real parts give $\mathfrak{t}(q_3, q_1, q_2) = \mathfrak{t}(q_1, q_2, q_3)$:

$$\operatorname{Re}((q_3 q_1) q_2) = \operatorname{Re}(q_3 (q_1 q_2)) = \operatorname{Re}((q_1 q_2) q_3).$$

Consequently $\mathfrak{t}_N(rX) = \mathfrak{t}_N(X)$, and the whole determinant in (44) is r-invariant with λ fixed. Equivalently, for the real permutation matrix $Pe_i = e_{i+1}$, indices modulo three,

$$P\big(\lambda I + E(q)\big)P^T = \lambda I + E(q_3, q_1, q_2).$$

This is also a Jordan automorphism: multiplication by a real permutation matrix merely reindexes the two-factor matrix products, so it preserves $(AB + BA)/2$. It therefore carries the positive cone and its determinant boundary to themselves. The cyclic bridge is simultaneously linear, isometric on $\mathbb{O}^3$, integral on the stated lattice image, and compatible with the actual Albert cubic and positive cone.

Dependence and the four-dimensional lattice summand. The map uses precisely the four specified origins f_i, the generator 289, the even/odd triplet order, the displayed order of b, the ordered D-coordinates, and the chosen orthonormal octonion basis. Changing one of these changes the displayed map; the construction makes no claim of independence from these data. Its restriction to the actual D-summand is

$$d \longmapsto \frac{1}{\sqrt{3}}(z(d), z(d), z(d)), \qquad z(d) = d_1 u_4 + d_2 u_5 + d_3 u_6 + d_4 u_7. \tag{46}$$

This is an isometry on D, and on $D^\vee$ in its real span. In particular its 24 shortest nonzero dual vectors become 24 specified points in this four-dimensional diagonal subspace. The separate Cartan-weight map $d \mapsto \sum_k d_k \varepsilon_k$ is related to it by the exact isometry $\sum_k d_k \varepsilon_k \mapsto (z(d), z(d), z(d))/\sqrt{3}$.

3.8 The Leech neighbour in the same three octonion coordinates

The preceding map also transports the entire neighbour construction, not only the real span of N. Put $N_{\mathbb{O}} = \mathcal{L}(N)$ and retain the actual ρ, α, K, Λ of Theorem 3.1. In the already fixed basis let

$$U = \sum_{a=0}^{7} u_a, \qquad V = \frac{u_0 + u_1 + u_2 + u_3}{\sqrt{2}} + \sqrt{3}u_4 + \frac{2u_5 + u_6}{\sqrt{3}}.$$

Substitution in the original isometry gives the explicit vectors

$$\begin{aligned} \varrho &= \mathcal{L}\rho = (V + 2U, V, V - 2U), \\ \delta &= \mathcal{L}\alpha = \left(\frac{(2+\sqrt{2})u_0 - 2u_1}{3}, \frac{(\sqrt{2}-1)u_0 + u_1}{3}, \frac{(\sqrt{2}-1)u_0 + u_1}{3} \right). \end{aligned} \tag{47}$$

Indeed each even triplet of ρ_A has mean $1/2$, each odd triplet has mean $-1/2$, and both residual triplets are $(2, 0, -2)$. The four mean-difference coordinates are $\sqrt{6}/2$, while the four D coordinates are $(3, 2, 1, 0)$. For α, the two residual triplets in the first fibre are $(2, -1, -1)/3$ and $(-2, 1, 1)/3$; its only nonzero mean-difference coordinate is $2/\sqrt{6}$. These facts prove every coefficient in (47). In particular $|\varrho|^2 = 84$, $|\delta|^2 = 2$, and $\langle \varrho, \delta \rangle = 1$ in the product octonion metric.

Define

$$\begin{aligned} K_{\mathbb{O}} &= \{ q \in N_{\mathbb{O}} : \langle \varrho, q \rangle \equiv 0 \pmod 6 \}, \\ \Lambda_{\mathrm{Leech},\mathbb{O}} &= K_{\mathbb{O}} + \mathbb{Z}(\varrho/6 - \delta). \end{aligned} \tag{48}$$

The isometry and its inverse prove, in both directions, $\mathcal{L}(K) = K_{\mathbb{O}}$ and $\mathcal{L}(\Lambda) = \Lambda_{\mathrm{Leech},\mathbb{O}}$. Thus (48) is a full coordinate description of the Leech lattice inside the very same three sectors. The glue test above, followed by its displayed height congruence, is an exact membership test for $K_{\mathbb{O}}$; adjoining the six cosets $k(\varrho/6 - \delta)$, $0 \le k < 6$, gives all of the Leech lattice. The common-sublattice indices, discriminant form, and order-twelve quotient are transported without alteration.

The cyclic action also acts on the marked family of neighbours. Retain $r = \rho(361)$ and its corresponding map $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$. For any transported Weyl vector $\rho' = T\rho$, $T \in O(N)$, the neighbour is defined using $T\alpha$, so that $\langle \rho', T\alpha \rangle = 1$. Then

$$c\,\mathcal{L}(K_{\rho'}) = \mathcal{L}(K_{r\rho'}), \qquad c\,\mathcal{L}(\Lambda_{\rho'}) = \mathcal{L}(\Lambda_{r\rho'}).$$

This follows directly from $\mathcal{L}r = c\mathcal{L}$ and (NL18). The inverse maps use r^{-1} and c^{-1}. It specifies exactly which marked neighbour is the target; it does not silently assume that the chosen ρ is fixed by r. Finally the common cubic evaluates on every one of these lattice points as

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X, X) + 2\,\mathrm{Re}(((\mathcal{L}X)_1(\mathcal{L}X)_2)(\mathcal{L}X)_3).$$

This identity holds for all $X \in N_{\mathbb{R}}$, hence also on the explicitly embedded N, K and Λ.

3.9 Exactly which root isometries preserve this glue

Retain the coordinate convention $(P_p x)_i = x_{p(i)}$. An arbitrary root-lattice isometry has the original form

$$T(X_1, X_2, X_3, X_4; d) = \big((\varepsilon_i Q_{\sigma_i} X_{p(i)})_{i=1}^4;\ T_D d \big),$$

where $\varepsilon_i \in \{\pm 1\}$, $\sigma_i \in S_6$, $p \in S_4$, and $T_D \in O(D)$. If $B = d_D(T_D)$, the necessary and sufficient condition for $T(N) = N$ is

$$B\tau(u) = \tau(P_p u)\ (u \in E), \qquad (\varepsilon_i \ell_{p(i)})_{i=1}^4 = (\ell_i A)_{i=1}^4 \quad \text{for some } A \in GL_2(\mathbb{F}_3), \tag{49}$$

where $\ell_1 = (1,1), \ell_2 = (1,2), \ell_3 = (1,0), \ell_4 = (0,1)$. The A, and hence $B = B_p$, are unique. Indeed the two-primary part of the glue is the graph of τ, and signs are trivial modulo two. Its three-primary part is the image of the injective map $j(a,b) = (a+b, a+2b, a, b)$, so preserving it is equivalent to $M_\varepsilon P_p j = jA$. These are exactly the two conditions in (49); the primary decomposition proves their sufficiency as well as necessity. Since the four ℓ_i represent the four distinct projective row lines over $\mathbb{F}_3$, each A gives exactly one signed permutation by this equation. This recovers the precise 48-element glue quotient, not merely its action on three labels.

The canonical full-automorphism theorem proves, with these same coordinates and section, that

$$O(N) = (S_6^4 \times W(D_4)) \rtimes L(G), \qquad G \cong GL_2(\mathbb{F}_3).$$

Conjugation by the actual map $\mathcal{L}$ now gives the entire orthogonal stabilizer of the embedded lattice, with explicit matrices obtained from (39) and its inverse:

$$O(\Lambda_{\mathbb{O}}) = \mathcal{L}O(N)\mathcal{L}^{-1}, \qquad \Gamma_G(g) = \mathcal{L}L_g\mathcal{L}^{-1}, \qquad \mathcal{L}L_g = \Gamma_G(g)\mathcal{L}. \tag{50}$$

Both inclusions in the first equality follow by conjugating an isometry preserving the specified lattice. This is a faithful action on the whole 24-dimensional space. Each $\Gamma_G(g)$ commutes with c, because the canonical equation $L_g\rho(289^2) = \rho(289^2)L_g$ conjugates to that equation.

For completeness, the previously displayed D_4-triality section and geometric frame permutations have a precise comparison even though that particular proposed linear conjugacy fails. On V_N, each of $\widehat{R}, \widehat{H}$ has trace $5 - 5 + 2 = 2$: the transposed A_5 pair contributes zero, the other two components contribute $5, -5$, and the D_4 reflection contributes 2. Their three-cycle has trace $5 + 1 = 6$. A geometric Albert-frame transposition acts on $\mathbb{O}^3$ by swapping two sectors with conjugation and by conjugating the fixed sector; its trace is $1 - 7 = -6$. Its three-cycle has trace zero. Therefore an invertible map cannot conjugate these two particular S_3 representations, since trace is invariant under conjugation. The complete real character decompositions are, respectively,

$$V_N|_{\langle \widehat{R}, \widehat{H} \rangle} \cong \mathbf{71} \oplus 5\,\mathrm{sgn} \oplus 6\,\mathrm{Std}, \qquad \mathbb{O}^3|_{S_3^{\mathrm{frame}}} \cong \mathbf{1} \oplus 7\,\mathrm{sgn} \oplus 8\,\mathrm{Std}\,.$$

These multiplicities follow by taking inner products with the characters $(1,1,1), (1,-1,1), (2,0,-1)$, on classes of sizes $1, 3, 2$. The exact whole-space relation for the original section is (50); the exact determinant-compatible cyclic relation is (42). In the latter relation the residue element has trivial discriminant action while its octonion action cycles the sectors. Those are two explicitly calculated actions of the same operator on two explicitly different constructions; no identification between their quotient homomorphisms is assumed.

Finally, the connected group Spin(8) acts on the ambient triality module, and its subgroup preserving $\Lambda_{\mathbb{O}}$ is exactly its intersection with (50). It is finite: the columns of a lattice isometry in a fixed lattice basis have prescribed bounded lengths, so each column has finitely many possible lattice values. A connected group preserving a discrete full lattice pointwise as a set acts trivially, since each orbit of a lattice vector is connected and discrete and the lattice spans. The nontrivial connected triality action therefore does not preserve the whole lattice. The proved cyclic subgroup $\langle c \rangle$ does preserve both the lattice and its Albert cubic. Equations (49) and (50) specify the full glue stabilizer; they do not assert that every element of that finite orthogonal group also preserves the cubic.

4 The arithmetic of the retained Albert cubic on both lattices

4.1 Original data and the exact polynomial

Retain the ordered coordinates $X = (x_{i,j}; d_k)$, $1 \le i \le 4$, $0 \le j \le 5$, $1 \le k \le 4$, with $\sum_{j=0}^{5} x_{i,j} = 0$, and their original Euclidean metric. Let

$$A_5 = \{x \in \mathbb{Z}^6 : \sum_j x_j = 0\}, \qquad D_4 = \{d \in \mathbb{Z}^4 : \sum_k d_k \equiv 0 \pmod 2\}, \qquad R = A_5^{\perp 4} \perp D_4.$$

Write $\lambda = (5, -1, -1, -1, -1, -1)/6$, $v_D = (1, 0, 0, 0)$, and $s_D = (1, 1, 1, 1)/2$. The original discriminant glue is generated by

$$\begin{aligned} &(3, 3, 3, 3; 00), \quad (3, 3, 0, 0; 10), \quad (3, 0, 3, 0; 01), \\ &(2, 2, 2, 0; 00), \quad (2, 4, 0, 2; 00) \end{aligned}$$

in $(\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2$, with the same marking $10 = v_D + D_4$, $01 = s_D + D_4$. Its order is 72: the first three generators have independent two-primary reductions, and the last two independent three-primary reductions. The lattice N is its inverse image in $R^\vee$.

Retain

$$\rho_A = (5, 3, 1, -1, -3, -5)/2, \qquad \rho_D = (3, 2, 1, 0), \qquad \rho = \rho_A^{\oplus 4} \perp \rho_D, \qquad \alpha = (e_0 - e_1, 0, 0, 0; 0),$$

and the exact neighbour

$$K = \{X \in N : (\rho, X) \equiv 0 \pmod 6\}, \qquad v = \rho/6 - \alpha, \qquad \Lambda = K + \mathbb{Z}v.$$

Here $\rho \in N$, $\rho^2 = 84$, and $(\rho, \alpha) = 1$, as follows immediately from these coordinates and the first glue generator. The name Λ always denotes this neighbour.

Fix an ordered orthonormal octonion basis $u_0 = 1, u_1, \ldots, u_7$. For the first part of the calculation no multiplication table is imposed beyond the actual multiplication of these basis elements. Define, in the original coordinates,

$$s_i = \sum_{j=0}^{5} (-1)^j x_{i,j}, \tag{51}$$

$$A_{2(i-1)+p} = x_{i,p} - \frac{x_{i,p} + x_{i,p+2} + x_{i,p+4}}{3}, \qquad p = 0, 1, \tag{52}$$

$$B_{2(i-1)+p} = x_{i,p+2} - \frac{x_{i,p} + x_{i,p+2} + x_{i,p+4}}{3}, \qquad p = 0, 1. \tag{53}$$

Write $A = A_0 + \mathbf{a}$, $B = B_0 + \mathbf{b}$, where $\mathbf{a} = \sum_{h=1}^{7} A_h u_h$ and $\mathbf{b} = \sum_{h=1}^{7} B_h u_h$. The common mean octonion is

$$M = \frac{\sqrt{2}}{6} \sum_{a=0}^{3} s_{a+1} u_a + \frac{\sqrt{3}}{3} \sum_{k=1}^{4} d_k u_{k+3}. \tag{54}$$

The retained isometry therefore has precisely the ordered form

$$\mathcal{L}X = (A + M, B + M, -A - B + M).$$

This is an identity with its original three triplets: the residuals sum to zero, the first four common mean coefficients are $s_i/\sqrt{18} = \sqrt{2}s_i/6$, and the last four are $d_k/\sqrt{3} = \sqrt{3}d_k/3$. For completeness the original squared norm and the output norm are both

$$|A|^2 + |B|^2 + |A + B|^2 + \frac{1}{6}\sum_i s_i^2 + \sum_k d_k^2.$$

In the original fibres this follows by separating the means $s_i/6, -s_i/6$ from their zero-sum triplets; in the output it follows from $A + B + (-A - B) = 0$ and $3|M|^2 = \sum_i s_i^2/6 + \sum_k d_k^2$. The inverse recovers $M = (q_1 + q_2 + q_3)/3$, then $s_i = 3\sqrt{2}M_{i-1}$, $d_k = \sqrt{3}M_{k+3}$, and $A = q_1 - M, B = q_2 - M$. Adding $(-1)^p s_i/6$ to the p-th residual triplet in the i-th fibre recovers each original $x_{i,p+2j}$. Thus the map is onto and preserves the original pairing by polarization.

For imaginary octonions put

$$\mathbf{a} \times \mathbf{b} = \frac{\mathbf{ab} - \mathbf{ba}}{2}, \qquad \phi(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \langle \mathbf{a} \times \mathbf{b}, \mathbf{c} \rangle.$$

Let $\phi_{ijk} = \phi(u_i, u_j, u_k)$. Thus ϕ is the actual alternating octonion three-form in the chosen frame, including all signs. Define

$$D = A_0^2 + A_0 B_0 + B_0^2 - \langle \mathbf{a}, \mathbf{a} \rangle - \langle \mathbf{a}, \mathbf{b} \rangle - \langle \mathbf{b}, \mathbf{b} \rangle, \tag{55}$$

$$Z = (2A_0 + B_0)\mathbf{a} + (A_0 + 2B_0)\mathbf{b} - 3\mathbf{a} \times \mathbf{b} = \sum_{h=1}^{7} Z_h u_h. \tag{56}$$

Equivalently, the component in the final formula is the exact quadratic polynomial

$$Z_h = (2A_0 + B_0)A_h + (A_0 + 2B_0)B_h - 3 \sum_{i,j=1}^{7} \phi_{ijh} A_i B_j.$$

Theorem 4.1 (Exact pullback with every radical and coefficient)**.** *The original cubic* $F(X) = \mathrm{Re}(((\mathcal{L}X)_1(\mathcal{L}X)_2)(\mathcal{L}X)_3)$ *has the complete polynomial expression*

$$F(X) = P_0(X) + \sqrt{2}P_2(X) + \sqrt{3}P_3(X), \tag{57}$$

where every variable is the displayed linear function of the original $x_{i,j}, d_k$*, and*

$$P_0 = -A_0 B_0(A_0 + B_0) + B_0\langle \mathbf{a}, \mathbf{a} \rangle + A_0\langle \mathbf{b}, \mathbf{b} \rangle + 2(A_0 + B_0)\langle \mathbf{a}, \mathbf{b} \rangle, \tag{58}$$

$$P_2 = \frac{-s_1 D + s_2 Z_1 + s_3 Z_2 + s_4 Z_3}{6} + \frac{s_1(4s_1^2 - 3\sum_{i=1}^{4} s_i^2)}{108} - \frac{s_1 \sum_{k=1}^{4} d_k^2}{6}, \tag{59}$$

$$P_3 = \frac{d_1 Z_4 + d_2 Z_5 + d_3 Z_6 + d_4 Z_7}{3}. \tag{60}$$

These identities hold throughout the original real vector space, in particular on N*,* K*, and* Λ*. They retain the ordered Albert determinant as*

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X, X) + 2F(X).$$

Proof. For $a = a_0 + \mathbf{a}$, $b = b_0 + \mathbf{b}$, $c = c_0 + \mathbf{c}$, binary octonion multiplication gives

$$\mathrm{Re}((ab)c) = a_0 b_0 c_0 - a_0\langle \mathbf{b}, \mathbf{c} \rangle - b_0\langle \mathbf{a}, \mathbf{c} \rangle - c_0\langle \mathbf{a}, \mathbf{b} \rangle - \phi(\mathbf{a}, \mathbf{b}, \mathbf{c}).$$

Call the first four, symmetric, terms $S(a, b, c)$. Substitute $(a, b, c) = (A + M, B + M, -A - B + M)$. Terms with exactly two copies of M in S sum to $S(M, M, A + B - A - B) = 0$. Terms with exactly one copy sum to $-S(M, A, A) - S(M, A, B) - S(M, B, B)$. The term with no copy is $S(A, B, -A - B) = P_0$, by multiplying its four displayed terms. The alternating term with no copy vanishes; terms with two or three copies vanish. Its three terms with one copy all equal $\phi(\mathbf{a}, \mathbf{b}, \mathrm{Im}\, M)$, since

$$\phi(\mathbf{a}, \mathrm{Im}\, M, -\mathbf{a} - \mathbf{b}) = \phi(\mathbf{a}, \mathbf{b}, \mathrm{Im}\, M), \quad \phi(\mathrm{Im}\, M, \mathbf{b}, -\mathbf{a} - \mathbf{b}) = \phi(\mathbf{a}, \mathbf{b}, \mathrm{Im}\, M).$$

Thus all one-copy terms together are $-M_0 D + \langle \operatorname{Im} M, Z\rangle$. Finally the term with three copies is

$$\operatorname{Re}(M^3) = 4M_0^3 - 3M_0|M|^2, \qquad |M|^2 = \frac{\sum_i s_i^2}{18} + \frac{\sum_k d_k^2}{3}, \qquad M_0 = \frac{\sqrt{2}s_1}{6}.$$

Substitution proves (58)–(60) with their displayed factors. In particular it proves the absence of a $\sqrt{6}$ term without discarding any mixed term. The determinant formula uses the original isometry and its unchanged squared norm. □

4.2 Finite generators of both original lattices

Order the 24 original simple roots as

$$g_{5(i-1)+j+1} = a_{i,j} = e_{i,j} - e_{i,j+1} \quad (1 \le i \le 4,\ 0 \le j \le 4),$$

followed by

$$g_{21} = e_1 - e_2, \quad g_{22} = e_2 - e_3, \quad g_{23} = e_3 - e_4, \quad g_{24} = e_3 + e_4$$

in the original D_4 coordinates. All unspecified components are zero. Adjoin the five actual glue lifts

$$\begin{aligned} g_{25} &= (3\lambda, 3\lambda, 3\lambda, 3\lambda; 0), & g_{26} &= (3\lambda, 3\lambda, 0, 0; v_D),\\ g_{27} &= (3\lambda, 0, 3\lambda, 0; s_D), & g_{28} &= (2\lambda, 2\lambda, 2\lambda, 0; 0),\\ g_{29} &= (2\lambda, 4\lambda, 0, 2\lambda; 0). \end{aligned}$$

Then

$$N = \sum_{i=1}^{29} \mathbb{Z} g_i, \qquad H_i = (\rho, g_i) = \begin{cases} 1, & 1 \le i \le 24,\\ 30, 18, 18, 15, 20, & i = 25, 26, 27, 28, 29 \text{ respectively.} \end{cases} \tag{61}$$

Indeed the first 24 vectors span R, the last five generate the original glue, and subtraction of the corresponding glue lift leaves every element of N in R. The height calculation uses $(\rho_A, \lambda) = 5/2$, $(\rho_D, v_D) = (\rho_D, s_D) = 3$, and the simple-root heights one.

Define the following 30, fully specified neighbour generators:

$$h_1 = 6\alpha, \qquad h_i = g_i - H_i\alpha \quad (2 \le i \le 29), \qquad h_{30} = v = \rho/6 - \alpha. \tag{62}$$

Then $K = \sum_{i=1}^{29} \mathbb{Z} h_i$ and $\Lambda = \sum_{i=1}^{30} \mathbb{Z} h_i$. For if $X = \sum_i n_i g_i \in K$, then

$$X = \sum_{i=2}^{29} n_i (g_i - H_i\alpha) + \left(\sum_{i=1}^{29} n_i H_i \right) \alpha,$$

and the last coefficient is divisible by six. Conversely every displayed generator before h_{30} has height divisible by six. Adjoining h_{30} is exactly the original neighbour definition.

4.3 A complete finite calculation of the cubic value modules

For a lattice L and a homogeneous real cubic F, define its additive value module by

$$\mathcal{V}_F(L) = \sum_{X \in L} \mathbb{Z} F(X) \subset \mathbb{R}.$$

This definition retains every value and takes its generated additive group. It does not claim that every element of the group is one individual value of F.

Theorem 4.2 (Finite cubic-value generators, including negative integers)**.** *If $L = \sum_{i=1}^{m} \mathbb{Z} b_i$, allowing dependent generators, then $\mathcal{V}_F(L)$ is generated by the following finite list:*

$$A_i = F(b_i), \qquad 1 \le i \le m, \tag{63}$$

$$D_{ij}^{+} = F(b_i + b_j) - F(b_i) - F(b_j), \qquad i < j, \tag{64}$$

$$D_{ij}^{-} = F(b_i - b_j) - F(b_i) + F(b_j), \qquad i < j, \tag{65}$$

$$\begin{aligned} D_{ijk} = {} & F(b_i + b_j + b_k) - F(b_i + b_j) - F(b_i + b_k) \\ & - F(b_j + b_k) + F(b_i) + F(b_j) + F(b_k), \qquad i < j < k. \end{aligned} \tag{66}$$

Consequently (61) *gives exactly 4495 specified generators of $\mathcal{V}_F(N)$, and* (62) *gives exactly 4960 specified generators of $\mathcal{V}_F(\Lambda)$, for every fixed orthonormal octonion frame and its actual ϕ_{ijk}.*

Proof. Let B_F be the symmetric trilinear polarization, with $B_F(X, X, X) = F(X)$; explicitly its value is $1/6$ times the seven-term expression in (66), with its three inputs replacing the three generators. For every integer tuple n, expansion gives

$$F\left(\sum_i n_i b_i\right) = \sum_i n_i^3 A_i + \sum_{i<j} (n_i^2 n_j C_{ij} + n_i n_j^2 E_{ij}) + \sum_{i<j<k} n_i n_j n_k D_{ijk},$$

where $C_{ij} = 3B_F(b_i, b_i, b_j)$ and $E_{ij} = 3B_F(b_i, b_j, b_j)$. The two pair generators are $D_{ij}^{+} = C_{ij} + E_{ij}$ and $D_{ij}^{-} = -C_{ij} + E_{ij}$. Therefore their contribution is

$$\frac{n_i n_j (n_i + n_j)}{2} D_{ij}^{+} + \frac{n_i n_j (n_j - n_i)}{2} D_{ij}^{-}.$$

Both coefficients are integers, for all positive, zero, and negative integers n_i, n_j: if either integer is even the claim is immediate, and if both are odd their sum and difference are even. This proves that the finite list generates all values. Conversely each entry of the list is an integer linear combination of actual values on L. The proof does not use linear independence of the generators. Finally $29 + 2\binom{29}{2} + \binom{29}{3} = 4495$ and $30 + 2\binom{30}{2} + \binom{30}{3} = 4960$. □

For numerical arithmetic now specify the additional octonion frame choice completely. Use $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$ in ordered coordinates (a, b), with

$$\begin{gathered} (a, b)(c, d) = (ac - \overline{d}\, b,\ da + b\overline{c}), \\ u_0 = (1, 0), \quad u_1 = (i, 0), \quad u_2 = (j, 0), \quad u_3 = (k, 0), \\ u_4 = (0, 1), \quad u_5 = (0, i), \quad u_6 = (0, j), \quad u_7 = (0, k). \end{gathered} \tag{67}$$

Here $ij = k$, $jk = i$, $ki = j$, and the reversed products are negative. The positive oriented three-form triples are exactly

$$(123),\ (145),\ (176),\ (246),\ (257),\ (347),\ (365). \tag{68}$$

Every ϕ_{ijk} is obtained from these by alternation, or is zero. This is a specified frame for the retained, previously frame-dependent isometry, without a change to any lattice coordinates or to the Albert matrix convention.

Theorem 4.3 (Exact value modules and their quotient of order twelve)**.** *For the frame* (67)*, the exact additive modules are*

$$\mathcal{V}_F(N) = \frac{1}{27}\mathbb{Z} \ \oplus\ \frac{\sqrt{2}}{54}\mathbb{Z} \ \oplus\ \frac{\sqrt{3}}{18}\mathbb{Z}, \tag{69}$$

$$\mathcal{V}_F(\Lambda) = \frac{1}{27}\mathbb{Z} \ \oplus\ \frac{\sqrt{2}}{216}\mathbb{Z} \ \oplus\ \frac{\sqrt{3}}{54}\mathbb{Z}. \tag{70}$$

In particular the inclusion of these two additive subgroups of $\mathbb{R}$ *has the exact quotient*

$$\mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/3 \cong \mathbb{Z}/12. \tag{71}$$

This is an inclusion of cubic value modules; it does not assert the false lattice inclusion $N \subset \Lambda$.

Proof. The polynomial coefficients in (58)–(60) are rational on both lattices for this exact multiplication table. The numbers $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$. For example, isolating one radical in a rational relation and squaring either gives a rational multiple of another irrational radical, or a rational square equal to two, three, or 3/2, which is impossible by parity of prime exponents. Thus a value has a unique rational coefficient triple (P_0, P_2, P_3).

Apply the complete finite list in Theorem 4.2 to the displayed original generators. Exact substitution gives the following least common multiples of coefficient denominators. The denominators in each entry are ordered as (P_0, P_2, P_3), and a zero coefficient has denominator one:

	A_i	D^+_{ij}	D^-_{ij}	D_{ijk}
N	$(27, 27, 1)$	$(9, 18, 18)$	$(9, 18, 18)$	$(9, 18, 18)$
Λ	$(27, 216, 27)$	$(27, 108, 54)$	$(27, 108, 54)$	$(9, 36, 54)$

. (72)

This table is a finite rational calculation, with no bound on lattice points or heuristic sampling. The complete arithmetic certificate is the accompanying `verify_cubic.py`: its `residual_data` and `cubic` functions are exactly (51)–(60); it forms every $A_i, D^+_{ij}, D^-_{ij}, D_{ijk}$ for the 29 or 30 explicitly defined vectors, using rational arithmetic. The `cross` function executes the three cyclic positive and three negative entries for each of the seven triples (68). Hence the full finite calculation is specified by these equations, the finite index bounds in Theorem 4.2, and integer arithmetic. Its 4495 and 4960 entries verify the upper inclusions in (69)–(70).

Here are short certificates for the reverse inclusions. Coordinates in the next two tables are relative to the respective three proposed generators on the right sides of (69) and (70). In the first table all pair symbols use the g_i:

$F(g_1)$	-4	4	0
$F(g_2)$	-5	-4	0
$F(g_7)$	0	18	0
$F(g_{28})$	-80	52	0
$D^+_{21,28}$	0	-36	16
$D^+_{2,27}$	-27	27	3
$D^+_{2,25}$	-81	45	0

. (73)

The determinants of the three columns obtained by transposing rows $(1, 4, 5)$, $(2, 3, 6)$, and $(2, 7, 6)$ are, respectively, $1792, -270, -1647$. Their greatest common divisor is one. In the second table all pair symbols use the h_i:

$F(h_2)$	14	-272	0
$F(h_4)$	5	-128	0
$F(h_{28})$	13780	-74672	0
$F(h_{30})$	14	-63	-2
$D^+_{2,27}$	7425	-60624	-9

. (74)

Rows $(1, 3, 4)$, $(1, 4, 5)$, and $(2, 4, 5)$ give determinants $-5405504, 2315394, 1281267$, whose greatest common divisor is also one.

For clarity, the elementary integer argument using these determinants does not require a normal-form theorem. Let $W \subset \mathbb{Z}^3$ be the group generated by one displayed table. For any three of its

columns forming a matrix C, the adjugate identity $C\,\mathrm{adj}(C) = \det(C)I$ implies $\det(C)\mathbb{Z}^3 \subset W$. Bezout's identity for the three displayed determinants, whose gcd is one, therefore gives $\mathbb{Z}^3 \subset W$. The opposite inclusion is immediate from the integer entries. Every table entry is an integer combination of actual values by its definition, proving the two reverse inclusions. Finally the coordinates of the inclusion of the first module in the second are $(a, b, c) \mapsto (a, 4b, 3c)$. Its cokernel is the displayed product of cyclic groups, with order twelve. This also proves the asserted exact quotient map. □

4.4 Frame dependence and exact symmetry transport

The finite-list theorem is valid for the actual coefficients of every orthonormal frame. The numerical modules (69)–(70) refer to the frame (67). This dependence has an exact comparison map. If $u'_a = Tu_a$ for a real isometry T fixing 1, then

$$\mathcal{L}_{u'} = (T, T, T)\mathcal{L}_u.$$

The symmetric part S of the trilinear tensor is unchanged, because T preserves the scalar coordinate and inner product. Writing $\mathbf{m} = \mathrm{Im}\, M$, its exact change is

$$F_{u'}(X) - F_u(X) = -3\big(\phi(T\mathbf{a}, T\mathbf{b}, T\mathbf{m}) - \phi(\mathbf{a}, \mathbf{b}, \mathbf{m})\big). \tag{75}$$

This follows from the complete expansion in the polynomial proof. If T is an octonion automorphism it preserves ϕ, so the cubic and its value modules are unchanged.

The dependence on a general orthonormal frame is substantive. Take the actual lattice vector with the only nonzero fibre coordinates

$$x_{1,1} = 1,\ x_{1,5} = -1, \quad x_{2,2} = 1,\ x_{2,4} = -1, \qquad d = (1, 1, 0, 0).$$

It lies in $R \subset N$, has $A = u_1$, $B = u_2$, and $M = (u_4 + u_5)/\sqrt{3}$. In the Cayley frame its value is zero. Replace only

$$u'_3 = (2u_3 - \sqrt{5}u_4)/3, \qquad u'_4 = (\sqrt{5}u_3 + 2u_4)/3.$$

This is an exact orthonormal frame change; the same original lattice vector now has value $-\sqrt{15}/3$, because $\phi(u_1, u_2, u'_4) = \sqrt{5}/3$ and $\phi(u_1, u_2, u_5) = 0$. This value is outside $\mathbb{Q} + \sqrt{2}\mathbb{Q} + \sqrt{3}\mathbb{Q}$. Indeed the independent sign changes of $\sqrt{2}, \sqrt{3}$ are automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. If an element z in that field had $z^2 = 15$, each sign change would send it to z or $-z$. In the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ this makes z a rational multiple of a single basis vector. Its square would force one of $15, 15/2, 5, 5/2$ to be a rational square, impossible by parity of its prime exponents. Thus the numerical specialization is not silently extended to an unspecified orthonormal frame.

Return to the explicit Cayley frame for the following numerical witnesses. Let $\sigma = \rho(289)$ act simultaneously by $e_{i,j} \mapsto e_{i,j+1 \bmod 6}$, and trivially on D_4. The exact isometry gives $\mathcal{L}\sigma^2 = c\mathcal{L}$ for $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$. Cyclic real associativity therefore proves $F(\sigma^{2k}X) = F(X)$ for every real X. These maps preserve N by their original discriminant action. The stabilizer of this cubic within the six-element residue group is exactly its three-element subgroup: direct substitution at $\alpha = g_1$ gives

$$F(\alpha) = \frac{-4 + 2\sqrt{2}}{27}, \qquad F(\sigma\alpha) = \frac{-5 - 2\sqrt{2}}{27}, \tag{76}$$

which differ. All odd powers have the same effect on the cubic as σ, by its proved invariance under even powers.

The original reflection exchanging $e_{1,0}, e_{1,1}$ sends α to $-\alpha$, hence changes its nonzero cubic value to its negative. The original full glue-preserving lifts

$$\widehat{R}(X_1, X_2, X_3, X_4; d) = (X_2, X_1, X_3, -X_4; R_D d),$$
$$\widehat{H}(X_1, X_2, X_3, X_4; d) = (X_1, X_3, X_2, -X_4; H_D d).$$

also have explicit witnesses:

$$F(\widehat{R}\alpha) = 0, \qquad F(g_7) = \frac{\sqrt{2}}{3}, \qquad F(\widehat{H}g_7) = 0.$$

Their original matrices remain

$$R_D = \mathrm{diag}(-1, 1, 1, 1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

These maps preserve the original lattice, as can be checked on its displayed generators. Both R_D and H_D preserve D_4: the former changes one sign, while the latter has integral coordinates on every even-sum integer vector and has output sum twice its first input coordinate. They are orthogonal involutions. Their marked discriminant actions fix v_D and exchange s_D, c_D, and exchange v_D, s_D while fixing c_D, respectively. On the binary glue generators, $\widehat{R}$ fixes the first two and sends the third to their second-plus-third sum; $\widehat{H}$ fixes the first and exchanges the second and third. On the ternary code generators $z_1 = (1, 1, 1, 0)$, $z_2 = (1, 2, 0, 1)$, the two signed component permutations act as $(z_1, z_2) \mapsto (z_1, 2z_2)$ and $(z_1, z_2) \mapsto (z_1, 2z_1 + 2z_2)$, respectively. Thus both preserve the full original glue and its inverse image. Every within-fibre coordinate permutation also preserves that inverse image, since its action on $\lambda + A_5$ is trivial: $Q\lambda - \lambda = Qe_0 - e_0 \in A_5$. These are concrete failures to preserve this cubic, without discarding their actual lattice actions or glue.

Negation preserves each lattice N and Λ, and its exact polynomial law is $F(-X) = -F(X)$. In the neighbour the nonzero witness is its displayed norm-four generator:

$$F(\rho) = 15\sqrt{2} + 16\sqrt{3}, \qquad F(v) = \frac{14}{27} - \frac{7\sqrt{2}}{24} - \frac{\sqrt{3}}{27}. \tag{77}$$

Both equalities follow by substituting the original coordinates of ρ and v into the proved polynomial.

Every original lattice isometry $T \in O(N)$, whether or not it preserves the cubic, has the exact replacement transport

$$\Gamma_T = \mathcal{L}T\mathcal{L}^{-1}, \qquad F^T(Y) = F(T^{-1}Y), \qquad F^T(TX) = F(X), \tag{78}$$

and

$$T\Lambda_\rho = \Lambda_{T\rho}, \qquad \mathcal{V}_{F^T}(\Lambda_{T\rho}) = \mathcal{V}_F(\Lambda_\rho), \qquad \mathcal{V}_{F^T}(N) = \mathcal{V}_F(N).$$

For the neighbour transport, write its exact elements as $x + k\rho/6$, where $x \in N$ and $(\rho, x) + k \equiv 0 \pmod 6$; application of T gives the identical congruence with $(T\rho, Tx)$. The inverse is T^{-1}. The value-module equalities then follow by the bijections and the displayed identity $F^T(TX) = F(X)$. In particular for $T = \sigma^{2k}$, one has $F^T = F$, so the same retained cubic has exactly the same value module on every neighbour $\Lambda_{\sigma^{2k}\rho}$ in this cyclic orbit. This statement names its target neighbour and does not assume the chosen ρ is fixed.

5 The residue cubic action and the common integral Leech correspondence

Retain the exact lattices $R = A^{\perp 4} \perp D \subset N$ and the binary and ternary codes from the index-72 construction. The four A coordinates remain the ordered six-element residue fibres. To avoid confusing a vector with a representation, write ϱ for the Weyl vector

$$\varrho_A = (5,3,1,-1,-3,-5)/2, \qquad \varrho_D = (3,2,1,0), \qquad \varrho = \varrho_A^{\oplus 4} \perp \varrho_D.$$

Thus $\varrho \in N$ and $\varrho^2 = 84$. The original residue representation is still denoted by ρ. The specific element studied here is

$$r = \rho(361) = \rho(289)^2, \qquad r|_{A_i} = C^2, \quad Ce_j = e_{j+1 \bmod 6}, \qquad r|_D = I. \tag{RF1}$$

In particular $r^3 = I$. This is the actual residue coordinate permutation, with no conjugation or replacement.

For each $\sigma \in \Omega = O(N)\varrho$, choose an original root α_σ with $(\sigma, \alpha_\sigma) = 1$ and set

$$\begin{aligned} \ell_\sigma(x) &= (\sigma, x) \bmod 6, \qquad K_\sigma = \ker(\ell_\sigma : N \to \mathbb{Z}/6), \\ \Lambda_\sigma &= K_\sigma + \mathbb{Z}(\sigma/6 - \alpha_\sigma) \\ &= \{x + k\sigma/6 : x \in N,\ k \in \mathbb{Z},\ (\sigma, x) + k \equiv 0 \pmod 6\}. \end{aligned} \tag{RF2}$$

The preceding neighbor proof shows that the definition is independent of α_σ, every Λ_σ is the specified rootless even unimodular lattice, and $N \cap \Lambda_\sigma = K_\sigma$ with both indices six. The family considered in this section is precisely $\mathscr{F} = \{\Lambda_\sigma : \sigma \in \Omega\}$.

5.1 A necessary and sufficient equality criterion

Theorem 5.1 (Equality of the marked Weyl neighbors)**.** *For $\sigma, \tau \in \Omega$, the following three statements are equivalent:*

$$\Lambda_\sigma = \Lambda_\tau, \qquad K_\sigma = K_\tau, \qquad \tau - \epsilon\sigma \in 6N \text{ for some } \epsilon \in \{1,-1\}. \tag{RF3}$$

Proof. Equality of the two neighbors gives equality of the kernels by intersection with N. Each ℓ_σ is onto because $(\sigma, \alpha_\sigma) = 1$. If the kernels are equal, their induced isomorphisms $N/K_\sigma \to \mathbb{Z}/6$ differ by an automorphism of $\mathbb{Z}/6$. Its two units are $1, -1$. Therefore $(\tau - \epsilon\sigma, N) \subset 6\mathbb{Z}$. Unimodularity $N^\vee = N$ gives $(\tau - \epsilon\sigma)/6 \in N$. Conversely this congruence makes the two kernels equal.

It remains to show that this kernel criterion actually gives equality of the chosen unimodular extensions. The exact formula (RF2), after replacing k by $-k$, gives $\Lambda_{-\sigma} = \Lambda_\sigma$. We may consequently suppose $\tau = \sigma + 6t$ with $t \in N$. Both Weyl vectors have squared norm 84, so

$$0 = \tau^2 - \sigma^2 = 12(\sigma, t) + 36t^2, \qquad (\sigma, t) = -3t^2 \in 6\mathbb{Z}. \tag{RF4}$$

The last inclusion uses the actual evenness of N. For $x + k\tau/6$ in (RF2), put $x' = x + kt \in N$. Then $x + k\tau/6 = x' + k\sigma/6$, and

$$(\sigma, x') + k \equiv (\sigma, x) + k \equiv (\tau, x) + k \equiv 0 \pmod 6.$$

This proves $\Lambda_\tau \subset \Lambda_\sigma$. Interchanging σ, τ proves the reverse inclusion. Thus no additional unverified choice of an extension is being identified merely from equality of its common kernel. □

It follows in particular that the map

$$\Omega/\!\sim \ \longrightarrow \ \mathscr{F}, \qquad [\sigma] \longmapsto \Lambda_\sigma, \qquad \sigma \sim \tau \iff \tau \equiv \pm\sigma \pmod{6N}, \tag{RF5}$$

is a bijection. It is $O(N)$-equivariant: the original formula $T\Lambda_\sigma = \Lambda_{T\sigma}$ follows by substitution in (RF2), and T preserves the congruence relation since $T(N) = N$.

5.2 The exact tetracode obstruction to a fixed neighbor

Theorem 5.2 (No fixed member of this specified family)**.** *The actual residue element $r = \rho(361)$ fixes no lattice of $\mathscr{F}$. Every orbit of its action on $\mathscr{F}$ has exactly three members.*

Proof. Suppose first that $r\sigma \equiv -\sigma \pmod{6N}$. Applying r successively, using $r(N) = N$ and $r^3 = I$, gives $\sigma \equiv -\sigma \pmod{6N}$. Then $\sigma/3 \in N$, contradicting its pairing $1/3$ with the root $\alpha_\sigma \in N$. Thus (RF3) permits fixedness only if $t = (r\sigma - \sigma)/6 \in N$.

Every isometry of N preserves the original root lattice, permutes its four A_5 components, and acts in each of them by a signed coordinate permutation, as proved in the full automorphism theorem. Negation permutes the six entries of ϱ_A. Consequently each A block $s = (s_0, \dots, s_5)$ of every $\sigma \in \Omega$ is a permutation of the six distinct numbers $5/2, 3/2, 1/2, -1/2, -3/2, -5/2$. In this block put $\Delta_j = s_{j-2 \bmod 6} - s_j$. If the corresponding block $\Delta/6$ of t belongs to $A^\vee$, its coordinate differences are integral, so

$$\Delta_j \equiv c \pmod{6} \qquad \text{for every } j \tag{RF6}$$

for one integer c modulo six. On either three-cycle of the index permutation, the three Δ_j sum to zero. Hence $3c \equiv 0 \pmod 6$ and $c \in \{0, 2, 4\}$. Each Δ_j is an integer between -5 and 5. If $c = 0$, (RF6) forces all $\Delta_j = 0$, contrary to the distinctness of the entries on each three-cycle. Therefore $c = 2$ or 4.

In the original discriminant marking, a vector of the A class $t_i\lambda + A$ has common fractional part $-t_i/6$. The corresponding class of $\Delta/6$ is therefore $t_i = -c \pmod 6$, also in $\{2, 4\}$. The difference on the D block is exactly zero because $r|_D = I$. Thus membership of the full vector $(r\sigma - \sigma)/6$ in N would require a glue word

$$(t_1, t_2, t_3, t_4; 00) \in H, \qquad t_i \in \{2, 4\} \text{ for every } i. \tag{RF7}$$

In $H = \{(3u + 2z; y)\}$ this forces $u = 0$ and all four z_i nonzero. But the exact ternary code is

$$C_3 = \{(a + b, a + 2b, a, b) : a, b \in \mathbb{F}_3\}. \tag{RF8}$$

If exactly one parameter is zero, exactly one coordinate is zero. If both are nonzero, precisely one of $a + b, a + 2b$ is zero. Thus every nonzero word has weight three, and the zero word has weight zero. No word in (RF7) belongs to H. This contradicts $t \in N$.

Neither sign in (RF3) can occur. Since $r^3 = I$, every orbit therefore has cardinality three. □

The conclusion concerns exactly $\mathscr{F}$ and the action (RF1). The integral correspondence for these three-member orbits is constructed next; no assertion about other embeddings or constructions of a Leech lattice follows from this obstruction.

5.3 The canonical orbit and its common integral lattice

Put $\sigma_j = r^j\varrho$, $\Lambda_j = \Lambda_{\sigma_j}$ and $K_j = K_{\sigma_j}$, with j read modulo three. Keep the same original root $\alpha = e_0 - e_1$ in the first A block. Directly $(\sigma_j, \alpha) = 1$ for all three j. Define

$$v_j = \sigma_j/6 - \alpha, \quad v = v_0, \qquad q_1 = (\sigma_1 - \sigma_0)/6, \quad q_2 = (\sigma_2 - \sigma_0)/6. \tag{RF9}$$

The complete original coordinates are

$$\begin{aligned} q_1 &= \big((-2, -2, 1, 1, 1, 1)/3\big)^{\oplus 4} \perp 0, \\ q_2 &= \big((-1, -1, -1, -1, 2, 2)/3\big)^{\oplus 4} \perp 0. \end{aligned} \tag{RF10}$$

Their classes in $R^\vee/R$ are respectively $(4,4,4,4;00)$ and $(2,2,2,2;00)$. Neither is in H, while $3q_1, 3q_2 \in R$. In particular each $q_i + N$ has exact order three. Direct pairings give

$$q_1^2 = q_2^2 = 16/3, \quad (q_1, q_2) = 8/3, \quad (\varrho, q_1) = (\varrho, q_2) = -16, \qquad q_1 + q_2 \in R. \tag{RF11}$$

For $x \in N$, write its original glue class as $(3u + 2z; y)$, with $z = (a + b, a + 2b, a, b) \in C_3$, and define the homomorphism

$$\beta : N \longrightarrow \mathbb{F}_3, \qquad \beta(x) = b = z_4. \tag{RF12}$$

This map is defined through the original quotient N/R. This definition uses the retained residue marking without changing it. The sum of the four z coordinates is also b modulo three. Write $\ell_j(x) = (\sigma_j, x) \bmod 6$. Then the exact three character formulas are

$$\ell_1(x) = \ell_0(x) + 4\beta(x), \qquad \ell_2(x) = \ell_0(x) + 2\beta(x) \quad \text{in } \mathbb{Z}/6. \tag{RF13}$$

The maps $b \mapsto 2b, 4b$ from $\mathbb{F}_3$ to $\mathbb{Z}/6$ are well-defined homomorphisms. To prove (RF13), the discriminant bilinear form gives

$$(q_1, x) \equiv \tfrac{5}{6}\, 4 \sum t_i \equiv \tfrac{1}{3} \sum t_i \equiv \tfrac{2}{3}\beta(x) \pmod{\mathbb{Z}}, \qquad t_i = 3u_i + 2z_i \pmod 6.$$

For q_2 the same calculation gives $(q_2, x) \equiv \beta(x)/3 \pmod{\mathbb{Z}}$. Multiplication by six and (RF9) prove both formulas.

Using the original dominant representatives $\omega_t = (1^t, 0^{6-t}) - \frac{t}{6}(1, \ldots, 1)$, put

$$u_* = (\omega_2, \omega_4, 0, \omega_2; 0) \in N. \tag{RF14}$$

Its ternary class is exactly the original generator $z_2 = (1, 2, 0, 1)$. It has $u_*^2 = 4$, $(\varrho, u_*) = 12$ and $\beta(u_*) = 1$. The map

$$(\ell_0, \beta) : N \longrightarrow \mathbb{Z}/6 \oplus \mathbb{F}_3 \tag{RF15}$$

is onto: α maps to $(1, 0)$ and u_* to $(0, 1)$. Consequently

$$\begin{gathered} I = \ker(\ell_0, \beta) = K_0 \cap K_1 = K_1 \cap K_2 = K_2 \cap K_0 = \bigcap_j K_j, \\ [N : I] = 18, \qquad [K_j : I] = 3, \qquad \det I = 324. \end{gathered} \tag{RF16}$$

The kernel equalities follow from (RF13), because either $2b = 0$ or $4b = 0$ in $\mathbb{Z}/6$ forces $b = 0$ in $\mathbb{F}_3$. The action r permutes the K_j, so it preserves I. Also $3q_i \in I$: these are in R, hence have $\beta = 0$, and $(\varrho, 3q_i) = -48$ is divisible by six.

Define the full-rank lattice

$$J = I + \mathbb{Z}(3v). \tag{RF17}$$

It is contained in Λ_0. The element $6v = \varrho - 6\alpha$ lies in I: its ℓ_0 value is zero, and its glue has zero ternary part. The element $3v$ is not in N, because $(3v, \alpha) = -11/2$ is not an integer. Thus

$$N \cap J = I, \qquad [J : I] = 2, \qquad \det J = 324/4 = 81, \qquad [\Lambda_0 : J] = 9. \tag{RF18}$$

This lattice is positive definite, even, and rootless, because it is a sublattice of Λ_0.

All exact root corrections in the cyclic action are retained. Put $\kappa_j = \alpha - r^j\alpha \in R$. The vectors $r\alpha = e_2 - e_3$ and $r^2\alpha = e_4 - e_5$ are original simple roots with ϱ-pairing one. Hence $\kappa_j \in I$. The precise formulas, rather than literal identification of different generators, are

$$r^j v = v + q_j + \kappa_j \quad (j = 1, 2), \qquad r(3v) = 3v + 3q_1 + 3\kappa_1. \tag{RF19}$$

Since $rI = I$, $3q_1 \in I$ and $\kappa_1 \in I$, this proves $rJ \subset J$. Applying $r^3 = I$ gives equality $rJ = J$.

Theorem 5.3 (All three pairwise Leech intersections)**.** *The actual three-member orbit satisfies*

$$\Lambda_0 \cap \Lambda_1 = \Lambda_1 \cap \Lambda_2 = \Lambda_2 \cap \Lambda_0 = \bigcap_j \Lambda_j = J, \qquad [\Lambda_j : J] = 9. \tag{RF20}$$

The element r is an integral isometry of J, and induces isometries $\Lambda_j \to \Lambda_{j+1}$ with this common restriction.

Proof. First $J \subset \Lambda_0 \cap \Lambda_1$: I lies in both, and $3v = 3v_1 - 3q_1$ with $3q_1 \in I$. Suppose $y \in \Lambda_0 \cap \Lambda_1$. Using the same α, write $y = x_0 + k_0 v_0 = x_1 + k_1 v_1$, $x_j \in K_j$. The fractional part of (v_j, α) is $1/6$ for both j, so pairing with $\alpha \in N$ gives $k_0 \equiv k_1 \pmod 6$. Absorbing multiples of $6v_j \in K_j$, take the same $k \in \{0, \dots, 5\}$ in both expressions. Then $x_0 - x_1 = kq_1 \in N$. The exact order three of $q_1 + N$ forces $k = 0$ or 3. For these values $kq_1 \in I \subset K_1$, so $x_0 = x_1 + kq_1 \in K_0 \cap K_1 = I$. Thus $y = x_0 + kv \in J$, proving the first equality. Apply r and r^2, using $rJ = J$, to prove the other pairwise equalities. Their common index is nine by (RF18) and the isometries induced by r. □

For comparison with the original index-72 glue, let $R_0 = R \cap K_0$. Since $\beta|_R = 0$, (RF13) gives $R \cap K_j = R_0$ for every j, and $R \cap I = R_0$. The reduction map $K_0/R_0 \cong H$ identifies I/R_0 with the kernel of the original ternary coefficient b. Thus

$$[I : R_0] = 24, \qquad [J : R_0] = 48, \qquad [K_0 : R_0] = 72, \qquad [\Lambda_j : R_0] = 432. \tag{RF21}$$

All these are inclusions of the displayed original lattices.

5.4 The exact discriminant form and its cyclic three planes

Theorem 5.4 (The integral correspondence over $\mathbb{F}_3^4$)**.** *One has*

$$J^\vee = \Lambda_i + \Lambda_j \quad (i \neq j), \qquad J^\vee/J \cong \mathbb{F}_3^4. \tag{RF22}$$

In the ordered basis

$$e_1 = u_* + J, \quad e_2 = v + J, \quad e_3 = ru_* + J, \quad e_4 = rv + J, \tag{RF23}$$

the discriminant quadratic form is

$$q_J(a, b, c, d) = \frac{2}{3}(2ad + bc + bd) \pmod{2\mathbb{Z}}, \tag{RF24}$$

and three times its discriminant bilinear form has matrix

$$B = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_3. \tag{RF25}$$

The three original Leech extensions are exactly the inverse images in $J^\vee$ of the isotropic planes

$$\begin{aligned} P_0 &= \{(a, b, 0, 0) : a, b \in \mathbb{F}_3\}, \\ P_1 &= \{(0, 0, c, d) : c, d \in \mathbb{F}_3\}, \\ P_2 &= \{(a, b, a - b, b) : a, b \in \mathbb{F}_3\}. \end{aligned} \tag{RF26}$$

The induced action of the actual r is the matrix

$$\mathcal{R} = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 0 & 2 & 2 \\ 0 & 1 & 0 & 2 \end{pmatrix}, \qquad \mathcal{R}^3 = I, \quad \mathcal{R}^T B \mathcal{R} = B, \quad \mathcal{R} P_j = P_{j+1}. \tag{RF27}$$

All entries in (RF23)–(RF27) use the original glue generator z_2, root α, pairing, and residue permutation.

Proof. Since $J \subset \Lambda_i = \Lambda_i^\vee$, each Λ_i lies in $J^\vee$. By (RF20), the two subgroups Λ_i/J and Λ_j/J have order nine and intersect trivially for $i \neq j$. Their sum has order 81, equal to $|J^\vee/J| = \det J = 81$, proving the first equality in (RF22).

The homomorphism $\beta : K_0/I \to \mathbb{F}_3$ is an isomorphism, with generator $u_* + I$. Hence Λ_0/J is generated by $u_* + J, v + J$. Both have order dividing three because $3u_* \in I$ and $3v \in J$. They are independent: if $au_* + bv \in J$, reduce first modulo N; the subgroup $J + N$ has the two cosets represented by $0, 3v$, whereas $v + N$ has order six. For $a, b \in \{0, 1, 2\}$ this forces $b = 0$. Then membership of au_* in $I = N \cap J$ forces $a = 0$. Thus $\Lambda_0/J \cong \mathbb{F}_3^2$ with the first two generators in (RF23). Applying r proves the corresponding statement for Λ_1/J and the last two generators. Their trivial intersection and total order 81 prove that (RF23) is indeed a basis of the full discriminant group.

Here are the full Euclidean Gram entries before any passage to a quotient:

$$\mathrm{Gram}(u_*, v, ru_*, rv) = \begin{pmatrix} 4 & 2 & -2 & -4/3 \\ 2 & 4 & -2/3 & -2/3 \\ -2 & -2/3 & 4 & 2 \\ -4/3 & -2/3 & 2 & 4 \end{pmatrix}. \tag{RF28}$$

For an explicit verification, each of the three nonzero A blocks of u_* pairs to $-2/3$ with its shifted block, giving $(u_*, ru_*) = -2$. One has $(u_*, \alpha) = (u_*, r\alpha) = (ru_*, \alpha) = 0$, $(\varrho, u_*) = 12$, $(\sigma_1, u_*) = -8$, and $(\sigma_2, u_*) = -4$, directly from (RF10) and (RF14). These give respectively $(u_*, v) = 2$, $(u_*, rv) = -4/3$, and $(v, ru_*) = -2/3$. Also $(\varrho, \sigma_1) = -12$, $(\varrho, r\alpha) = (\sigma_1, \alpha) = 1$, and $(\alpha, r\alpha) = 0$. Thus

$$(v, rv) = -12/36 - 1/6 - 1/6 = -2/3.$$

The four diagonal entries equal four, and $(ru_*, rv) = (u_*, v) = 2$ by the exact isometry r. This proves (RF28). Reducing its entries modulo $\mathbb{Z}$ gives (RF25). The diagonal and integral cross terms in the squared norm are even. Reducing the remaining cross terms modulo $2\mathbb{Z}$ gives exactly (RF24). Changing any coordinate by three changes (RF24) by an even integer, so it is well-defined on the asserted field basis.

It remains to calculate the action without identifying a family permutation with a fixed Leech automorphism. By definition $re_1 = e_3$, $re_2 = e_4$. The vector $u_* + ru_* + r^2u_*$ lies in I: its β value is $1 + 1 + 1 = 0$, because r acts trivially on N/R, and its ℓ_0 value is $12 - 4 - 8 = 0$. Consequently

$$r^2 u_* \equiv -u_* - ru_* \pmod{J}. \tag{RF29}$$

By (RF19),

$$v + rv + r^2 v = 3v + q_1 + q_2 + \kappa_1 + \kappa_2.$$

The κ_j belong to I, and $3v \in J$. Furthermore $q_1 + q_2 - u_* + ru_* \in I$. Indeed $q_1 + q_2 \in R$, so its β value is zero; subtracting u_* and adding ru_* keeps that value zero. Its pairing with ϱ is $-32 - 12 - 4 = -48$, divisible by six. Thus

$$r^2 v \equiv u_* - v - ru_* - rv \pmod{J}. \tag{RF30}$$

The four images in (RF29)–(RF30) give the four columns of $\mathcal{R}$ in (RF27). The matrix has cube the identity and preserves B either by multiplication, or by the already proved facts $r^3 = I$ and $rJ = J$ with its original isometric pairing.

The first two planes in (RF26) are respectively Λ_0/J and Λ_1/J by their proved bases. Their next image is the span of $(-1, 0, -1, 0)$ and $(1, -1, -1, -1)$, from (RF29)–(RF30). Solving their two linear parameters gives exactly $(a, b, a-b, b)$, proving the stated $P_2 = \Lambda_2/J$. Each plane is isotropic: the first two vanish in (RF24), and the third has $2ab + b(a-b) + b^2 = 3ab = 0$ in $\mathbb{F}_3$. They have nine elements and pairwise intersection zero, also following directly from (RF26). Finally each Λ_j is exactly the inverse image of its quotient plane because it contains J. This proves the claimed integral correspondence and all its maps, indices, pairings, and cyclic actions. □

5.5 The fixed fourth plane and its exact A_2^{12} extension

The same finite discriminant form has the additional plane

$$P_\infty = \{(a, b, -a+b, -b) : a, b \in \mathbb{F}_3\}. \tag{RF31}$$

Put

$$p = u_* - ru_*, \qquad q = v + ru_* - rv, \qquad M = J + \mathbb{Z}p + \mathbb{Z}q. \tag{RF32}$$

These are literal vectors in the original Euclidean space.

Theorem 5.5 (The fixed fourth extension)**.** *The plane P_∞ is isotropic and is fixed by the induced residue action. Its inverse image M is a positive definite even unimodular lattice with exactly 72 roots, of root system A_2^{12} and minimum squared norm two. The original residue element r is an integral isometry of M. It fixes four of these A_2 components pointwise and acts as a coordinate three-cycle on each of the other eight.*

The exact intersection and index relations are

$$\begin{gathered} M \cap \Lambda_j = J, \qquad [M : J] = [\Lambda_j : J] = 9, \\ M \cap N = L_0 := \{x \in N : \beta(x) = 0,\ (\varrho, x) \equiv 0 \pmod 2\}, \\ [M : L_0] = [N : L_0] = 6. \end{gathered} \tag{RF33}$$

If R_M denotes the lattice generated by all its roots, then

$$R_M \cong A_2^{\perp 12}, \qquad \det R_M = 3^{12} = 531441, \qquad [M : R_M] = 3^6 = 729. \tag{RF34}$$

Proof. Substitution of (RF31) in (RF24) gives the numerator $2a(-b) + b(-a+b) + b(-b) = -3ab = 0$ in $\mathbb{F}_3$. Thus P_∞ is isotropic. The matrix (RF27) maps its vector with parameters (a, b) to

$$(a+b, b, -a, -b), \tag{RF35}$$

which is again in P_∞, now with parameters $(a+b, b)$. The images of p, q in (RF23) are $(1, 0, -1, 0)$ and $(0, 1, 1, -1)$, a basis of P_∞. Consequently (RF32) is its exact inverse image under $J^\vee \to J^\vee/J$, and $[M : J] = 9$. Isotropy gives even norms on M; polarization gives integral pairings. Its determinant is $\det J/9^2 = 1$, and its positive definiteness and rank are inherited from the unchanged ambient real space. Equation (RF35), together with $rJ = J$, proves $rM = M$. Each of P_0, P_1, P_2 intersects P_∞ in zero, as follows by equating their coordinates in (RF26) and (RF31). Taking inverse images proves $M \cap \Lambda_j = J$.

For its relation with N, (RF19) gives the exact expression

$$q = ru_* - q_1 - \kappa_1. \tag{RF36}$$

Therefore

$$N + M = N + \mathbb{Z}(\varrho/2) + \mathbb{Z}q_1. \tag{RF37}$$

The two indicated cosets modulo N have orders two and three. For the first, pair with α; for the second, use (RF10). Subgroups of coprime orders intersect trivially, so the quotient in (RF37) has order six. Also $p \in N$ has $\beta(p) = 0$ and $(\varrho, p) = 12 - (-4) = 16$, hence image $(4, 0)$ in (RF15). Thus $p + I$ has order three. The vector $3q$ lies in J because $q + J$ has order three, and lies in N by (RF36) and $3q_1 \in N$. It is therefore in $N \cap J = I$. We also have $6v \in I$. Write an element of M using generators $I, 3v, p, q$ and reduce the coefficient of $3v$ modulo two and of q modulo three. If this element is in N, their two coprime-order cosets in (RF37) force both reduced coefficients to vanish. Hence

$$M \cap N = I + \mathbb{Z}p. \tag{RF38}$$

Its image in (RF15) is precisely $2(\mathbb{Z}/6)\oplus 0$, which proves the displayed formula for L_0 and $[N : L_0] = 6$. Since $\det N = \det M = 1$, the equal covolumes give $[M : L_0] = 6$, proving (RF33).

We next prove that the following root count is exhaustive. Every vector in $N + M$ has one of the six forms

$$x + a\varrho/2 - bq_1, \qquad x \in N,\ a \in \{0, 1\},\ b \in \{0, 1, 2\}. \tag{RF39}$$

When $a = 0$, the vector is in $R^\vee$. Every norm-two vector of this particular $R^\vee$ is already a root of R. Indeed, the A class minima are $0, 5/6, 4/3, 3/2, 4/3, 5/6$, and the three nonzero D classes have minimum one. In a nonzero class of $R^\vee/R$ with minimum at most two, the possible minima are exactly among

$$\{5/6, 1, 4/3, 3/2, 5/3, 11/6\}. \tag{RF40}$$

This exhausts them because at most two A classes can be nonzero under the bound; if two are nonzero, each has minimum $5/6$; with a nonzero D class there is at most one nonzero A class and its minimum must be $5/6$. No number in (RF40) is congruent to two modulo $2\mathbb{Z}$. All norms within a fixed discriminant class differ by an even integer, so no such class has a norm-two vector. This proves the claim about $R^\vee$.

All original roots have $\beta = 0$, so (RF33) says that those lying in M are exactly the ones with even ϱ height. In each original A block they are $e_i - e_j$ with $i \neq j$ and $i \equiv j \pmod 2$. These give two orthogonal A_2 systems per block, on the coordinate triples $\{0, 2, 4\}$ and $\{1, 3, 5\}$, and 48 roots in total. In D they are the eight roots $\pm e_1 \pm e_3,\ \pm e_2 \pm e_4$. They form four pairwise orthogonal A_1 systems. This accounts for exactly 56 roots with $a = 0$ in (RF39).

For $a = 1$, absorb the A dual vector $-bq_1$ into the marked A discriminant classes. If the original class of x is $(t_1, t_2, t_3, t_4; y) \in H$, the shifted class in each A is $t_i - 4b$. The exact half-Weyl-shift minima proved in the neighbor calculation are

$$\min_{w \in t\lambda + A} |w + \varrho_A/2|^2 = \begin{cases} 3/8 & t \equiv 0 \pmod 3, \\ 17/24 & t \not\equiv 0 \pmod 3, \end{cases} \qquad \min_{w+D=y} |w + \varrho_D/2|^2 = 1/2. \tag{RF41}$$

For completeness the first line follows by choosing centered quarter coordinates when $t = 0, 3$; for other t, the nearest twelfth coordinates have square sum $13/24$ and coordinate sum ± 1, and restoring zero sum costs $1/6$. The second line follows from the integer coordinate distances $(1/2, 0, 1/2, 0)$ and half-integer distances $(0, 1/2, 0, 1/2)$, with their ties meeting every D marking. Thus the total minimum in (RF39) with $a = 1$ is at least two. Equality requires $t_i - 4b \equiv 0 \pmod 3$ in every component. Since $t_i \equiv 2z_i \pmod 3$, this says $z_i = 2b$ for all four coordinates. For $b = 1, 2$ this is excluded by the exact code (RF8). For $b = 0$ it requires $z = 0$. The eight binary codewords and the

two D ties then give exactly sixteen equality vectors $y = x + \varrho/2$. Every one belongs to M: the norm equation gives

$$2 = x^2 + (\varrho, x) + 21, \qquad (\varrho, x) = -19 - x^2, \tag{RF42}$$

which is odd because N is even. Also $\beta(x) = 0$ because $z = 0$. Therefore $x + 3\alpha \in L_0$ and $y = (x + 3\alpha) + 3v \in M$. This proves exhaustively that M has exactly $56 + 16 = 72$ roots.

Here are the full root coordinates and the resulting exact orthogonal decomposition. Let ζ_i be supported in the ith original A block with entries

$$\zeta = (1, -1, 1, -1, 1, -1)/4, \qquad \zeta^2 = 3/8.$$

In the following table every Z_y is an actual sum of those four mutually orthogonal vectors, and every δ_y is an original D root:

y	Z_y	δ_y
00	$\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4$	$e_1 + e_3$
10	$-\zeta_1 - \zeta_2 + \zeta_3 + \zeta_4$	$e_1 - e_3$.
01	$-\zeta_1 + \zeta_2 - \zeta_3 + \zeta_4$	$e_2 + e_4$
11	$\zeta_1 - \zeta_2 - \zeta_3 + \zeta_4$	$e_2 - e_4$

(RF43)

The sixteen new roots are exactly

$$\{\epsilon Z_y + \eta\delta_y/2 : y \in \mathbb{F}_2^2,\ \epsilon, \eta \in \{1, -1\}\}. \tag{RF44}$$

To verify the signs from the original glue, the A half-shift minimizer for binary coordinate $u_i = 0$ is ζ_i and for $u_i = 1$ it is $-\zeta_i$. For $y = 00$ the two codewords are $0000, 1111$; for 10 they are $1100, 0011$; for 01 they are $1010, 0101$; and for 11 they are $0110, 1001$. These give respectively the four opposite pairs $\pm Z_y$ in (RF43). The two shifted D minimizers in these exact original markings are respectively $\pm(e_1 + e_3)/2, \pm(e_1 - e_3)/2, \pm(e_2 + e_4)/2, \pm(e_2 - e_4)/2$. This proves (RF44) with every sign and marking retained.

The four Z_y are mutually orthogonal and have squared norm $3/2$, as follows by dotting their four displayed sign rows. The δ_y are mutually orthogonal of squared norm two. The Z_y are orthogonal to the entire original D space, and to the eight old A_2 systems because their coordinates are constant on each parity triple. For each y, the six roots

$$\{\pm\delta_y\} \ \cup \ \{\epsilon Z_y + \eta\delta_y/2 : \epsilon, \eta = \pm 1\} \tag{RF45}$$

are an A_2 system: take simple roots δ_y and $Z_y - \delta_y/2$, whose Gram matrix is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$; their sum is $Z_y + \delta_y/2$. The four such systems are mutually orthogonal and orthogonal to the eight old systems. They exhaust the 72 proved roots. Thus their root lattice is $A_2^{\perp 12}$ of determinant 3^{12}, and unimodularity of M gives the index 3^6. This proves (RF34) and the Coxeter number three of this rooted Niemeier lattice directly from its twelve components.

Finally r fixes each ζ_i because it permutes coordinates within their parity triples; it fixes each δ_y because it is the identity on D. Hence it fixes the four systems (RF45) pointwise. On each of the eight old A_2 systems it cyclically permutes its three coordinate positions, giving its exact order-three Coxeter action. This completes the asserted classification of the fixed fourth extension and its precise relation to the three Leech lattices through J. □

6 The cubic on the common lattice and the fourth Niemeier extension

Retain the original coordinates, Cayley multiplication, F, the generators g_i and their heights H_i from Section 4, and the exact common lattice and fourth extension of Section 5. The symbol u below is the unchanged vector u_* in (RF32).

6.1 Integral generators of the common lattice and the fixed extension

Define the unchanged residue action by

$$(rX)_{i,j} = x_{i,j-2 \pmod 6}, \qquad (rX)_{D,k} = d_k.$$

The two primary parts of the original glue have the exact descriptions

$$H_2 = \{(\varepsilon_1+\varepsilon_2+\varepsilon_3, \varepsilon_1+\varepsilon_2, \varepsilon_1+\varepsilon_3, \varepsilon_1; \varepsilon_2, \varepsilon_3) : \varepsilon_i \in \mathbb{F}_2\},$$
$$H_3 = \{(t+s, t+2s, t, s; 00) : t, s \in \mathbb{F}_3\}.$$

In the original $(\mathbb{Z}/6)^4$ coordinates this means $k = 3w + 2z$, with w the four binary entries of H_2 and z the four ternary entries of H_3. The binary D_4 marking is retained. Chinese remainders recover w, z uniquely, and the displayed parameterizations are injective. Consequently

$$\beta : N \longrightarrow \mathbb{F}_3, \qquad \beta(X) = z_4 = s$$

is a well-defined additive homomorphism. The actual lifts (61) satisfy

$$\beta(g_i) = 0 \quad (1 \le i \le 28), \qquad \beta(g_{29}) = 1.$$

Indeed roots have zero glue, the first three lifts are two-primary, and the last two have $(t, s) = (1, 0), (0, 1)$, respectively.

Set, with no change of representatives,

$$I = \{X \in N : (\rho, X) \equiv 0 \pmod 6,\ \beta(X) = 0\}, \qquad J = I + \mathbb{Z}(3v), \tag{79}$$
$$\omega_t = (\underbrace{1, \dots, 1}_{t}, 0, \dots, 0) - \frac{t}{6}(1,1,1,1,1,1), \qquad u = (\omega_2, \omega_4, 0, \omega_2; 0),$$
$$p = u - r(u), \qquad q = v + r(u) - r(v),$$
$$M = J + \mathbb{Z}p + \mathbb{Z}q. \tag{80}$$

Thus J and M are precisely the common-lattice and residue-fixed extension presentations being evaluated here; the following value-group proof needs no root count, lattice determinant, or classification label for M. For explicitness, the two extra vectors in the original five blocks are

$$p = ((1,1,-1,-1,0,0), (1,1,0,0,-1,-1), 0, (1,1,-1,-1,0,0); 0),$$
$$q = \tfrac{1}{3}((-2,4,4,-2,-2,-2), 0, (2,2,-1,-1,-1,-1), (1,1,1,1,-2,-2); 0).$$

The zero blocks have their original sizes six and four. These coordinates follow by substituting $v = \rho/6 - \alpha$ into (80); in particular the shifted α term in $r(v)$ is retained. Also $u \in N$, $\beta(u) = 1$, $(\rho, u) = 12$, and $(u, u) = 4$: its class is $(2, 4, 0, 2; 00)$, $(\rho_A, \omega_t) = t(6-t)/2$, and $(\omega_t, \omega_t) = t(6-t)/6$.

Theorem 6.1 (Integral generating lists with both kernel conditions)**.** *Define the following 32 ordered vectors:*

$$\begin{aligned} b_1 &= 6\alpha, & b_i &= g_i - H_i\alpha \quad (2 \le i \le 28), \\ b_{29} &= 3g_{29} - 60\alpha, & b_{30} &= 3v = \rho/2 - 3\alpha, \\ b_{31} &= p, & b_{32} &= q. \end{aligned} \tag{81}$$

Then the integral, not merely rational, equalities are

$$I = \sum_{i=1}^{29} \mathbb{Z} b_i, \qquad J = \sum_{i=1}^{30} \mathbb{Z} b_i, \qquad M = \sum_{i=1}^{32} \mathbb{Z} b_i.$$

Proof. Every one of the first 29 vectors has $\beta = 0$ and height divisible by six, by the exact heights and characters already calculated. Conversely take any integral expression $X = \sum_{i=1}^{29} n_i g_i$ for $X \in I$. Applying β shows $n_{29} = 3a$ for an integer a, regardless of which expression was chosen. The exact equality

$$X = \sum_{i=2}^{28} n_i(g_i - H_i\alpha) + a(3g_{29} - 60\alpha) + \left(n_1 + \sum_{i=2}^{28} n_i H_i + 60a \right) \alpha$$

has last coefficient (ρ, X), which is divisible by six. It therefore expresses X as an integer combination of the first 29 b_i. The reverse containment was proved for each generator. The last two equalities follow by adjoining exactly the vectors in (79) and (80). No assertion of linear independence is used. □

6.2 Complete finite certificates for the two new value groups

Theorem 6.2 (Equality of the common-lattice and fixed-extension value-generated groups)**.** *For the retained Cayley multiplication and the original ordered coordinates, the two additive cubic-value groups are exactly*

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) = \frac{1}{27}\mathbb{Z} \ \oplus \frac{\sqrt{2}}{216}\mathbb{Z} \ \oplus \frac{\sqrt{3}}{18}\mathbb{Z}. \tag{82}$$

The equality concerns generated additive groups, not equality of individual-value sets $F(J)$ and $F(M)$.

Proof. Apply Theorem 4.2 to the first 30 and all 32 vectors in (81). The resulting list lengths are exactly

$$30 + 2\binom{30}{2} + \binom{30}{3} = 4960, \qquad 32 + 2\binom{32}{2} + \binom{32}{3} = 5984.$$

Write all pair and triple symbols in the next tables using these b_i, not the g_i or h_i. Substitution in the full polynomial gives the complete denominator certificate

	A_i	D^+_{ij}	D^-_{ij}	D_{ijk}
J	$(27, 216, 1)$	$(9, 36, 18)$	$(9, 36, 18)$	$(9, 36, 18)$
M	$(27, 216, 1)$	$(9, 36, 18)$	$(9, 36, 18)$	$(9, 36, 18)$

. (83)

As before each triple is the least common multiple of all reduced rational denominators in its full category, in the order (P_0, P_2, P_3); a zero has denominator one. The finite arithmetic is completely specified by (51)–(60), (81), and all indices in (63)–(66). The accompanying standard-library checker `check_jm_values.py`, using the included `cubic_engine.py`, executes every entry with

exact fractions and asserts this table. It reconstructs all 72 marked glue classes, checks all 29 $\beta(g_i)$, and checks both kernel conditions on all the displayed I generators. These finite computations establish the upper containment in (82), including all mixed polarization terms involving p, q, not just their individual cubic values.

Here is a short certificate for the reverse containment, using only J. Coordinates of each row are taken in the ordered proposed value-group basis $(1/27, \sqrt{2}/216, \sqrt{3}/18)$:

$$\begin{array}{c|rrr} F(b_2) & 14 & -272 & 0 \\ F(b_4) & 5 & -128 & 0 \\ F(b_{27}) & 23220 & -111672 & -90 \\ F(b_{28}) & 13780 & -74672 & 0 \\ F(b_{30}) & 378 & -1701 & -18 \\ D^+_{2,27} & 7425 & -60624 & -3 \\ D^+_{11,21} & 36 & 48 & -16 \end{array} . \tag{84}$$

Let C_1, C_2, C_3 have, respectively, columns obtained by transposing rows $(2, 3, 5)$, $(2, 5, 6)$, and $(1, 4, 7)$ of this table. Direct three-by-three determinants are

$$d_1 = -39859290, \qquad d_2 = 11531403, \qquad d_3 = -43244032.$$

The following explicit integer identity certifies their gcd one:

$$(-6882196252)d_1 + (-23788908901)d_2 + (-907)d_3 = 1. \tag{85}$$

Put $(c_1, c_2, c_3) = (-6882196252, -23788908901, -907)$. For each standard basis column $e_t \in \mathbb{Z}^3$ the exact integer reconstruction is

$$e_t = \sum_{j=1}^{3} C_j(c_j \operatorname{adj}(C_j)e_t). \tag{86}$$

All columns of every C_j represent integer combinations of actual cubic values on J; their integer coefficient vectors in (86) therefore give each of the three claimed generators as such a combination. This is an explicit reverse-containment morphism, not an index inference. The checker also expands every adjugate and verifies all three axis reconstructions, saving their integer coefficients in `EXACT_RESULTS.json`. Thus $\mathcal{V}_F(J)$ contains the right-hand side of (82). Since $J \subset M$, the same certificate proves its containment in $\mathcal{V}_F(M)$. The upper bounds already established finish both equalities. □

For additional exact checks on the unchanged representatives, the three individual values are

$$F(3v) = 14 - \frac{63}{8}\sqrt{2} - \sqrt{3}, \qquad F(p) = -2, \qquad F(q) = \frac{64}{27}.$$

They agree with the corresponding rows of the complete calculation; they are not substituted for the mixed-term certificate.

6.3 Exact inclusions, quotient maps, and the factor of the Albert determinant

Theorem 6.3 (The exact four-to-three chain of value groups)**.** *There is a strict chain of additive subgroups of the original real coefficient field*

$$\mathcal{V}_F(N) \subsetneq \mathcal{V}_F(J) = \mathcal{V}_F(M) \subsetneq \mathcal{V}_F(\Lambda).$$

The inclusion matrices in the ordered bases already displayed are

$$\mathcal{V}_F(N) \longrightarrow \mathcal{V}_F(J) = \mathcal{V}_F(M) : (a,b,c) \longmapsto (a,4b,c),$$

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) \longrightarrow \mathcal{V}_F(\Lambda) : (a,b,c) \longmapsto (a,b,3c).$$

Their exact quotient maps are

$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{18} \longmapsto b \pmod 4,$$
$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{54} \longmapsto c \pmod 3.$$

In particular

$$\mathcal{V}_F(J)/\mathcal{V}_F(N) \cong \mathbb{Z}/4, \qquad \mathcal{V}_F(M)/\mathcal{V}_F(N) \cong \mathbb{Z}/4,$$
$$\mathcal{V}_F(\Lambda)/\mathcal{V}_F(J) \cong \mathbb{Z}/3, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(M) \cong \mathbb{Z}/3,$$
$$\mathcal{V}_F(M)/\mathcal{V}_F(J) = 0.$$

Proof. The first basis comparison uses $\sqrt{2}/54 = 4(\sqrt{2}/216)$, with the other basis entries equal. The second uses $\sqrt{3}/18 = 3(\sqrt{3}/54)$, with its other entries equal. Uniqueness of rational radical coefficients proves that the displayed kernels are exactly the corresponding subgroups. Surjectivity is witnessed by $\sqrt{2}/216$ and $\sqrt{3}/54$, respectively, both actual elements of the proved additive groups. Their orders modulo the smaller groups are exactly four and three: the coefficient divisibility conditions are necessary and sufficient. This also proves strictness. The last quotient is zero by the equality of the two proved groups. In particular the composite inclusion is $(a,b,c) \mapsto (a,4b,3c)$, recovering the full quotient $\mathbb{Z}/4 \oplus \mathbb{Z}/3$, with explicit map

$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{54} \longmapsto (b \bmod 4, c \bmod 3).$$

Its identification with $\mathbb{Z}/12$ is $(\bar{b},\bar{c}) \mapsto 9b + 4c \pmod{12}$, whose inverse is reduction modulo four and modulo three. Every assertion follows from the proved cubic coefficient groups, not from an index of the original lattices. □

The factor two in the original Albert determinant is unchanged:

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X,X) + 2F(X).$$

Thus, if one instead asks for the additive group generated by the actual cubic term $2F$, it is exactly $2\mathcal{V}_F(L)$. For J, M this is

$$\frac{2}{27}\mathbb{Z} \ \oplus \frac{\sqrt{2}}{108}\mathbb{Z} \ \oplus \frac{\sqrt{3}}{9}\mathbb{Z}.$$

Multiplication by two is an injective additive map of the real coefficient field and restricts to the exact isomorphism $\mathcal{V}_F(L) \to \mathcal{V}_{2F}(L) = 2\mathcal{V}_F(L)$, $x \mapsto 2x$, with inverse $y \mapsto y/2$. It intertwines the same two inclusion matrices and the same quotients; no factor has been absorbed into F.

6.4 Reproduction and scope of the exact certificate

The accompanying evidence folder is

`supporting_materials/workbench/evidence/`
`period_lattice_continuation_20260906/jm_cubic/`

It contains both full source files `cubic_engine.py` and `check_jm_values.py`. Running the latter with Python's standard library performs the following finite proof certificate: all 20399 polarization entries for N, Λ, J, M, all displayed denominator and determinant assertions, the integer Bezout and adjugate axis reconstructions, and the exact inclusion coefficient comparisons. The output `EXACT_RESULTS.json` retains each selected column, determinant, Bezout coefficient, and every expanded integer linear combination reconstructing a value-group axis.

There is additionally an independent coefficient-identity check: the original 24 simple roots form a real basis of the original 24-dimensional space. The direct Cayley–Dickson implementation and the rational polynomial implementation are compared on the 24 basis vectors, their 552 ordered-by-index plus/minus pairs, and their 2024 three-element sums, a total of 2600 exact inputs. The finite polarization proof shows these evaluations determine every coefficient of a homogeneous cubic in those 24 variables. Consequently this is a full polynomial-identity certificate on the real ambient space, not sampling of lattice points. The field implementation retains all four rational coefficients in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and verifies that the $\sqrt{6}$ coefficient is identically zero. A further 65 direct checks cover the displayed new generating lists and u, p, q.

This computation proves the stated additive value groups and their morphisms. It asserts neither equality of individual-value sets nor that the cubic group reconstructs a lattice or classifies all cubic-preserving lattice isometries.

7 A cyclic Wilson Leech lattice in the original triality space

7.1 The two retained metrics and the octonion convention

Let $V = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ be the original ordered octonion space. Retain its metric and cubic

$$\begin{aligned} h_0(x,x) &= N(x_1) + N(x_2) + N(x_3), \\ \tau(x_1,x_2,x_3) = \mathrm{Re}((x_1x_2)x_3), &\qquad d(Q_{\mathbb{O}}(0;x)) = 2\tau(x_1,x_2,x_3), \end{aligned}$$

where

$$Q_{\mathbb{O}}(\lambda;x) = \begin{pmatrix} \lambda & x_3 & \bar{x}_2 \\ \bar{x}_3 & \lambda & x_1 \\ x_2 & \bar{x}_1 & \lambda \end{pmatrix}.$$

Wilson's norm on the same underlying ordered space is

$$q_{\mathrm{W}}(x) = \tfrac{1}{2} h_0(x,x).$$

The map

$$\iota : (V, q_{\mathrm{W}}) \longrightarrow (V, h_0), \qquad \iota(x_1,x_2,x_3) = (x_1/\sqrt{2}, x_2/\sqrt{2}, x_3/\sqrt{2}) \tag{87}$$

is an explicit isometry. Both metrics remain specified. In particular, for the original λ, which is not rescaled,

$$d(Q_{\mathbb{O}}(\lambda;\iota x)) = \lambda^3 - \lambda q_{\mathrm{W}}(x) + \frac{1}{\sqrt{2}}\tau(x_1,x_2,x_3). \tag{88}$$

Indeed the quadratic coordinates scale by $1/2$, while the trilinear term scales by $(1/\sqrt{2})^3 = 1/(2\sqrt{2})$ and its original coefficient is 2. Thus the off-diagonal cubic after ι is $1/(2\sqrt{2})$ times its value before ι.

For an explicit realization inside the same $\mathbb{O}$, choose a Cayley–Dickson basis $(1, \mathsf{i}, \mathsf{j}, \mathsf{k}, \ell, \mathsf{i}\ell, \mathsf{j}\ell, \mathsf{k}\ell)$, with

$$(a + b\ell)(c + d\ell) = (ac - \bar{d}b) + (da + b\bar{c})\ell, \qquad \mathsf{ij} = \mathsf{k}.$$

Wilson's ordered orthonormal basis is realized by

$$(i_\infty, i_0, i_1, i_2, i_3, i_4, i_5, i_6) = (1, \mathsf{i}, \mathsf{j}, \ell, \mathsf{k}, \mathsf{j}\ell, -\mathsf{k}\ell, \mathsf{i}\ell). \tag{89}$$

The seven oriented Fano triples are $(t, t+1, t+3)$ modulo 7: $i_t i_{t+1} = i_{t+3}$, with cyclic products within each triple and opposite signs on reversal. To verify (89), its seven corresponding products, in order $t = 0, \ldots, 6$, are

$$\mathsf{ij} = \mathsf{k}, \quad \mathsf{j}\ell = \mathsf{j}\ell, \quad \ell\mathsf{k} = -\mathsf{k}\ell, \quad \mathsf{k}(\mathsf{j}\ell) = \mathsf{i}\ell, \quad (\mathsf{j}\ell)(-\mathsf{k}\ell) = \mathsf{i}, \quad (-\mathsf{k}\ell)(\mathsf{i}\ell) = \mathsf{j}, \quad (\mathsf{i}\ell)\mathsf{i} = \ell.$$

Alternativity supplies cyclic products on each quaternionic line. The seven lines cover all pairs of different imaginary basis units; squares are -1 and 1 is the unit. This proves the algebra isomorphism on the full multiplication table, hence by bilinearity on all octonions. It also preserves conjugation and the norm. No conjugation of a coordinate of $Q_{\mathbb{O}}$ is performed.

7.2 The explicit eight-dimensional lattice and two sublattices

In the displayed Wilson basis write $e_0 = i_\infty, e_{t+1} = i_t$ and put

$$D_8 = \{(z_0, \ldots, z_7) \in \mathbb{Z}^8 : \textstyle\sum z_i \equiv 0 \pmod 2\},$$
$$s = \tfrac{1}{2}(-1 + i_0 + i_1 + i_2 + i_3 + i_4 + i_5 + i_6), \qquad L_{\mathrm{W}} = D_8 + \mathbb{Z}s.$$

This is Wilson's lattice L, renamed L_{W} to distinguish it from the project's original rank-24 Niemeier lattice. Since $2s \in D_8$ and $s \notin D_8$, it is exactly $D_8 \cup (s + D_8)$.

Lemma 7.1. *The lattice L_{W} is even and unimodular, has minimum norm 2, and has the explicit basis*

$$b_1 = s, \quad b_j = e_{j-1} - e_j \ (2 \le j \le 7), \quad b_8 = e_6 + e_7.$$

Its Gram matrix for the unchanged octonion norm is

$$G_8 = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \qquad \det G_8 = 1. \tag{90}$$

Its norm-two vectors are precisely the 112 vectors $\pm e_i \pm e_j$ with $i < j$ and the 128 vectors $\frac{1}{2}(\pm e_0 \pm \cdots \pm e_7)$ with an odd number of minus signs.

Proof. For $z \in D_8$, $N(z) \equiv \sum z_i = 0 \pmod 2$ and $h(s,z) = \frac{1}{2}\sum z_i - z_0 \in \mathbb{Z}$. As $N(s) = 2$, every norm in $D_8 + \mathbb{Z}s$ is even. The covolume of D_8 is 2, and its displayed extension has index 2, so L_{W} has covolume 1. The integral bilinear form follows by polarizing the even norm, and covolume one proves self-duality. The columns b_i all belong to L_{W}. Their determinant has absolute value one: expansion along the first coordinate gives $1/2$ times the determinant 2 of the indicated D_7 basis. They therefore form a basis. Their scalar products give the displayed matrix. Positivity and evenness give minimum at least 2, and $N(b_i) = 2$ attains it. Integer vectors of norm two have exactly two nonzero entries ± 1. Half-integer vectors have norm at least $8/4 = 2$, with equality exactly when all eight entries are $\pm 1/2$. Their membership in $s + D_8$ is exactly the stated odd-minus parity. □

Define, with the bars in the positions printed in Wilson,

$$\mathsf{M} = L_{\mathrm{W}}\bar{s}, \qquad \mathsf{N} = L_{\mathrm{W}}s.$$

All products are right products. Let B be the basis matrix with columns b_i. Right multiplication by s in this basis is the following integer matrix:

$$S = B^{-1}R_sB = \begin{pmatrix} 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ -2 & -1 & 0 & 1 & -1 & 1 & 0 & -2 \\ -4 & -1 & -1 & 1 & 0 & 0 & 1 & -3 \\ -6 & -1 & -1 & 0 & 0 & 1 & 0 & -4 \\ -8 & 0 & -1 & 0 & -1 & 1 & 1 & -5 \\ -10 & -1 & 0 & 0 & -1 & 0 & 1 & -5 \\ -5 & 0 & -1 & 1 & -1 & 0 & 0 & -2 \\ -7 & 0 & 0 & 0 & 0 & 0 & 0 & -4 \end{pmatrix}. \tag{91}$$

This is a direct multiplication-table calculation using (89); it can be checked by multiplying each column b_j by the displayed s. Set $T = -I_8 - S$. Since $s + \bar{s} = -1$, $s\bar{s} = 2$, and alternativity allows repeated s and $\bar{s} = -1 - s$, one has

$$T = B^{-1}R_{\bar{s}}B, \qquad S + T = -I_8, \qquad ST = TS = 2I_8, \qquad S^{\mathsf{T}}G_8S = T^{\mathsf{T}}G_8T = 2G_8. \tag{92}$$

Lemma 7.2. *Both* M, N *are sublattices of* L_{W} *of index* 16 *and minimum norm* 4*, and*

$$2L_{\mathrm{W}} \subset \mathsf{M}, \mathsf{N} \subset L_{\mathrm{W}}, \qquad \mathsf{M} + \mathsf{N} = L_{\mathrm{W}}, \qquad \mathsf{M} \cap \mathsf{N} = 2L_{\mathrm{W}}.$$

Their images $\overline{\mathsf{M}}, \overline{\mathsf{N}}$ *in* $\mathcal{E} = L_{\mathrm{W}}/2L_{\mathrm{W}}$ *are complementary four-dimensional* $\mathbb{F}_2$ *subspaces.*

Proof. The integer matrices S, T give the inclusions in L_{W}; $ST = 2I$ gives the inclusions of $2L_{\mathrm{W}}$. Norm composition shows that each right multiplication scales all squared lengths by $N(s) = N(\bar{s}) = 2$; thus its determinant has absolute value $(\sqrt{2})^8 = 16$, proving the indices and minimum norms. The identity $S + T = -I$ gives $\mathsf{M} + \mathsf{N} = L_{\mathrm{W}}$. Modulo $2L_{\mathrm{W}}$, each image has size $2^8/16 = 2^4$. Their sum is the eight-dimensional space $\mathcal{E}$, so their intersection is zero. This proves the asserted intersection upstairs. □

7.3 The binary glue code and its exact generators

Wilson's original lattice is exactly

$$\Lambda_{\mathrm{W}} = \{(x,y,z) \in L_{\mathrm{W}}^3 : x+y, x+z, y+z \in \mathsf{M}, \quad x+y+z \in \mathsf{N}\}. \tag{93}$$

It carries Wilson's norm q_{W}, as printed. Define the lattice inside the original triality metric by

$$\Lambda_{\mathrm{cyc}} = \iota(\Lambda_{\mathrm{W}}) \subset (V, h_0).$$

The simultaneous use of $\mathsf{M} = L_{\mathrm{W}}\bar{s}$ for pair sums and $\mathsf{N} = L_{\mathrm{W}}s$ for the total sum is essential.

Write any class in $\mathcal{E}$ uniquely as $m + n$, with $m \in \overline{\mathsf{M}}$ and $n \in \overline{\mathsf{N}}$. Then the image of (93) in $\mathcal{E}^3$ is the explicit binary code

$$\mathcal{C} = \{(m_1 + n, m_2 + n, m_1 + m_2 + n) : m_1, m_2 \in \overline{\mathsf{M}},\ n \in \overline{\mathsf{N}}\}. \tag{94}$$

Indeed the pair conditions say that the three N components are equal; the total-sum condition says that the sum of the three M components is zero. This proves both directions of the code description. Its dimension is $4 + 4 + 4 = 12$.

For a concrete lift, the first four columns of S and of T are bases of their images modulo 2. The determinant of the eight-column matrix $(T_1, T_2, T_3, T_4, S_1, S_2, S_3, S_4)$ is -7, so these columns are independent modulo 2; each of the two images already has dimension four. Put $m_j = b_j\bar{s}$, $n_j = b_j s$ for $1 \le j \le 4$. A complete set of 36 integer generators of Λ_{W} is

$$\begin{gathered}(2b_i, 0, 0),\ (0, 2b_i, 0),\ (0, 0, 2b_i) \quad (1 \le i \le 8),\\ (m_j, m_j, 0),\ (m_j, 0, m_j),\ (n_j, n_j, n_j) \quad (1 \le j \le 4).\end{gathered} \tag{95}$$

These vectors are all in (93). Their reductions span precisely (94); the first 24 generate its full kernel $(2L_{\mathrm{W}})^3$. This proves that they generate the entire lattice, not only a sublattice.

Theorem 7.3. *The explicitly generated lattice Λ_{W} with q_{W}, and therefore Λ_{cyc} with the unaltered h_0, is an even unimodular rank-24 lattice of minimum norm 4. In particular it is rootless. This is the actual Wilson Leech construction, through the explicit isometry* (87).

Proof. The code proves $[L_{\mathrm{W}}^3 : \Lambda_{\mathrm{W}}] = 2^{24-12} = 2^{12}$. Since L_{W} has covolume one for its original octonion metric, Λ_{W} has covolume 2^{12} for h_0. Changing to the separately recorded metric q_{W} multiplies volume in dimension 24 by $(1/\sqrt{2})^{24} = 2^{-12}$, so its covolume is exactly one.

For every triple of vectors in a real inner product space,

$$N(x) + N(y) + N(z) = N(x + y) + N(x + z) + N(y + z) - N(x + y + z).$$

For a vector of Λ_{W} the four terms on the right are norms in M or N, so each is divisible by 4: these are twice norms in the even lattice L_{W}. It follows that $q_{\mathrm{W}}(x, y, z)$ is even. Polarization of these even norms gives an integral bilinear form. An integral lattice of covolume one equals its dual, proving unimodularity.

If a nonzero vector had $q_{\mathrm{W}} = 2$, its three coordinate norms would sum to 4. Each nonzero coordinate in L_{W} has norm at least 2, so one coordinate is zero. The other two then lie in M, each of whose nonzero vectors has norm at least 4. At most one can be nonzero. That coordinate also lies in N by the total-sum condition, so it lies in $\mathsf{M} \cap \mathsf{N} = 2L_{\mathrm{W}}$ and has norm at least 8, a contradiction. Hence all nonzero norms are at least 4. The vectors $(2b_i, 0, 0)$ have $q_{\mathrm{W}} = 4$, so the minimum is exactly 4. The isometry ι transfers every statement to the original metric. □

7.4 Cyclic symmetry, the fixed lattice, and exact three-primary glue

Retain the canonical cycle $c(x, y, z) = (z, x, y)$. Denote the opposite cycle used in Wilson coordinates by $c_{\mathrm{W}}(x, y, z) = (y, z, x)$. Direct substitution gives

$$c_{\mathrm{W}} = c^{-1} = c^2, \qquad c_{\mathrm{W}}^2 = c, \qquad \langle c_{\mathrm{W}} \rangle = \langle c \rangle.$$

Both actions are retained with these directions; no action is renamed by silently reversing it. The defining three coordinate memberships, three pair conditions, and total-sum condition are all unchanged by c_{W}; hence both cycles preserve Λ_{W} and Λ_{cyc}. They preserve their metrics and the original cubic:

$$\mathrm{Re}((xy)z) = \mathrm{Re}((yz)x).$$

This follows from the octonion adjoint identities $h(ab,d) = h(b,\bar{a}d) = h(a,d\bar{b})$, which imply real associativity, and $\mathrm{Re}(ab) = \mathrm{Re}(ba)$. In the Albert algebra c_{W} is realized by conjugation with the *real* permutation matrix for $(1\,3\,2)$. It sends the coordinates (x_1, x_2, x_3) to (x_2, x_3, x_1) with no conjugations, and permutes the ordered diagonal frame accordingly. The inverse real permutation matrix, for $(1\,2\,3)$, realizes the canonical c and sends (x_1, x_2, x_3) to (x_3, x_1, x_2). Thus the fixed lattices of the two cycles are identical. For the retained residue isometry $\mathcal{L}r = c\mathcal{L}$, the opposite action satisfies $\mathcal{L}r^{-1} = c_{\mathrm{W}}\mathcal{L}$.

Theorem 7.4. *Before applying ι, the cyclic fixed lattice and its orthogonal complement in Λ_{W} are exactly*

$$F = \{(n,n,n) : n \in \mathsf{N}\}, \qquad C = \{(m_1, m_2, -m_1 - m_2) : m_1, m_2 \in \mathsf{M}\}.$$

Their ranks are 8 and 16. For the retained Wilson metric, and therefore after ι for the original triality metric, their Gram matrices in the displayed bases are

$$G_F = 3G_8, \qquad G_C = \begin{pmatrix} 2G_8 & G_8 \\ G_8 & 2G_8 \end{pmatrix}. \tag{96}$$

The fixed-lattice basis is $(b_j s, b_j s, b_j s)$, $1 \le j \le 8$; the coinvariant bases are $(b_j\bar{s}, 0, -b_j\bar{s})$ and $(0, b_j\bar{s}, -b_j\bar{s})$. In particular the fixed lattice has minimum norm 6 and determinant 3^8. The coinvariant lattice has determinant 3^8, and the map which negates its second L_{W} parameter identifies its displayed Gram matrix with the usual $A_2 \otimes E_8$ Gram matrix.

Proof. A fixed triple has the form (x,x,x). Its pair condition holds for every $x \in L_{\mathrm{W}}$ because $2L_{\mathrm{W}} \subset \mathsf{M}$. Its total condition is $3x \in \mathsf{N}$. As $L_{\mathrm{W}}/\mathsf{N}$ has exponent two, $3x \in \mathsf{N}$ is equivalent to $x \in \mathsf{N}$. This proves the first description. Orthogonality to the whole diagonal real eight-space says $x + y + z = 0$. The pair conditions are then exactly $x, y, z \in \mathsf{M}$, and the total condition is automatic. This proves the description of C.

Right multiplication by either s or $\bar{s}$ multiplies the octonion metric by 2. Thus

$$q_{\mathrm{W}}(as, as, as) = 3N(a),$$

and

$$q_{\mathrm{W}}(a\bar{s}, b\bar{s}, -(a+b)\bar{s}) = N(a) + N(b) + N(a+b) = 2N(a) + 2N(b) + 2h(a,b).$$

Polarization gives both full Gram matrices. Their determinants are

$$\det G_F = 3^8 \det G_8 = 3^8, \qquad \det G_C = \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^8 (\det G_8)^2 = 3^8.$$

Minimum norm 6 for F follows from the exact factor 3 and the minimum 2 in L_{W}. The sign map $b \mapsto -b$ changes both off-diagonal blocks from G_8 to $-G_8$, proving the stated exact comparison to $A_2 \otimes E_8$. □

There is a complete integral comparison, including the glue:

$$0 \longrightarrow F \oplus C \longrightarrow \Lambda_{\mathrm{W}} \xrightarrow{\Sigma} \mathsf{N}/3\mathsf{N} \longrightarrow 0, \qquad \Sigma(x,y,z) = x+y+z \pmod{3\mathsf{N}}. \tag{97}$$

To prove surjectivity, for every $a \in L_{\mathrm{W}}$ the explicit vector

$$v(a) = (as, a, -a)$$

lies in Λ_{W} and has sum as. Its pair sums are $a(s+1) = -a\bar{s} \in \mathsf{M}$, $a(s-1) = -a\bar{s} - 2a \in \mathsf{M}$, and 0; its total lies in N. If a lattice vector has total $3n$, subtracting $(n,n,n) \in F$ leaves a vector of C. This proves the kernel assertion. Thus

$$\Lambda_{\mathrm{W}}/(F \oplus C) \cong \mathsf{N}/3\mathsf{N} \cong L_{\mathrm{W}}/3L_{\mathrm{W}} \cong (\mathbb{Z}/3)^8, \qquad [\Lambda_{\mathrm{W}} : F \oplus C] = 3^8.$$

All maps commute with c_{W} and with its inverse c; both cycles act trivially on the quotient.

The discriminant anti-isometry is also explicit. Use the quadratic discriminant convention $q_D(u+D) = h_D(u,u) \pmod{2\mathbb{Z}}$ for an even lattice D with its retained metric. The orthogonal projections of the lift $v(a)$ are

$$\begin{aligned} f(a) &= \tfrac{1}{3}(as, as, as), \\ \gamma(a) &= \left(\tfrac{2}{3}as,\ a - \tfrac{1}{3}as,\ -a - \tfrac{1}{3}as\right). \end{aligned} \tag{98}$$

They belong to F^* and C^*, respectively: their pairings with F or C are the pairings of the integral vector $v(a)$ with the same lattice, because the other projection is orthogonal. The classes $f(a) + F$ identify $L_{\mathrm{W}}/3L_{\mathrm{W}}$ with F^*/F. If $\gamma(a) \in C$, then $f(a) = v(a) - \gamma(a) \in \Lambda_{\mathrm{W}} \cap F_{\mathbb{R}} = F$, so $a \in 3L_{\mathrm{W}}$. Since C^*/C has order $\det G_C = 3^8$, the map

$$f(a) + F \longmapsto \gamma(a) + C$$

is an isomorphism of the two discriminant groups. Its quadratic sign is determined without an unspecified convention:

$$q_F(f(a)) = \tfrac{1}{3}N(a), \qquad q_C(\gamma(a)) = q_{\mathrm{W}}(v(a)) - \tfrac{1}{3}N(a) = 2N(a) - \tfrac{1}{3}N(a).$$

As $N(a)$ is even, their sum $2N(a)$ is zero modulo $2\mathbb{Z}$. This is the required anti-isometry. Equations (97) and (98) specify the overlattice itself, including its actual coset representatives.

7.5 Finite lattice symmetries and exact cubic tests

Independent sign changes of the three sectors preserve Λ_{W}. Changing the sign of x changes a pair sum by $-2x \in 2L_{\mathrm{W}} \subset \mathsf{M}$ and changes the total by $-2x \in 2L_{\mathrm{W}} \subset \mathsf{N}$. The same argument applies to each sector. All six sector permutations preserve the defining conditions too. Under signs $(\epsilon_1, \epsilon_2, \epsilon_3)$ the cubic τ is multiplied by $\epsilon_1\epsilon_2\epsilon_3$; hence exactly the even sign changes in this subgroup preserve it. Cyclic permutations preserve τ as already proved.

The odd permutation $(x,y,z) \mapsto (y,x,z)$ need not preserve it, even on the lattice itself. In Wilson's basis put

$$a = 2(1+i_0), \qquad b = 2(1+i_1), \qquad d = 2(i_2+i_3).$$

Every coordinate belongs to $2L_{\mathrm{W}}$, so $(a,b,d) \in \Lambda_{\mathrm{W}}$. The real part of $((1+i_0)(1+i_1))(i_2+i_3)$ is -1, because $i_0 i_1 = i_3$ and the i_3^2 term is the only real contribution. Reversing the first two factors changes this contribution to $+1$. Consequently

$$\tau(a,b,d) = -8, \qquad \tau(b,a,d) = 8. \tag{99}$$

After the specified scalar isometry the original determinant cubic has values $-4\sqrt{2}$ and $4\sqrt{2}$.

The simultaneous Fano-index operations

$$A : i_t \longmapsto i_{t+1}, \qquad B_0 : i_t \longmapsto i_{2t}, \qquad i_\infty \longmapsto i_\infty$$

are octonion algebra automorphisms, as the defining oriented Fano products show. They are orthogonal permutations preserving D_8 and the odd-minus half coset, and they fix the displayed s and $\bar{s}$. They consequently preserve $\mathsf{M}, \mathsf{N}, \Lambda_{\mathrm{W}}$. Applied simultaneously to all three sectors, they preserve τ. They satisfy $A^7 = B_0^3 = 1$ and $B_0 A B_0^{-1} = A^2$; their permutations are the distinct maps $t \mapsto 2^j t + k$ with $j = 0, 1, 2$ and $k \in \mathbb{F}_7$. Thus they form an explicitly realized $C_7 \rtimes C_3$ of order 21.

The even sign changes and the cycle c_{W} form $(\mathbb{Z}/2)^2 \rtimes C_3 \cong A_4$. They commute with the simultaneous Fano operations, and their intersection is trivial: a simultaneous Fano operation fixes the real unit in every sector and does not permute sectors. Hence an explicit finite group

$$A_4 \times (C_7 \rtimes C_3)$$

of order 252 preserves both the literal lattice and the original cubic. This is a proved subgroup; no assertion that it is the full cubic stabilizer is needed.

Wilson also gives the lattice symmetries

$$r_t(x, y, z) = (x, yi_t, zi_t) \qquad (t = 0, \ldots, 6).$$

For completeness their lattice action can be checked solely from the explicit finite matrices above. Let $U_t = B^{-1} R_{i_t} B$. The signed permutation multiplication table and (91) give

$$U_t \in GL_8(\mathbb{Z}), \qquad T^{-1}(U_t - I_8) \in M_8(\mathbb{Z}), \qquad \tfrac{1}{2}(U_t - I_8)T \in M_8(\mathbb{Z}) \quad (0 \le t \le 6). \tag{100}$$

These are finite exact matrix assertions, not assumptions about octonionic matrix associativity. The full multiplication rule and the exact rational algorithm checking every entry for all seven values are included in the accompanying `check_wilson.py`. Concretely, start with the 64 products $1 i_t = i_t 1 = i_t$, $i_t^2 = -1$, and the seven oriented Fano triples, form the eight columns $R_{i_t}(e_j)$, conjugate by the explicitly displayed B, and substitute the integer $T = -I_8 - S$. For each of the three matrices in (100) the reduced denominators of every entry are 1, for each $t = 0, 1, 2, 3, 4, 5, 6$. Also $\det U_t = 1$. This is a reproducible $7 \cdot 3 \cdot 64$-entry certificate. The first condition is also immediate from the signed permutation: right multiplication by i_t flips four coordinate signs, preserving the odd-minus parity of L_{W}.

The second matrix assertion says $L_{\mathrm{W}}(i_t - 1) \subset \mathsf{M}$, and the third says $\mathsf{M}(i_t - 1) \subset 2L_{\mathrm{W}}$. Thus $x + yi_t$ and $x + zi_t$ differ from the original pair sums by elements of M. The pair $(y + z)i_t$ remains in M, and the transformed total differs from $x + y + z$ by $(y + z)(i_t - 1) \in 2L_{\mathrm{W}} \subset \mathsf{N}$. All defining conditions are preserved. The inverse follows from $r_t^2(x, y, z) = (x, -y, -z)$ and the proved sign symmetries. Norm composition proves that r_t is an isometry.

Nevertheless none of these seven r_t preserves the original cubic. For $a_t = 2(1 + i_t)$, the triple (a_t, a_t, a_t) lies in $(2L_{\mathrm{W}})^3 \subset \Lambda_{\mathrm{W}}$. All products below occur in the associative real subalgebra generated by i_t, and give

$$\tau(a_t, a_t, a_t) = 8 \operatorname{Re}((1 + i_t)^3) = -16,$$

whereas $(1 + i_t)i_t = -1 + i_t$ and

$$\tau(a_t, a_t i_t, a_t i_t) = 8 \operatorname{Re}((1 + i_t)(-1 + i_t)^2) = 16.$$

Therefore the original determinant cubic on the isometric lattice Λ_{cyc} changes from $-8\sqrt{2}$ to $8\sqrt{2}$. This supplies actual lattice vectors as counterexamples, not only ambient real vectors outside the lattice.

7.6 Exact comparison with coordinate lattices and the marked lane

The construction has the explicit integral sandwich

$$(\sqrt{2}L_{\mathrm{W}})^3 \;\subset\; \Lambda_{\mathrm{cyc}} \;\subset\; (L_{\mathrm{W}}/\sqrt{2})^3, \qquad [\Lambda_{\mathrm{cyc}} : (\sqrt{2}L_{\mathrm{W}})^3] = [(L_{\mathrm{W}}/\sqrt{2})^3 : \Lambda_{\mathrm{cyc}}] = 2^{12}. \tag{101}$$

This is (94) transported by the exact isometry (87). The inclusions are literal inclusions in the fixed three octonion sectors, and each commutes with both c_{W} and the canonical $c = c_{\mathrm{W}}^{-1}$. Together with the explicit bases, the code, and the three-primary glue (98), it determines actual commensurability maps and cosets rather than only an abstract isometry class.

Under (89), the single negative column takes the Wilson odd-minus half roots to even-minus half roots in the current ordered Cayley–Dickson basis. The element s remains exactly

$$s = \tfrac{1}{2}(-1 + \mathsf{i} + \mathsf{j} + \mathsf{k} + \ell + \mathsf{i}\ell + \mathsf{j}\ell - \mathsf{k}\ell),$$

with coordinate vector $(-1,1,1,1,1,1,1,-1)/2$. The signed permutation has determinant $+1$ and all 64 basis products agree, as independently checked in the accompanying code. Thus the lattice, its cyclic action, and the triality cubic live in one specified multiplication convention.

7.7 Literal intersections with both retained lattice embeddings

Let $U = \mathrm{span}_{\mathbb{R}}(1, \mathsf{i}, \mathsf{j}, \mathsf{k})$ be the first four coordinates in the *current* ordered Cayley–Dickson basis, and let

$$D_4 = \{a = (a_0, a_1, a_2, a_3) \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\} \subset U.$$

This quaternionic D_4 denotes the displayed four-coordinate lattice; the original Niemeier D summand below remains in its original place.

Let J_{W} denote the signed permutation matrix in (89). If $U_0 : \mathbb{R}^4 \to \mathbb{R}^8$ is the coordinate inclusion in the current basis, then

$$S^{-1}B^{-1}J_{\mathrm{W}}^{-1}U_0 = \begin{pmatrix} 1/2 & -1/2 & -1/2 & -1/2 \\ -1/2 & 0 & 0 & 1/2 \\ -1 & 1/2 & 0 & 1/2 \\ -3/2 & 1 & 1/2 & 1/2 \\ -2 & 1 & 1 & 1/2 \\ -5/2 & 3/2 & 1 & 1 \\ -5/4 & 3/4 & 3/4 & 3/4 \\ -7/4 & 3/4 & 3/4 & 3/4 \end{pmatrix}. \tag{102}$$

All matrices on the left have already been explicitly specified. Thus the equation is a direct rational multiplication certificate.

Lemma 7.5. *In this fixed coordinate frame,*

$$\mathsf{N} \cap U = 2D_4.$$

Proof. Since $\mathsf{N} \subset L_{\mathrm{W}}$, an element of $\mathsf{N} \cap U$ has integral first four coordinates: the half-integer coset of the current E_8 description has no zero coordinate among its last four. Write those integral coordinates as k_0, k_1, k_2, k_3. Membership in N is exactly integrality of the matrix (102) applied to k. Subtracting its last two rows gives $k_0/2$, so k_0 is even. The second row then forces k_3 even, the third forces k_1 even, and the fourth forces k_2 even. Write $k = 2a$. The first six rows are now integral. The seventh and eighth rows are integral precisely when $-5a_0 + 3a_1 + 3a_2 + 3a_3$ is even, equivalently $a_0 + a_1 + a_2 + a_3$ is even. Conversely these conditions make all eight rows integral. This proves both inclusions with no unstated lattice identification. □

We now retain the original lattices and their actual map $\mathcal{L}$. The original coordinates are $X = (x_{i,j}; d_k)$, $1 \le i \le 4$, $0 \le j \le 5$, $1 \le k \le 4$, with $\sum_j x_{i,j} = 0$. The root lattice is $R = A_5^{\perp 4} \perp D$, with its unchanged Euclidean metric. Its index-72 overlattice N is the inverse image of the original glue

$$(k_1, k_2, k_3, k_4; y) = \big(3u + 2(a+b, a+2b, a, b);\ (u_2+u_4, u_3+u_4)\big),$$

where $u \in \mathbb{F}_2^4$ has even weight and $a, b \in \mathbb{F}_3$; the first four entries are modulo 6. Here $k_i = -6x_{i,0} \pmod 6$ and y is the original $D^\vee/D$ marking, so $d = 0$ has $y = 00$. Retain the original neighbour

$$\rho_A = (5,3,1,-1,-3,-5)/2, \quad \rho_D = (3,2,1,0), \quad \rho = \rho_A^{\oplus 4} \perp \rho_D,$$

$$K = \{X \in N : (\rho, X) \in 6\mathbb{Z}\}, \qquad \alpha = (e_0 - e_1, 0, 0, 0; 0), \qquad \Lambda = K + \mathbb{Z}(\rho/6 - \alpha).$$

Thus $\rho \in N$, $\rho^2 = 84$, $(\rho, \alpha) = 1$. Every vector of N or Λ has rational original coordinates.

For exact use of the retained isometry, set

$$\sigma_i = \sum_{j=0}^{5} (-1)^j x_{i,j}, \qquad t_{i,p,j} = x_{i,p+2j} - \frac{1}{3}\sum_{h=0}^{2} x_{i,p+2h}.$$

Write $u_0 = 1, u_1 = \mathsf{i}, u_2 = \mathsf{j}, u_3 = \mathsf{k}$ and retain the remaining current Cayley–Dickson basis vectors $u_4, \dots, u_7$. The already specified map has the exact coordinate form

$$(\mathcal{L}X)_{j+1,a} = \begin{cases} t_{i,p,j} + \sqrt{2}\,\sigma_{a+1}/6, & 0 \le a \le 3, \\ t_{i,p,j} + \sqrt{3}\,d_{a-3}/3, & 4 \le a \le 7, \end{cases} \qquad a = 2(i-1) + p.$$

The placement of σ_{a+1} is retained, including its difference from the index i of the residual. No coordinate is reassigned.

Theorem 7.6. *The literal intersections in the original metric space are exactly*

$$\Lambda_{\mathrm{cyc}} \cap \mathcal{L}(N) = \Lambda_{\mathrm{cyc}} \cap \mathcal{L}(\Lambda) = \mathcal{I} := \{\sqrt{2}(a,a,a) : a \in D_4 \subset U\}. \tag{103}$$

In original Niemeier coordinates the inverse image of this common lattice is

$$\mathcal{L}^{-1}(\mathcal{I}) = \{x_{i,j} = (-1)^j a_{i-1},\ d = 0 :\ a \in D_4\} \subset K.$$

It has rank four, minimum norm 12*, and Gram matrix*

$$G_{\mathcal{I}} = 6\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}, \qquad \det G_{\mathcal{I}} = 4 \cdot 6^4 = 5184. \tag{104}$$

An explicit basis is $\sqrt{2}(\delta_j, \delta_j, \delta_j)$*, where* $\delta_1 = (1,-1,0,0)$*,* $\delta_2 = (0,1,-1,0)$*,* $\delta_3 = (0,0,1,-1)$*,* $\delta_4 = (0,0,1,1)$*.*

Proof. Each coordinate of Λ_{cyc} belongs to $\mathbb{Q}/\sqrt{2} = \sqrt{2}\mathbb{Q}$ in the current frame. For a rational original vector X, the first four coordinate formulas above therefore force $t_{i,p,j} = 0$ at the corresponding positions, by linear independence of $1, \sqrt{2}$ over $\mathbb{Q}$. The last four force both $t_{i,p,j} = 0$ and $d_{a-3} = 0$, by linear independence of $1, \sqrt{2}, \sqrt{3}$. For the latter independence, a relation with nonzero $\sqrt{3}$ coefficient would place $\sqrt{3}$ in $\mathbb{Q}(\sqrt{2})$; squaring $A + B\sqrt{2}$ forces $AB = 0$ and then would make 3 or

$3/2$ a rational square, a contradiction. All residuals therefore vanish and $d = 0$. The zero fibre sums give

$$x_{i,j} = (-1)^j \sigma_i/6.$$

First let $X \in N$. Its $A_5^\vee$ integral-difference conditions are exactly $k_i := \sigma_i/3 \in \mathbb{Z}$, yielding $x_{i,j} = (-1)^j k_i/2$. Their discriminant labels are $-3k_i = 3k_i \pmod 6$. Their three-primary parts are zero, so the injectivity of $(a, b) \mapsto (a + b, a + 2b, a, b)$ forces $a = b = 0$. Since $d = 0$, the binary condition is $u_2 + u_4 = u_3 + u_4 = 0$ and $\sum u_i = 0$. These equations give $u_1 = u_2 = u_3 = u_4$. Thus all four k_i have the same parity. Conversely these conditions satisfy every original glue requirement. The image is the diagonal triple

$$\mathcal{L}X = (k/\sqrt{2}, k/\sqrt{2}, k/\sqrt{2}), \qquad k \in U.$$

By the fixed-lattice description of Wilson's construction it belongs to Λ_{cyc} exactly when $k \in \mathsf{N}$. The previous lemma makes this equivalent to $k = 2a$, $a \in D_4$, which proves the first intersection equality.

For the second equality, a vector of Λ can be written $X = n + m\rho/6$ with $n \in N$, $m \in \mathbb{Z}$. The already proved necessary condition $d_X = 0$ gives $d_n = -m(3, 2, 1, 0)/6 \in D^\vee$. All coordinates of a vector of $D^\vee$ are integral or all half-integral; the fourth coordinate is zero, so all are integral. Its third coordinate forces $6 \mid m$, and hence $X \in N$. In fact $\Lambda \cap N = K$: in $k + m(\rho/6 - \alpha) \in N$ the pairing with the original root α forces $6 \mid m$, and $(\rho, \rho - 6\alpha) = 84 - 6 = 78 \in 6\mathbb{Z}$. Thus an intersection vector belongs to K. For the diagonal-fibre vectors just found,

$$(\rho, X) = \frac{3}{2} \sum_{i=1}^{4} k_i.$$

Consequently the condition $X \in K$ is $\sum k_i \equiv 0 \pmod 4$. Every $k = 2a$ with $a \in D_4$ satisfies it, proving equality of the two intersections and the displayed inverse image.

Finally $h_0(\sqrt{2}(a, a, a), \sqrt{2}(b, b, b)) = 6h(a, b)$. The four δ_j form a basis of D_4, since their determinant has absolute value two and D_4 has index two in $\mathbb{Z}^4$. Their scalar products give (104). The least nonzero norm in D_4 is two: a norm-one integer vector has odd coordinate sum, while $(1, -1, 0, 0)$ has norm two. This proves the exact minimum and determinant. □

The two slice indices are also literal. Define the common four-dimensional real subspace $\mathcal{U} = \{(u, u, u) : u \in U\}$. The proof gives

$$\mathcal{L}(N) \cap \mathcal{U} = \{(k, k, k)/\sqrt{2} : k \in P\}, \quad P = 2\mathbb{Z}^4 + \mathbb{Z}(1, 1, 1, 1),$$

$$\mathcal{L}(\Lambda) \cap \mathcal{U} = \{(k, k, k)/\sqrt{2} : k \in P_0\}, \quad P_0 = \{k \in P : \textstyle\sum k_i \equiv 0 \pmod 4\} = 2D_4 + \mathbb{Z}(1, 1, 1, 1).$$

The first proof does not need Wilson membership to obtain these slice equations. The lattice P has basis $2e_0, 2e_1, 2e_2, (1, 1, 1, 1)$ and covolume 8. The lattice $2D_4$ has covolume $2^4 \cdot 2 = 32$, so $[P : 2D_4] = 4$. The class of $(1, 1, 1, 1)$ has exact order two modulo $2D_4$, yielding $[P_0 : 2D_4] = 2$. Therefore

$$[\mathcal{L}(N) \cap \mathcal{U} : \mathcal{I}] = 4, \qquad [\mathcal{L}(\Lambda) \cap \mathcal{U} : \mathcal{I}] = 2.$$

The rank-four intersection, strictly below rank 24, also proves that these full lattices are not commensurable *in the retained embedding*: a common finite-index subgroup would have full rank. Their exact existing relation is the common lattice with the explicit embeddings, inverse map, metrics, and indices just proved.

7.8 The cubic value group on the literal common lattice

Retain $F(y) = \tau(y_1, y_2, y_3)$, so the Albert off-diagonal determinant is $2F$. For $y = \sqrt{2}(a, a, a) \in \mathcal{I}$ write $a = a_0 + \mathbf{a}$ in the same quaternionic U. Since $\mathbf{a}^2 = -N(\mathbf{a})$ and a_0 is real, direct multiplication inside this associative subalgebra gives

$$\mathrm{Re}(a^3) = a_0^3 - 3a_0N(\mathbf{a}) = 4a_0^3 - 3a_0N(a).$$

Every scalar and factor is therefore

$$F(\sqrt{2}(a, a, a)) = 2\sqrt{2}(4a_0^3 - 3a_0N(a)). \tag{105}$$

For $a \in D_4$, $N(a)$ is even, because $N(a) \equiv \sum a_i = 0 \pmod 2$. The integer in parentheses is therefore even, so every value belongs to $4\sqrt{2}\mathbb{Z}$. For the actual lattice vector $a = (1, 1, 0, 0)$ it equals $4 - 3 \cdot 2 = -2$, giving $F = -4\sqrt{2}$. Consequently the *additive subgroup generated by all values* is exactly

$$\langle F(\mathcal{I})\rangle_{\mathbb{Z}} = 4\sqrt{2}\mathbb{Z}, \qquad \langle d(Q_{\mathbb{O}}(0; \mathcal{I}))\rangle_{\mathbb{Z}} = 8\sqrt{2}\mathbb{Z}.$$

This assertion concerns the generated additive groups; it does not claim that every element of either group is an individually attained cubic value. For an explicit distinction, put $P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2))$. If $3 \mid P(a)$, reduction modulo three forces $3 \mid a_0$, and substitution $a_0 = 3b$ gives $P(a) = 9b(3b^2 - a_1^2 - a_2^2 - a_3^2)$. Thus $3 \mid P(a)$ implies $9 \mid P(a)$. The value $P(a) = 6$ is impossible, so $F = 12\sqrt{2}$ is not attained, although it belongs to the proved additive value group.

7.9 Primary source and exact reproducibility data

Wilson's octonionic construction is the source of the pair and total conditions [8, Sections 2–4, pp. 2–6]. His page 2 uses the odd-minus lattice L and $R = \bar{L}$; page 3 gives pair sums in $L\bar{s}$, the total in Ls, and the metric equal to one half of the sum of octonion norms. The displayed basis map and (87) retain these conjugations and the exact scale in the present coordinates. The distinct coordinate change on Wilson's page 6 is not substituted for this isometry.

The complete integer-arithmetic program and its output are supplied in the following directory (the two displayed lines form one path):

```
supporting_materials/workbench/evidence/
lattice_foundation_20260906/wilson/
```

The files are `check_wilson.py` and `EXACT_CHECKS.json`. They record the generating-lattice index 4096, Gram determinant 1, even diagonal, integral cyclic action, the fixed and coinvariant Gram matrices, and the index 6561 of their sum. The same program checks all 64 multiplication-table products for (89) and the stated exact cubic tests. The proofs above establish rootlessness and the global descriptions; these conclusions are not inferred from a finite search.

8 The exact cubic image of the original common lattice

Retain the coordinates a_0, a_1, a_2, a_3 and the original lattice

$$D_4 = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\}. \tag{D4I1}$$

The polynomial already used in section 7.8 is

$$P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2)). \tag{D4I2}$$

Here and throughout this section F is the original triality cubic $F = \tau$, with its ordered octonionic arguments as in sections 7 and 7.8. In particular, on the literal common lattice of theorem 7.6,

$$\mathcal{I} = \{\sqrt{2}(a,a,a) : a \in D_4\}, \qquad F(\sqrt{2}(a,a,a)) = 2\sqrt{2}\,P(a). \tag{D4I3}$$

No coordinate permutation, scaling of P, or alteration of the lattice is made. The witnesses below are actual points of this original lattice. The image proof is elementary and requires no external arithmetic existence theorem.

8.1 Necessary restrictions in the original coordinates

For every integer b, one has $b^2 \equiv b \pmod 2$. Consequently, for $a \in D_4$,

$$a_0^2 + a_1^2 + a_2^2 + a_3^2 \equiv a_0 + a_1 + a_2 + a_3 \equiv 0 \pmod 2.$$

Writing $s = a_1^2 + a_2^2 + a_3^2$, the factor $a_0^2 - 3s$ is congruent to $a_0^2 + s \equiv 0 \pmod 2$. Thus $P(a)$ is even.

Modulo three the original polynomial satisfies $P(a) \equiv a_0^3$. If $3 \mid P(a)$, the field $\mathbb{Z}/3\mathbb{Z}$ has no nonzero element with zero cube, so $3 \mid a_0$. Substitution of $a_0 = 3b$ into the original formula gives

$$P(a) = 3b(9b^2 - 3s) = 9b(3b^2 - s). \tag{D4I4}$$

Therefore

$$P(a) \in 2\mathbb{Z}, \qquad 3 \mid P(a) \implies 9 \mid P(a). \tag{D4I5}$$

8.2 Three explicit affine lines and their exact values

Define the common direction

$$v = (-3,1,1,1). \tag{D4I6}$$

Its coordinate sum is zero, and direct substitution gives

$$P(v) = -3\big(9 - 3(1+1+1)\big) = 0.$$

For every $r \in \mathbb{Z}$ and $\varepsilon \in \{-1,1\}$, put

$$L_\varepsilon(r) = (-3r - \varepsilon, r, r, r + \varepsilon) = rv + (-\varepsilon, 0, 0, \varepsilon). \tag{D4I7}$$

Its coordinate sum is $-3r-\varepsilon+r+r+r+\varepsilon = 0$, so it belongs to D_4. The original last-three-coordinate square sum and first-coordinate square are respectively

$$s = r^2 + r^2 + (r+\varepsilon)^2 = 3r^2 + 2r\varepsilon + 1, \qquad (-3r-\varepsilon)^2 = 9r^2 + 6r\varepsilon + 1.$$

Thus the factor in the original cubic is exactly

$$a_0^2 - 3s = (9r^2 + 6r\varepsilon + 1) - (9r^2 + 6r\varepsilon + 3) = -2,$$

and therefore

$$P(L_\varepsilon(r)) = (-3r - \varepsilon)(-2) = 6r + 2\varepsilon. \tag{D4I8}$$

For every $h \in \mathbb{Z}$, define the third line

$$L_0(h) = (-3h, h+1, h-1, h) = hv + (0,1,-1,0). \tag{D4I9}$$

Its coordinate sum is zero, so it also belongs to D_4. Here

$$s = (h+1)^2 + (h-1)^2 + h^2 = 3h^2 + 2,$$

and consequently

$$P(L_0(h)) = (-3h)(9h^2 - 3(3h^2+2)) = (-3h)(-6) = 18h. \tag{D4I10}$$

These identities hold for all signed integer parameters. In particular, $L_0(0) = (0, 1, -1, 0)$ has cubic value zero, and the zero vector also has cubic value zero.

8.3 The global image and a proved reconstruction map

Set

$$S = \{n \in \mathbb{Z} : n \in 2\mathbb{Z} \text{ and } (3 \nmid n \text{ or } 9 \mid n)\}. \tag{D4I11}$$

Theorem 8.1 (Complete cubic image and explicit section)**.** *The original polynomial has the exact image*

$$P(D_4) = 18\mathbb{Z} \ \cup \ (6\mathbb{Z}+2) \ \cup \ (6\mathbb{Z}-2) = S. \tag{D4I12}$$

Equivalently,

$$P(D_4) = \{n \in \mathbb{Z} : n \bmod 18 \in \{0, 2, 4, 8, 10, 14, 16\}\}. \tag{D4I13}$$

The following map is an actual section $\sigma : S \to D_4$:

$$\sigma(n) = \begin{cases} (-3h, h+1, h-1, h), & n = 18h, \\ (-3r-\varepsilon, r, r, r+\varepsilon), & n = 2(3r+\varepsilon), \quad \varepsilon \in \{-1, 1\}. \end{cases} \tag{D4I14}$$

Its displayed parameters are uniquely determined in their respective cases, and $P \circ \sigma = \mathrm{id}_S$.

Proof. Equation (D4I5) proves $P(D_4) \subseteq S$. For the reverse inclusion let $n \in S$. If $3 \mid n$, then $9 \mid n$, and evenness gives $n = 18h$ for a unique $h \in \mathbb{Z}$. Equation (D4I10) gives $P(L_0(h)) = n$.

If $3 \nmid n$, write $n = 2q$. Then $3 \nmid q$, so there is a unique $\varepsilon \in \{-1, 1\}$ with $q \equiv \varepsilon \pmod 3$. The number $r = (q - \varepsilon)/3$ is consequently a unique integer. Equation (D4I8) gives

$$P(L_\varepsilon(r)) = 6r + 2\varepsilon = 2q = n.$$

The two displayed cases are disjoint and cover S, and both witnesses lie in D_4 by section 8.2. These calculations prove the asserted section identity and both inclusions in (D4I12). The first arithmetic progression contributes residue zero modulo eighteen; the second contributes $2, 8, 14$; and the third contributes $16, 4, 10$. This proves (D4I13). □

The classification includes every positive, zero, and negative value. The full excluded set is the union of the odd integers with $18\mathbb{Z} + 6$ and $18\mathbb{Z} + 12$. Let

$$A = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 = 0\} \tag{D4I15}$$

be the literal sum-zero sublattice in the original coordinates. The three witnesses lie in $A \subseteq D_4$, so the two inclusions $S \subseteq P(A) \subseteq P(D_4) = S$ prove

$$P(A) = P(D_4) = S. \tag{D4I16}$$

The image of the original cubic $F = \tau$ on the common lattice is therefore exactly

$$\begin{aligned} F(\mathcal{I}) &= 2\sqrt{2}\, S \\ &= 4\sqrt{2}\big((9\mathbb{Z}) \cup (3\mathbb{Z}+1) \cup (3\mathbb{Z}-1)\big). \end{aligned} \tag{D4I17}$$

Because the original off-diagonal Albert determinant is $2F$, its corresponding image is $4\sqrt{2}\, S$. The additive group generated by that determinant image is $8\sqrt{2}\mathbb{Z}$, while the additive group generated by $F(\mathcal{I})$ is $4\sqrt{2}\mathbb{Z}$. Indeed every P value is even, and the first two lines at parameter zero attain $P = 2$ and $P = -2$. Thus the generated additive group of the P values is exactly $2\mathbb{Z}$; applying the two original scalar factors proves both group assertions without identifying an image set with its generated group.

8.4 Exact local images with explicit local sections

For a prime p, let

$$D_{4,p} = \{a \in \mathbb{Z}_p^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}_p\}. \tag{D4I18}$$

This is precisely $D_4 \otimes_{\mathbb{Z}} \mathbb{Z}_p$ in the original coordinates. To verify the equality directly, use the integer basis

$$\begin{aligned} b_0 &= (1,-1,0,0), \quad b_1 = (0,1,-1,0), \\ b_2 &= (0,0,1,-1), \quad b_3 = (0,0,1,1). \end{aligned} \tag{D4I19}$$

For any a in the displayed definition of $D_{4,p}$, the unique coefficients of these basis vectors are

$$\begin{aligned} c_0 &= a_0, & c_1 &= a_0 + a_1, \\ c_3 &= (a_0 + a_1 + a_2 + a_3)/2, & c_2 &= c_3 - a_3. \end{aligned} \tag{D4I20}$$

All belong to $\mathbb{Z}_p$, and substitution gives the original four coordinates. Conversely a $\mathbb{Z}_p$-linear combination of the four vectors has coordinate sum $2c_3$. These two direct calculations prove the tensor-product identification, including at $p = 2$. Taking integral coefficients gives the same basis description of the original D_4.

The three line identities of section 8.2 hold over $\mathbb{Z}_p$, since they are integer polynomial identities. Each line stays in the sum-zero submodule of $D_{4,p}$.

Theorem 8.2 (All local images and the exact local-to-global statement)**.** *For the unchanged polynomial P,*

$$\begin{aligned} P(D_{4,2}) &= 2\mathbb{Z}_2, \\ P(D_{4,3}) &= \mathbb{Z}_3^{\times} \cup 9\mathbb{Z}_3, \\ P(D_{4,p}) &= \mathbb{Z}_p \qquad (p \geq 5). \end{aligned} \tag{D4I21}$$

An integer n belongs to every local image if and only if $n \in S$, and hence if and only if $n \in P(D_4)$. The global section is exactly (D4I14)*; explicit local sections are given in the proof.*

Proof. For $p = 2$, the parity calculation of section 8.1, reduced modulo two, gives $P(D_{4,2}) \subseteq 2\mathbb{Z}_2$. Conversely, given an arbitrary $n = 2q \in 2\mathbb{Z}_2$, put

$$r = (q-1)/3 \in \mathbb{Z}_2.$$

Division by three is valid since three is a unit in $\mathbb{Z}_2$. Then $L_1(r) \in D_{4,2}$ and

$$P(L_1(r)) = 6r + 2 = 2q = n.$$

Thus $n \mapsto L_1((n/2-1)/3)$ is an explicit section on $2\mathbb{Z}_2$, and proves the first equality.

For $p = 3$, reduction of the original polynomial modulo three and the substitution $a_0 = 3b$ in (D4I4) prove that any value divisible by three belongs to $9\mathbb{Z}_3$. Every unit $n \in \mathbb{Z}_3^\times$ is congruent modulo three to exactly one of 2ε, with $\varepsilon \in \{-1, 1\}$. Set

$$r = (n - 2\varepsilon)/6 \in \mathbb{Z}_3.$$

This quotient is integral because the numerator is divisible by three and two is a unit. The line $L_\varepsilon(r)$ then has P value $6r + 2\varepsilon = n$. Every $n \in 9\mathbb{Z}_3$ can be written uniquely as $18h$ with $h = n/18 \in \mathbb{Z}_3$, again because two is a unit. The third line has $P(L_0(h)) = 18h = n$. These two disjoint cases give the explicit local section

$$n \longmapsto \begin{cases} L_\varepsilon((n-2\varepsilon)/6), & n \in \mathbb{Z}_3^\times, \quad n \equiv 2\varepsilon \pmod 3, \\ L_0(n/18), & n \in 9\mathbb{Z}_3. \end{cases} \tag{D4I22}$$

They prove the second equality in (D4I21).

For every prime $p \geq 5$, six is a unit. Given any $n \in \mathbb{Z}_p$, choose $r = (n-2)/6 \in \mathbb{Z}_p$. The point $L_1(r)$ belongs to $D_{4,p}$ and has $P(L_1(r)) = 6r + 2 = n$. Thus $n \mapsto L_1((n-2)/6)$ is a section on all of $\mathbb{Z}_p$, proving the third equality.

For integers n, the first local condition is precisely evenness; the second is precisely $3 \nmid n$ or $9 \mid n$; and the other primes impose no condition. Membership in every local image is therefore exactly $n \in S$. The explicit integral section in theorem 8.1 proves that those conditions suffice for global integral representability. This proves the local-to-global assertion without approximation or inference from a finite modular calculation. □

8.5 The exact factorization and quotient metric on the sum-zero slice

Define the integral map into the original sublattice A of (D4I15) by

$$\Phi : \mathbb{Z}^3 \longrightarrow A, \qquad \Phi(w, u, v) = (-3w - u - v, w + u, w + v, w). \tag{D4I23}$$

Its coordinate sum is zero. For $a \in A$, its proposed inverse is

$$\Phi^{-1}(a) = (a_3, a_1 - a_3, a_2 - a_3). \tag{D4I24}$$

Substitution returns the original second, third, and fourth coordinates immediately; the first becomes $-a_1 - a_2 - a_3 = a_0$ by the defining equation of A. Substituting Φ into the inverse returns (w, u, v). These two equalities prove that Φ is an exact integral-module isomorphism.

For $a = (a_0, y, z, w) \in A$, put $T = y + z + w = -a_0$. Direct expansion of the original polynomial gives

$$\begin{aligned} P(a) &= -T(T^2 - 3(y^2 + z^2 + w^2)) \\ &= 2T(y^2 + z^2 + w^2 - yz - yw - zw) \\ &= T((y-z)^2 + (y-w)^2 + (z-w)^2). \end{aligned} \tag{D4I25}$$

Under Φ, one has $T = 3w + u + v$, and the three differences are $u - v, u, v$. Consequently

$$P(\Phi(w, u, v)) = 2(3w + u + v)(u^2 - uv + v^2). \tag{D4I26}$$

This identity explains the three original lines without changing their coordinates. On $L_\varepsilon(r)$, the quotient differences are $(u, v) = (-\varepsilon, -\varepsilon)$, so $u^2 - uv + v^2 = 1$; on $L_0(h)$, they are $(u, v) = (1, -1)$, so $u^2 - uv + v^2 = 3$. The original first coordinates remain $-3r - \varepsilon$ and $-3h$, respectively.

The exact quotient map is

$$\chi : A \longrightarrow \mathbb{Z}^2, \qquad \chi(a) = (a_1 - a_3, a_2 - a_3). \tag{D4I27}$$

It is onto because $\chi\Phi(0,u,v) = (u,v)$. An element lies in its kernel exactly when its last three coordinates all equal a common integer w; its first coordinate is then $-3w$. Thus $\ker\chi = \mathbb{Z}(-3,1,1,1)$. The section $(u,v) \mapsto \Phi(0,u,v)$ proves the split exact sequence

$$0 \longrightarrow \mathbb{Z}(-3,1,1,1) \longrightarrow A \xrightarrow{\chi} \mathbb{Z}^2 \longrightarrow 0. \tag{D4I28}$$

For $\omega = (-1+\sqrt{-3})/2$, one has $\omega + \overline{\omega} = -1$ and $\omega\overline{\omega} = 1$. Direct multiplication therefore proves

$$(u + v\omega)(u + v\overline{\omega}) = u^2 - uv + v^2. \tag{D4I29}$$

The quadratic factor in the original cubic is exactly this Eisenstein norm, with the preceding coefficient two unchanged.

The original Euclidean metric on these same coordinates is, by expansion,

$$\|\Phi(w,u,v)\|^2 = 12w^2 + 8w(u+v) + 2(u^2 + uv + v^2). \tag{D4I30}$$

Write $d = (-3,1,1,1)$ for the direction when used in the metric calculation. Its squared norm is twelve, and the exact pairing is

$$\langle \Phi(w,u,v), d\rangle = 12w + 4(u+v). \tag{D4I31}$$

Hence orthogonal projection away from the real line $\mathbb{R}d$ is given in the original coordinates by

$$\begin{aligned} \pi\Phi(w,u,v) &= \Phi(w,u,v) - \left(w + \frac{u+v}{3}\right) d \\ &= \left(0, \frac{2u-v}{3}, \frac{2v-u}{3}, -\frac{u+v}{3}\right). \end{aligned} \tag{D4I32}$$

This depends only on the quotient coordinate $\chi(a) = (u,v)$. Its squared norm is

$$\begin{aligned} \|\pi\Phi(w,u,v)\|^2 &= \frac{(2u-v)^2 + (2v-u)^2 + (u+v)^2}{9} \\ &= \frac{2}{3}(u^2 - uv + v^2). \end{aligned} \tag{D4I33}$$

The quotient metric on $A/\mathbb{Z}d$, realized by this exact projection, thus has Gram matrix

$$\frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{D4I34}$$

in the displayed quotient coordinates (u,v). The retained common-lattice map $a \mapsto \sqrt{2}(a,a,a)$ multiplies every squared norm by six, as proved in theorem 7.6. Consequently the corresponding quotient Gram matrix on the image of A in $\mathcal{I}$ is

$$\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}. \tag{D4I35}$$

This is the exact quotient-metric calculation on the indicated rank-three slice. It neither identifies that slice with the full D_4 nor changes the original common-lattice metric.

9 The marked Niemeier lattice, its twelve theta indices, and the umbral shadow

The original objects are

$$A = \{(x_0, \dots, x_5) \in \mathbb{Z}^6 : \sum_j x_j = 0\}, \qquad D = \{d \in \mathbb{Z}^4 : \sum_i d_i \equiv 0 \pmod 2\}, \qquad R = A^{\perp 4} \perp D.$$

The metrics are the displayed Euclidean metrics. The four copies of A retain their original fibre order $f_i 289^j$ for $0 \le j < 6$, with fibre labels $(00, 10, 01, 11)$. Their discriminant generator is $\lambda = (5, -1, -1, -1, -1, -1)/6$. The discriminant of D retains $10 \mapsto v = (1, 0, 0, 0)$, $01 \mapsto s = (1, 1, 1, 1)/2$, and $11 \mapsto c = (-1, 1, 1, 1)/2$. Consequently the original norm-valued quadratic refinement, with codomain $\mathbb{Q}/2\mathbb{Z}$, is

$$q_R(k_1, k_2, k_3, k_4; y) = \frac{5}{6} \sum_i k_i^2 + \mathbf{1}_{y \neq 0} \pmod{2\mathbb{Z}}.$$

The original glue and Niemeier lattice are

$$\begin{aligned} E &= \{u \in \mathbb{F}_2^4 : \sum_i u_i = 0\}, \qquad \tau(u) = (u_2 + u_4, u_3 + u_4), \\ j(a, b) &= (a + b, a + 2b, a, b), \qquad C_3 = j(\mathbb{F}_3^2), \\ H &= \{(3u + 2j(a, b); \tau(u)) : u \in E,\ (a, b) \in \mathbb{F}_3^2\}, \\ N &= \{x \in R^\vee : x + R \in H\}. \end{aligned}$$

Thus $|H| = 8 \cdot 9 = 72$ and $N/R = H$, with the root system $X = A_5^4 D_4$ proved in the retained lattice calculation. In particular, no norm-two vectors are introduced by the nonzero glue classes: their original minima are four or six. We use this proved literal root system, rather than a rank-only identification, throughout the construction below.

9.1 The primary prescription and its complete coefficient matrix

Write $e(t) = \exp(2\pi i t)$ and $q = e(\tau)$ for $\tau \in \mathbb{H}$. Cheng–Duncan–Harvey, arXiv:1307.5793v2, Section 3.2 [1, Sections 3.2–3.3], source labels `thetexp`, `transf_theta`, and `transf_theta_2`, use

$$\theta_{m,r}(\tau, z) = \sum_{k \equiv r \ (2m)} q^{k^2/(4m)} e(kz), \qquad r \in \mathbb{Z}/(2m).$$

Their Section 3.3, source label `def:OmegaMatrices`, defines

$$\Omega_m(n)_{r,s} = \begin{cases} 1, & r + s \equiv 0 \pmod{2n}, \quad r - s \equiv 0 \pmod{2m/n}, \\ 0, & \text{otherwise}, \end{cases} \qquad n \mid m.$$

Their ADE table, source label `ADE1`, assigns

$$\Omega^{A_{m-1}} = \Omega_m(1), \qquad \Omega^{D_{m/2+1}} = \Omega_m(1) + \Omega_m(m/2),$$

and extends by addition over components with the same Coxeter number. These are the primary authors' prescribed maps. The following calculation specializes and proves their values in the present marked coordinates; it does not assert a new ADE classification.

The six-cycle $e_j \mapsto e_{j+1 \bmod 6}$ on an A fibre has characteristic polynomial $(x^6-1)/(x-1)$ on its augmentation space. Its order is six. In D, take the original simple roots $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$, $\alpha_3 = e_3 - e_4$, $\alpha_4 = e_3 + e_4$. Their reflection product $s_{\alpha_1} s_{\alpha_2} s_{\alpha_3} s_{\alpha_4}$ sends

$$e_1 \mapsto e_2, \quad e_2 \mapsto e_3, \quad e_3 \mapsto -e_1, \quad e_4 \mapsto -e_4.$$

It has order six and characteristic polynomial

$$(x^3+1)(x+1) = (x^2-x+1)(x+1)^2 = \frac{(x^2-1)(x^6-1)}{(x-1)(x^3-1)}.$$

Thus the original five components have Coxeter number $h = m = 6$, and the primary theta prescription has precisely $2h = 12$ indices and denominator $4h = 24$ in its exponents. The root rank remains $4 \cdot 5 + 4 = 24$; the glue index remains 72.

Put $J = \Omega_6(3)$. Its two congruences are

$$r + s \equiv 0 \pmod 6, \qquad r - s \equiv 0 \pmod 4.$$

The simultaneous solutions modulo twelve consist of the single value $s \equiv 5r \pmod{12}$: this value satisfies both congruences, and the difference of any two solutions is divisible by both six and four, hence by twelve. Therefore

$$\boxed{\Omega^X = 5I_{12} + J, \qquad J_{r,s} = \mathbf{1}_{s \equiv 5r \ (12)}.}$$

In the full original index order $0, 1, \ldots, 11$ this is

$$\Omega^X = \begin{pmatrix}
6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 5 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 5
\end{pmatrix}.$$

The unary series and the actual shadow are

$$S_{6,r}(\tau) = \frac{1}{2\pi \mathrm{i}} \frac{\partial \theta_{6,r}}{\partial z}\Big|_{z=0} = \sum_{k \equiv r \ (12)} k q^{k^2/24}, \qquad S^X = \Omega^X S_6.$$

All series and their derivatives converge normally on compact subsets of their stated domains, by Gaussian decay. Substitution $k \mapsto -k$ proves $S_{6,-r} = -S_{6,r}$, so $S_{6,0} = S_{6,6} = 0$. Let $\mathcal{A} : \mathbb{C}^5 \to \mathbb{C}^{12}$ insert zero in positions zero and six and put $(\mathcal{A}u)_{12-r} = -u_r$ for $1 \le r \le 5$. This retains, and explicitly reconstructs, all twelve components. Then

$$\Omega^X \mathcal{A} = \mathcal{A} M^X, \qquad \boxed{M^X = \begin{pmatrix}
5 & 0 & 0 & 0 & 1 \\
0 & 4 & 0 & 0 & 0 \\
0 & 0 & 6 & 0 & 0 \\
0 & 0 & 0 & 4 & 0 \\
1 & 0 & 0 & 0 & 5
\end{pmatrix}.}$$

Indeed its (r,s) entry is $\Omega^X_{r,s} - \Omega^X_{r,12-s}$; direct substitution gives the matrix. In particular the fourth A-type multiplicity and the full D-type contribution have both been retained. The diagonal entries $(5,4,6,4,5)$ are exactly the Coxeter exponent multiplicities: four copies of $(1,2,3,4,5)$ and one copy of $(1,3,3,5)$.

9.2 Quadratic refinements and the finite theta transformations

Let $L = \mathbb{Z}\ell$ with $(\ell,\ell) = 12$. Its dual is $L^\vee = \mathbb{Z}\ell/12$. Hence

$$A_L = L^\vee/L = \mathbb{Z}/12, \qquad q_L(r) = r^2/12 \pmod{2\mathbb{Z}}, \qquad Q_L(r) = q_L(r)/2 = r^2/24 \pmod{\mathbb{Z}},$$

and the associated bilinear form is $B_L(r,s) = rs/12 \pmod{\mathbb{Z}}$. The symbols q_L and Q_L deliberately have different codomains and retain the factor two.

Define the unitary Fourier matrices and reflection by

$$(F_\pm)_{r,s} = 12^{-1/2} e(\pm rs/12), \qquad R_{r,s} = \mathbf{1}_{s \equiv -r \ (12)}.$$

Finite geometric-series summation gives

$$F_\pm^2 = R, \qquad F_- = RF_+ = F_+R, \qquad F_+F_- = I.$$

The substitution $k \mapsto k+12$ shows that $e(k^2/24)$ depends only on $r = k \bmod 12$. Thus, with $D_{r,s} = e(r^2/24)\delta_{r,s}$,

$$\theta_6(\tau+1, z) = D\theta_6(\tau, z).$$

Poisson summation of the Gaussian gives

$$\theta_6(-1/\tau, z/\tau) = \sqrt{-\mathrm{i}\tau}\, e(6z^2/\tau) F_- \theta_6(\tau, z),$$
$$\theta_6(-1/\tau, -z/\tau) = \sqrt{-\mathrm{i}\tau}\, e(6z^2/\tau) F_+ \theta_6(\tau, z).$$

Here $\sqrt{-\mathrm{i}\tau}$ is the holomorphic square root positive when τ is positive imaginary. For completeness, apply the Fourier transform convention $\widehat{f}(u) = \int_{\mathbb{R}} f(x) e(-ux)\, dx$ to $f(x) = \exp(\pi \mathrm{i}\tau x^2/12 + 2\pi \mathrm{i} z x)$. Completing the square evaluates its transform; the real positive Gaussian integral is obtained by squaring the integral and using polar coordinates, and holomorphic continuation gives the expression for $\Im\tau > 0$. Periodizing $f(r+12x)$, its n-th Fourier coefficient is $12^{-1} e(nr/12) \widehat{f}(n/12)$. Summing these coefficients at zero and grouping n by its residue modulo twelve gives the first displayed transformation. Absolute Gaussian convergence justifies the integration, grouping, and Fourier evaluation. Replacing z by $-z$ and using $\theta_{6,-r}(\tau, z) = \theta_{6,r}(\tau, -z)$ gives the second. The second, with its plus-sign Fourier matrix, is precisely CDH's convention; the first uses the usual generator $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ with elliptic argument z/τ.

The matrices carrying the finite multiplier for this first convention are

$$U(T) = D, \qquad U(S) = e(-1/8) F_-.$$

The square-root weight factor has not been included in U. Differentiation of the theta transformation at $z = 0$ gives

$$S_6(-1/\tau) = \tau \sqrt{-\mathrm{i}\tau} F_- S_6(\tau), \qquad S_6(\tau+1) = D S_6(\tau).$$

Equivalently, CDH's reflected convention gives $S_6(-1/\tau) = -\tau\sqrt{-\mathrm{i}\tau} F_+ S_6(\tau)$. The equality follows from $RS_6 = -S_6$. Thus the unary series has weight $3/2$ with the same finite multiplier; theta has weight $1/2$.

The permutation J preserves Q_L because $((5r)^2 - r^2)/24 = r^2 \in \mathbb{Z}$. Since $5^{-1} = 5$ modulo twelve,

$$(JF_\pm)_{r,s} = 12^{-1/2} e(\pm 5rs/12) = (F_\pm J)_{r,s}.$$

Therefore Ω^X commutes with both finite multiplier matrices. The shadow transformation follows by applying this proved intertwiner. Its component series vanish at infinity since every nonzero summand has strictly positive exponent; the modular transformation shows the same cuspidal behavior at every cusp of the full modular group.

There is also an exact comparison with the original A5 discriminant, without replacing its form. Reduction

$$p : \mathbb{Z}/12 \longrightarrow \mathbb{Z}/6, \qquad p(r) = r \bmod 6$$

has kernel $\{0, 6\}$ and satisfies

$$q_A(p(r)) = \frac{5}{6} r^2 = 10 q_L(r) \pmod{2\mathbb{Z}}.$$

Thus it is a surjective quadratic similitude with precisely multiplier ten, and its bilinear forms satisfy the same multiplier relation. It is not asserted to be an isometry. Applying p to four coordinates and the identity to the retained D discriminant gives

$$\widetilde{D} = (\mathbb{Z}/12)^4 \oplus \mathbb{F}_2^2 \longrightarrow D_R, \qquad (r; y) \longmapsto (p(r_1), \dots, p(r_4); y).$$

The pullback quadratic refinement is exactly $10 \sum_i q_L(r_i) + \mathbf{1}_{y \neq 0}$, its radical is $\{0, 6\}^4 \oplus 0$, and its quotient by that radical is the original (D_R, q_R). To prove the radical claim, pairing with each coordinate generator forces $5r_i/6 \in \mathbb{Z}$, hence $6 \mid r_i$; the nondegenerate D form forces $y = 0$. The vectors so obtained have quadratic value 30 in each nonzero coordinate, hence zero modulo two. The inverse image of the original H has order $16 \cdot 72 = 1152$ and is isotropic for this pullback form. This gives the exact relation of the theta indices to all original glue coordinates, with the form and multiplier specified on both sides.

9.3 The full Fourier determinant and the retained order-twelve character

Let $\chi : \mathrm{SL}_2(\mathbb{Z}) \to \mathbb{Z}/12$ be the retained marking

$$\chi(T) = -1, \qquad \chi(S) = 3.$$

These values respect the presentation $S^4 = I$, $(ST)^3 = S^2$: in the abelian target the required relations are $4 \cdot 3 = 0$ and $3(3 - 1) = 2 \cdot 3$ modulo twelve. They therefore define a character, and its value at T makes it onto. We now calculate the determinant of the entire twelve-dimensional finite theta representation, rather than that of a reduced component list.

Put $\zeta = e(1/12)$. The Vandermonde determinant gives

$$\det F_+ = 12^{-6} \prod_{0 \le j < k \le 11} (\zeta^k - \zeta^j).$$

For every such pair,

$$\zeta^k - \zeta^j = 2\mathrm{i} \sin\!\left(\frac{\pi(k-j)}{12}\right) \exp\!\left(\frac{\pi \mathrm{i}(j+k)}{12}\right),$$

whose sine is strictly positive. There are $\binom{12}{2} = 66$ pairs, and each index occurs in eleven of them, so

$$\sum_{j<k} (j + k) = 11 \sum_{j=0}^{11} j = 11 \cdot 66 = 726.$$

The phase of the determinant is consequently

$$\mathrm{i}^{66}\exp(726\pi\mathrm{i}/12) = (-1)\mathrm{i} = -\mathrm{i}.$$

Its remaining factor is positive. Unitarity, proved by the finite geometric sum above, makes its absolute value one. It follows exactly that

$$\boxed{\det F_+ = -\mathrm{i}, \qquad \det F_- = \mathrm{i}.}$$

In particular retaining CDH's reflection changes the Fourier determinant by $\det R = (-1)^5 = -1$; it cannot be dropped.

The sum of the squares of all twelve indices is

$$\sum_{r=0}^{11} r^2 = \frac{11\cdot 12\cdot 23}{6} = 506.$$

Consequently

$$\boxed{\det U(T) = e(506/24) = e(1/12), \qquad \det U(S) = e(-12/8)\det F_- = -\mathrm{i} = e(-3/12).}$$

For an explicit check that the finite matrices have the relevant metaplectic relations, let $G = \sum_{r=0}^{11} e(r^2/24)$. Listing the squares modulo twenty-four gives

$$G = 1 + 4e(1/24) + 2e(1/6) + 2e(3/8) + 2e(2/3) + e(1/2).$$

The first and last terms cancel, as do the terms with arguments $1/6$ and $2/3$. The elementary sine and cosine values at $\pi/12$ and $3\pi/4$ give

$$4e(1/24) + 2e(3/8) = \sqrt{6}(1+\mathrm{i}) = \sqrt{12}\,e(1/8).$$

Completing the square in the finite sum proves

$$(F_-D)^2 = e(1/8)D^{-1}F_-, \qquad (F_-D)^3 = e(1/8)R.$$

For the first equality, its (r,s) entry before evaluation is $12^{-1}e(s^2/24)\sum_t e(t^2/24 - t(r+s)/12)$; shifting t by $r+s$ produces exactly the stated Gauss sum and matrix entry. The second follows by multiplication and $F_-^2 = R$. It follows that

$$(U(S)U(T))^3 = U(S)^2 = -\mathrm{i}R, \qquad U(S)^4 = -I_{12}.$$

Taking determinants kills this metaplectic central sign because the dimension twelve is even. The determinant therefore descends to an ordinary character of $\mathrm{SL}_2(\mathbb{Z})$. Its two computed generator values, together with generation by S, T, prove

$$\boxed{\det U(\gamma) = e(-\chi(\gamma)/12).}$$

The dual determinant is its inverse and is exactly $e(\chi(\gamma)/12)$, with the retained signs $T \mapsto e(-1/12)$, $S \mapsto \mathrm{i}$.

This is a statement about the determinant multiplier, with its domain and codomain specified. It is not a statement that a vector of weight $1/2$ has scalar weight one after taking a determinant. The separate Dedekind-eta identities

$$\eta(\tau+1) = e(1/24)\eta(\tau), \qquad \eta(-1/\tau) = \sqrt{-\mathrm{i}\tau}\,\eta(\tau)$$

show that η^2 has weight one and multiplier $\det U$, whereas η^{-2} has weight minus one and multiplier $(\det U)^{-1}$. Indeed $\eta^2(S\tau) = (-\mathrm{i})\tau\eta^2(\tau)$ and $\eta^{-2}(S\tau) = \mathrm{i}\tau^{-1}\eta^{-2}(\tau)$. The integer-weight factors $(c\tau+d)^{\pm 1}$ obey the cocycle identity by matrix multiplication, so the two generator formulas prove the statements for every γ. At $-I$ the character and odd integer weight each contribute a minus sign, whose product is one.

9.4 The residue Fourier map and what the Weyl quotient retains

The original marking is $\kappa(289^j) = j \pmod 6$, with the full ordered residue list

$$(289^j)_{j=0}^{5} = (1, 289, 361, 169, 121, 529) \pmod{840}.$$

On the six-dimensional complex coordinate carrier define

$$E_2 : \mathbb{C}[S_{840}] \longrightarrow \mathbb{C}[\mathbb{Z}/12], \qquad E_2(e_{289^j}) = e_{2j},$$

and on a separately indexed $\mathbb{C}^6$ define

$$B(e_k) = \frac{e_k + e_{k+6}}{\sqrt{2}}, \qquad 0 \le k < 6.$$

Both are injective and preserve the indicated coordinate Hermitian products. Retain the original negative-sign normalized Fourier matrix $F_6^- = (6^{-1/2} e(-kj/6))_{k,j}$. Entry-by-entry calculation gives

$$\boxed{F_- E_2 = B F_6^-.}$$

Indeed the (r, j) entry of the left side is $12^{-1/2} e(-rj/6)$, and the right side has the same value, using r mod 6. Thus the original unnormalized sample transform is carried into this formula by its exact factor $\sqrt{6}$. Translation $j \mapsto j + 1$ intertwines under E_2 with translation $r \mapsto r + 2$ modulo twelve. This gives a literal Fourier and translation map, not an identification of different quadratic forms. In particular $Q_L(6) = 3/2 = 1/2 \pmod{\mathbb{Z}}$, so the quotient $\mathbb{Z}/12 \to \mathbb{Z}/6$ does not carry Q_L itself to a quadratic form on the quotient. Its exact multiplier-ten comparison with the original A5 form was proved above.

For the original lattice action, $\rho(289)$ is the same six-cycle in each of the four A coordinate sets and is the identity on D. Each coordinate transposition of A is reflection in the corresponding root $e_i - e_j$, so the full residue action is in $W(R) = W(A)^4 \times W(D)$. It acts trivially on every A discriminant because $Q_\sigma \lambda - \lambda = e_{\sigma(0)} - e_0 \in A$, and it fixes the D discriminant. Consequently

$$S_{840} \xrightarrow{\rho} O(N) \longrightarrow O(N)/W(R)$$

is the trivial homomorphism. This proves exactly which residue-group action is killed, without discarding its already constructed Fourier map.

The original marked quotient is

$$G = O(N)/W(R) \cong \mathrm{GL}_2(\mathbb{F}_3).$$

Here its actual coordinate map uses $\ell_1 = (1, 1)$, $\ell_2 = (1, 2)$, $\ell_3 = (1, 0)$, $\ell_4 = (0, 1)$. For $A \in \mathrm{GL}_2(\mathbb{F}_3)$ determine the unique signs and permutation by

$$\ell_i A = \varepsilon_i \ell_{p(i)}, \qquad (P_p u)_i = u_{p(i)}, \qquad B_p \tau(u) = \tau(P_p u).$$

The four row lines exhaust the projective line, so the signs and permutation exist uniquely. The binary equation defines B_p uniquely since τ is onto. Then $(k; y) \mapsto (\varepsilon_i k_{p(i)}; B_p y)$ preserves precisely the original two glue codes, and its ternary restriction is $M_g j = jA$. Conversely a marked glue stabilizer gives this same A. Multiplication of these identities proves the group isomorphism, with the original row/column convention retained.

To compare its chosen signed lifts with CDH's simple-root permutation section, let $Q_{\mathrm{rev}} e_j = e_{5-j}$ in an A fibre. When $\varepsilon_i = -1$, replace the representative $-I$ by $-Q_{\mathrm{rev}}$; their quotient is $Q_{\mathrm{rev}} \in W(A)$

and $-Q_{\mathrm{rev}}(e_j - e_{j+1}) = e_{4-j} - e_{5-j}$. Thus the resulting map permutes the original simple roots by diagram reversal. When $\varepsilon_i = 1$ no change is needed. For D, put $\Delta = (\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4)$. Associate the original classes v, s, c with outer nodes $1, 4, 3$, respectively; their fundamental weights are e_1, $(1,1,1,1)/2$, and $(1,1,1,-1)/2$, the last of which differs from the retained c by $(1,0,0,-1) \in D$. For B_p, let P_{B_p} permute these outer nodes exactly as B_p permutes v, s, c and fix node two. Then

$$d_{B_p} = \Delta P_{B_p} \Delta^{-1}$$

is the exact original-coordinate diagram representative. Its orthogonality follows because permuting the outer nodes preserves the original Gram matrix. It preserves D, and its action on fundamental weights proves its discriminant action is exactly B_p. Its quotient with the originally chosen D lift has trivial discriminant action, hence belongs to the proved Weyl kernel $W(D)$. This specifies the actual change of section on every summand.

Define the following four class functions using those retained maps:

$$a_g = \#\{i : p(i) = i\}, \quad b_g = \sum_{p(i)=i} \varepsilon_i, \quad s_g = \mathrm{sgn}(B_p|_{\{v,s,c\}}), \quad c_g = \#\,\mathrm{Fix}(B_p|_{\{v,s,c\}}) - 1.$$

The first two are the traces on the four-component permutation and signed permutation representations. The last is the trace on the two-dimensional sum-zero subspace of the three outer-node permutation representation, and s_g is its sign character. Thus these are precisely CDH's $(\bar{\chi}^{X_A}, \chi^{X_A}, \chi^{X_D}, \check{\chi}^{X_D})$; the remaining $\bar{\chi}^{X_D}$ is one because the single D component is fixed. The comparison of sections above proves this identification for the actual marked quotient.

CDH Section 5.1 [1, Section 5.1] defines the full twisted coefficient matrix by

$$(\Omega_g^{X_A})_{r,s} = \big(b_g \mathbf{1}_{r \text{ even}} + a_g \mathbf{1}_{r \text{ odd}}\big)\delta_{r,s},$$

and, for its exceptional $m = 6$ D4 case, by

$$(\Omega_g^{X_D})_{r,s} = \begin{cases} c_g\delta_{r,s}, & r \equiv 0,3 \pmod 6, \\ \delta_{r,s} + s_g J_{r,s}, & r \equiv 1,5 \pmod 6, \\ s_g\delta_{r,s} + s_g J_{r,s}, & r \equiv 2,4 \pmod 6. \end{cases}$$

All deltas in this formula have indices modulo twelve. The exact maps from the marked glue quotient to the prescribed twisted shadows are

$$g \longmapsto \Omega_g^X = \Omega_g^{X_A} + \Omega_g^{X_D}, \qquad S_g^X = \Omega_g^X S_6.$$

They are class-function constructions from traces of representations; the assignment $g \mapsto \Omega_g^X$ is not asserted to be a group representation. On the odd five-component presentation the explicit matrix is

$$\boxed{M_g^X = \begin{pmatrix} a_g+1 & 0 & 0 & 0 & s_g \\ 0 & b_g & 0 & 0 & 0 \\ 0 & 0 & a_g+c_g & 0 & 0 \\ 0 & 0 & 0 & b_g & 0 \\ s_g & 0 & 0 & 0 & a_g+1 \end{pmatrix}.}$$

This follows by the same signed component reconstruction $\mathcal{A}$: in rows two and four the two D terms cancel, in row three the contribution is c_g, and rows one and five pair with one another. For

reference, all possible character tuples are

$[g]$	a_g	b_g	s_g	c_g	number of elements
$1A$	4	4	1	2	1
$2A$	4	-4	1	2	1
$2B$	2	0	-1	0	12
$4A$	0	0	1	2	6
$3A$	1	1	1	-1	8
$6A$	1	-1	1	-1	8
$8A, 8B$	0	0	-1	0	12

with the CDH class names of their table `tab:chars:eul:6`. The formulas apply to each actual A, so their validity does not depend on a choice of conjugacy-class names. The finite checker obtains the table from all 48 invertible matrices over $\mathbb{F}_3$ and the original row forms.

The distinguished residue action has quotient $g = 1$ and therefore produces the untwisted shadow, while a product $\rho(r)L_g$ produces the same S_g^X for every r. The four-component signed action and the D4 triality quotient survive and enter the displayed matrix. The central element $-I \in \mathrm{GL}_2(\mathbb{F}_3)$, for example, changes b_g from four to minus four while fixing a_g, s_g, c_g; it changes the even-index odd-space entries with exactly that sign.

The already marked composite

$$\mathrm{SL}_2(\mathbb{Z}) \xrightarrow{\chi} \mathbb{Z}/12 \longrightarrow \mathbb{Z}/6 \xrightarrow{j\mapsto 289^j} S_{840}$$

is onto, has kernel $\chi^{-1}(\{0,6\})$, and sends T to $289^5 = 529$. Following it by the Weyl quotient kills it, by the proved residue calculation. Following χ into the dual determinant multiplier retains its complete order twelve, by the determinant theorem. These are explicit different outgoing morphisms from the same marking, including their kernels and signs.

9.5 The shadow-to-Fricke map with the original scalar seventy-two

The Coxeter calculation above gives the original formal frame exponent

$$\pi_X = -5[1] + [2] - [3] + 5[6].$$

Let $\mathcal{E}(\sum a_d[d]) = \prod_d \eta(d\tau)^{a_d}$. It is a group homomorphism from addition of formal exponents to multiplication of nonzero holomorphic functions, by the exponent laws. The retained scalar function is

$$\boxed{T_6(\tau) = \mathcal{E}(-\pi_X) = \frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5}.}$$

The minus sign on π_X is essential: $\mathcal{E}(\pi_X) = T_6^{-1}$.

Here is a direct coefficient map from the actual shadow matrix to this eta function. Its diagonal difference vector is $\alpha = (5, 4, 6, 4, 5)$, and set $\alpha_0 = 0$ modulo six. Define the Lambert-series map at this marked matrix by

$$\mathscr{L}(\Omega^X)(\tau) = 1 + \sum_{n\geq 1} \alpha_{n \bmod 6} \frac{nq^n}{1-q^n}.$$

Convergence is absolute and locally uniform since $|q| < 1$ and α is bounded. The coefficient of q^N is exactly $\sum_{d|N} d\alpha_{d \bmod 6}$; thus this specifies every coefficient, not only its initial terms. For all positive integers n the retained multiplicities satisfy

$$\alpha_{n \bmod 6} = 5 - \mathbf{1}_{2|n} + \mathbf{1}_{3|n} - 5\mathbf{1}_{6|n}.$$

Checking the six residues proves this identity. The product definition $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ and termwise logarithmic differentiation then give

$$\boxed{f^X := \mathscr{L}(\Omega^X) = -q\frac{d}{dq}\log T_6 = 5\lambda_6 + \lambda_2 - \lambda_3,}$$

where the original functions on the right are

$$\lambda_d = q\frac{d}{dq}\log\frac{\eta(d\tau)}{\eta(\tau)} = \frac{d-1}{24} + \sum_{n\geq 1}\sigma_1(n)(q^n - dq^{dn}).$$

The constant is $[5(6-1)+(2-1)-(3-1)]/24 = 1$, and the other coefficients agree by the six-residue identity. This proves the shadow-to-eta calculation directly for every coefficient.

The precise primary connecting map is CDH Section 4.2 [1, Section 4.2]:

$$\sigma^X(\tau,z) = \sum_{r\ (12)} \overline{S_r^X(\tau)}\theta_{6,r}(\tau,z), \qquad \mathscr{S}_{1,1}(\sigma^X) = f^X.$$

Its source labels are `eqn:sigmaX`, `def:weight2`, `weight2:Aseries`, and `weight2_EZ`. The displayed Lambert map is its explicit specialization at this root-system shadow. It verifies the resulting function independently of the general Shimura-lift theorem. In particular the shadow and the retained Fricke function are connected by this complete coefficient map.

The original modular subgroup here is $\Gamma_0(6)$, and $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(6)] = 12$. To compute the index, reduction modulo six gives $\mathrm{SL}_2(\mathbb{Z}/6) \cong \mathrm{SL}_2(\mathbb{F}_2)\times\mathrm{SL}_2(\mathbb{F}_3)$ of order $6\cdot 24 = 144$. Reduction is onto: elementary upper and lower unipotent matrices generate over each field, and CRT choices of their entries realize each factor while acting trivially on the other. The upper triangular subgroup modulo six has $\varphi(6)\cdot 6 = 12$ elements (choose its first unit diagonal entry and its upper-right entry; the other diagonal entry is its inverse). Its inverse image is $\Gamma_0(6)$, proving index $144/12 = 12$. This modular-group index and the twelve theta indices have their specified constructions; neither replaces the lattice index seventy-two.

Under the exact Fricke transformation $W_6\tau = -1/(6\tau)$, the eta inversion formula sends each $\eta(dW_6\tau)$ to $\sqrt{-\mathrm{i}(6/d)\tau}\,\eta((6/d)\tau)$. The retained exponent vector at $d = 1,2,3,6$ is $(5,-1,1,-5)$; its reversed vector is its negative and its exponent sum is zero. The τ factors therefore cancel. Since each $6/d$ is positive, the consistent square-root branch gives the positive scalar

$$\prod_{d|6}(6/d)^{a_d/2} = 6^{5/2}3^{-1/2}2^{1/2} = 72.$$

Consequently

$$\boxed{T_6(W_6\tau) = 72/T_6(\tau).}$$

For the weight-two slash action of the determinant-six representative,

$$(f|_2 W_6)(\tau) = (6\tau^2)^{-1}f(-1/(6\tau)).$$

Differentiating the last identity and using $d(W_6\tau)/d\tau = (6\tau^2)^{-1}$ gives

$$\boxed{f^X|_2 W_6 = -f^X.}$$

On the Jacobi side, the primary Eichler–Zagier operator $\mathcal{W}_6(6)$ acts on theta by $\Omega_6(6) = R$. This is verified by its defining congruences, $r+s = 0 \pmod{12}$, $r-s = 0 \pmod 2$. On the odd shadow

and on the corresponding skew-Jacobi form its eigenvalue is minus one. The specialized coefficient map just proved carries that negative eigenline to the negative Fricke eigenline of f^X. This proves an exact equivariant linear comparison on these lines; it does not evaluate an $\mathrm{SL}_2(\mathbb{Z})$ character on the determinant-six matrix.

Finally, the original sheet coordinates remain

$$J_6 = T_6 + 5 + 72/T_6, \qquad Y_6 = T_6 - 72/T_6.$$

Direct substitution and differentiation prove

$$J_6(W_6\tau) = J_6(\tau), \quad Y_6(W_6\tau) = -Y_6(\tau), \quad Y_6^2 = (J_6-5)^2-288, \quad dJ_6 = Y_6\, dT_6/T_6 = -2\pi\mathrm{i}Y_6 f^X\, d\tau.$$

Thus the original scalar 72, the additive shift 5, the branch constant 288, and the differential factor $-2\pi\mathrm{i}$ all survive the complete shadow-to-sheet calculation.

9.6 Established scope and reproducibility

The constructed maps are the ADE coefficient specialization, the exact finite-theta intertwiner, the multiplier determinant and its dual, the quadratic similitude to the original discriminants, the marked glue-group trace prescription, the original six-to-twelve Fourier map, and the coefficient-level shadow-to-Fricke map. Their kernels, weights, signs, and constants were specified and proved above. They establish concrete relations even where a direct isometry or an identical group action is absent. They do not construct a new graded moonshine module or a new moonshine theorem. The name "umbral shadow" here refers to precisely the CDH prescription, with its source provenance retained.

The standard-library program `check_shadow.py` uses exact arithmetic in $\mathbb{Q}[z]/(z^8 - z^4 + 1)$ with $z = e(1/24)$. It verifies both full and odd matrices, the Fourier determinant by the Vandermonde product, the Gauss sum, the character signs, the Fourier embedding, all 48 original-code character records, the first 240 Lambert coefficients, and the squared positive Fricke scalar 72^2. The proofs above establish the unbounded identities independently of those finite checks.

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